

The quantum Hall effect (QHE) is one of the most remarkable condensed matter phenomena discovered in the second half of the twentieth century. It rivals superconductivity in its fundamental significance as a manifestation of quantum mechanics on macroscopic scales. Like superconductivity, there is dissipationless transport and it has fundamental implications for electrical metrology. Surprisingly, it provides a new method with which to measure the fine-structure constant with extreme precision in a dirty and disordered solid-state system. In this chapter we will study the integer version of the quantum Hall effect (IQHE), which takes place in a 2D electron gas (2DEG) in a high magnetic field. The universal values of the Hall resistance observed in transport were a total experimental surprise, but can now be understood by invoking disorder and single-particle quantum dynamics, but largely ignoring electron–electron interactions. In Chapter 16 we will study the fractional quantum Hall effect (FQHE), where the physics is that of strongly correlated interacting electrons, which bizarrely exhibit states described by fractional quantum numbers!

Furthermore, we will learn in Chapters 13 and 14 that the remarkable physics and universality of the transport properties is a manifestation of the appearance of topology in the mathematical description of these systems. The new theoretical ideas associated with the role of topology have implications in many different fields of physics beyond condensed matter.

We turn now to an introduction to the phenomenology of the IQHE.

## 12.1 Hall-Effect Transport in High Magnetic Fields

As noted in Chapter 7, the Lorentz force associated with an applied magnetic field causes a component of the electric field to develop which is perpendicular both to the current and to the magnetic field. The conductivity and resistivity therefore become tensors:

$$J_\mu = \sigma_{\mu\nu} E_\nu, \quad (12.1)$$

$$E_\mu = \rho_{\mu\nu} J_\nu. \quad (12.2)$$

For magnetic field  $\vec{B} = B\hat{z}$  and current density  $\vec{J} = J\hat{x}$ , the requirement that the  $\hat{y}$  component of the electric force cancels out the corresponding component of the Lorentz force (so that there is no current in the  $\hat{y}$  direction) yields the standard classical expression for the Hall resistivity<sup>1</sup>

<sup>1</sup> The corresponding expression in SI units has no factor of  $c$ .

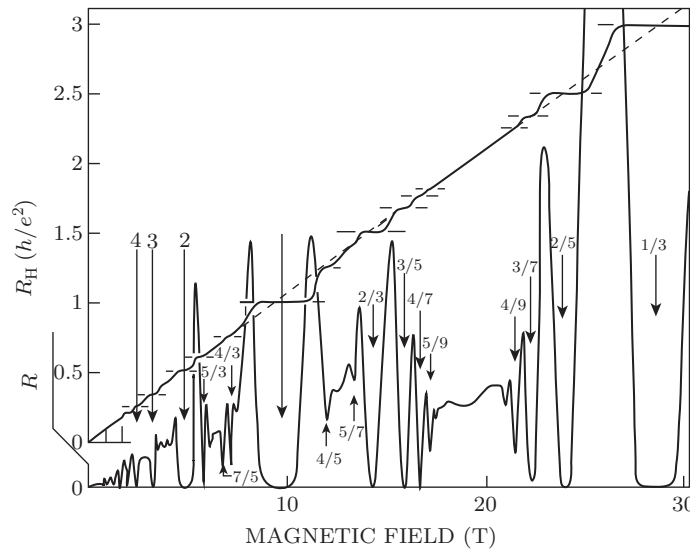
$$\rho_{yx} = \frac{B}{nqc}, \quad (12.3)$$

where  $q$  is the charge of the current-carrying particles. This expression shows that  $\rho_{yx}$  grows linearly with the magnetic field  $B$ , with a coefficient that depends only on the carrier density  $n$  and charge  $q$ , but *not* on the scattering time or other measures of randomness. As a result, the Hall effect can be very useful for measuring the density of carriers and the sign of their charge (electron-like or hole-like), and is indeed widely used for this purpose in ordinary metals and doped semiconductors.

The basic experimental observation of the quantum Hall effect in a high-mobility 2DEG at very low temperature is dramatically different from the classical result given above; see Fig. 12.1. While following the linear field dependence of (12.3) at weak field, beyond several tesla one observes instead a series of so-called **quantum Hall plateaux** on which the Hall resistance  $R_H$  is quantized:<sup>2</sup>

$$R_H = \frac{V_y}{I_x} = \rho_{yx} = \frac{1}{\nu} \frac{h}{e^2}, \quad (12.4)$$

where  $\nu$  is a quantum number whose meaning will be made clear soon. This quantization is universal and independent of all microscopic details such as the type of semiconductor material, the purity of the sample, the precise value of the magnetic field on a specific plateau, and so forth. As a result, the effect is now used to maintain<sup>3</sup> the standard of electrical resistance by metrology laboratories



**Figure 12.1** Transport data of a high-mobility two-dimensional electron gas (2DEG), showing the Hall resistance  $R_H$  and the (dissipative) longitudinal resistance  $R$  as functions of the magnetic field, at low temperature. Plateaux in  $R_H$ , and corresponding deep minima in  $R$ , are labeled by the quantum number  $\nu$ . Integer and fractional values of  $\nu$  correspond to the integer and fractional quantum Hall effects, respectively. Figure reprinted from [79] with permission from Elsevier.

<sup>2</sup> See the discussion further below about the fact that resistance and resistivity have the same units in 2D.

<sup>3</sup> Maintain does *not* mean *define*. The SI ohm is defined in terms of the kilogram, the second, and the speed of light (formerly the meter). It is best realized using the reactive impedance of a capacitor whose capacitance is computed from first principles. This is an extremely tedious procedure, and the QHE is a very convenient method for realizing a fixed, reproducible impedance to check for drifts of resistance standards. It does not, however, *define* the ohm. Rather it is a measure of  $h/e^2$  in units of the SI ohm. At the time of writing a proposal is under active consideration to redefine the SI system of units in such a way that, like the speed of light, both Planck's constant and the electron charge will be

around the world. In addition, since the speed of light now has a defined value,<sup>4</sup> measurement of  $h/e^2$  is equivalent to a measurement of the fine-structure constant of fundamental importance in quantum electrodynamics.

Associated with each of these plateaux is a dramatic decrease in the (dissipative) longitudinal resistance  $R \propto \rho_{xx} \rightarrow 0$  which drops by as much as 13 orders of magnitude in the plateau regions, in sharp contrast to the Drude theory which, as we saw in Eq. (7.22), predicts a  $B$ -independent  $\rho_{xx}$  because its effect (the Lorentz force) is canceled out by the Hall electric field. Clearly the system is undergoing a sequence of phase transitions into highly idealized dissipationless states. As we will discuss later, a vanishingly small dissipation ( $\rho_{xx}$ ) is crucial to the observability of the quantization of  $R_H$ .

In the so-called integer quantum Hall effect (IQHE) discovered by von Klitzing, Dorda, and Pepper in 1980 [80], the quantum number  $\nu$  is a simple integer to within an experimental precision of about  $10^{-11}$  and an absolute accuracy of about  $10^{-9}$  (both being limited by our ability to do resistance metrology [81]).<sup>5</sup> We will discuss the origin of this precise quantization in this chapter, using a non-interacting-electron model.

In 1982, Tsui, Störmer, and Gossard [82] discovered that, in certain devices with reduced (but still non-zero) disorder, the quantum number  $\nu$  could take on rational fractional values. This so-called fractional quantum Hall effect (FQHE) is the result of quite different underlying physics involving strong Coulomb interactions and correlations among the electrons. The particles condense into special quantum states whose excitations have the bizarre property of carrying *fractional* quantum numbers, including fractional charge and fractional statistics that are intermediate between ordinary Bose and Fermi statistics. The FQHE has proven to be a rich and surprising arena for the testing of our understanding of strongly correlated quantum systems. With a simple twist of a dial on her apparatus, the quantum Hall experimentalist can cause the electrons to condense into a bewildering array of new “vacua,” each of which is described by a different topological quantum field theory. The novel “topological order” describing these phases is completely unprecedented and has revolutionized our understanding of the types of order that can occur in quantum fluids.

To take a first peek into the fascinating physics we are going to explore, and to get an idea about the physical meaning of the quantum number  $\nu$ , we compare Eq. (12.4) with the classical expression for the Hall resistivity given in Eq. (12.3). Clearly the classical expression knows nothing about Planck’s constant and thus quantum mechanics. Nevertheless, we can insist on using  $h/e^2$ , the quantum unit of resistance, in the expression above:

$$R_H = \frac{B}{nec} = \frac{h}{e^2} \frac{Be}{nhc} = \frac{h}{e^2} \frac{BA}{hc/e} \frac{1}{nA} = \frac{h}{e^2} \frac{\Phi}{\Phi_0} \frac{1}{N_e} = \frac{h}{e^2} \frac{N_\Phi}{N_e}, \quad (12.5)$$

where  $A$  is the area of the system,  $N_e = nA$  is the number of electrons,  $\Phi = BA$  is the flux enclosed in the system,

$$\Phi_0 = \frac{hc}{e} \quad (12.6)$$

exactly defined. This will have the benefit that the kilogram will no longer be defined in terms of the mass of a lump of platinum–iridium alloy that was cast in a foundry in 1879 and has been slowly losing weight every time it is cleaned.

<sup>4</sup> With the speed of light defined, the meter is no longer defined by an artifact (the meter bar) but rather by the distance light travels in a certain defined fraction of the second.

<sup>5</sup> As von Klitzing noted in his 1985 Nobel lecture, several experimentalists in addition to himself had seen flat Hall plateaux on many occasions over several years prior to his February 1980 measurement showing that the value of the Hall resistance on the plateaux was very precisely quantized in units of  $h/e^2$  independently of the sample geometry and the precise value of the magnetic field. Any of these people could have made the discovery, but it appears that it was von Klitzing’s early training in metrology that made the decisive difference.

is the flux quantum, and  $N_\Phi$  is the number of flux quanta in the system. Comparing Eqs. (12.3) and (12.5), we can identify the quantum number

$$\nu = \frac{N_e}{N_\Phi} \quad (12.7)$$

as the number of electrons per flux quantum in the system. For reasons that will become apparent, this ratio is commonly referred to as the (Landau-level) filling factor. The observation of the QHE suggests that when the ratio between the electron and flux numbers lies close to certain special numbers, the 2DEG forms very special collective quantum ground states whose transport properties are robust against changes of system parameters such as the magnetic field. The purpose of this chapter and Chapter 16 is to reveal the nature of such states.

We begin with a brief description of why two-dimensionality is important to the universality of the result. We then give a review of the classical and semiclassical theories of the motion of charged particles in a magnetic field. Next, we consider the limit of low temperatures and strong fields where a full quantum treatment of the dynamics is required. After that we will be in a position to understand the importance of Anderson localization in the observed quantization, as well as quantum phase transitions in the IQHE. In Chapter 13 we will show that the Hall conductance is actually equal to a topological invariant known as the **first Chern number** of the filled (magnetic) Bloch bands, thus revealing the topological nature of its quantization. This further elucidates the robustness of the quantization against (non-universal) microscopic details. In Chapter 16 we will study the physics described by the novel wave function invented by Robert Laughlin [83] to describe the special condensed state of the electrons on the  $\nu = 1/3$  plateau, as well as the physics behind the many other fractional quantum Hall plateaux visible in Fig. 12.1.

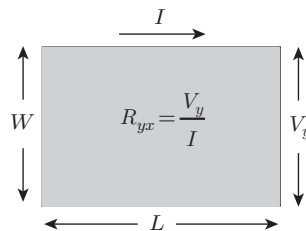
## 12.2 Why 2D Is Important

As we learned in our study of scaling in the localization transition, resistivity (which is what theorists calculate) and resistance (which is what experimentalists measure) for *classical* systems (in the shape of a hypercube) of size  $L$  are related by

$$R = \rho L^{(2-d)}. \quad (12.8)$$

Two dimensions (2D) is therefore special since, in this case, the resistance of the sample is *scale-invariant*. Closely related to this is the fact that, in 2D,  $R$  and  $\rho$  have the *same* dimensionality as  $h/e^2$ . However,  $R$  is still *geometry-dependent*. This is illustrated in Fig. 12.2, where we have a rectangular-shaped sample of size  $L \times W$ . In this case a correction factor enters the expression for  $R$ :

$$R = \rho \frac{L}{W}. \quad (12.9)$$



**Figure 12.2** Illustration of scale- and geometry-independence of the Hall resistance for a rectangular-shaped sample.

This factor is the aspect ratio  $L/W$ , a dimensionless geometric property of the sample (describing its shape<sup>6</sup> but *not* its size).

The situation is very different in a Hall measurement. Assuming current flows in the horizontal ( $x$ ) direction, we have

$$R_{yx} = V_y/I_x = E_y W/I_x = E_y/(I_x/W) = E_y/J_x = \rho_{yx}. \quad (12.10)$$

Thus the aspect ratio  $L/W$  (or any other geometric parameter) does *not* enter the relation between  $R_{yx}$  and  $\rho_{yx}$ . This turns out to be crucial to the universality of the result. In particular, it means that one does not have to measure the physical dimensions of the sample to one part in  $10^{10}$  in order to obtain the Hall resistivity to that precision. Since the locations of the edges of the sample are not well-defined enough for one to even contemplate such a measurement, this is a very fortunate feature.

The above arguments are essentially based on dimensional analysis specific to 2D. It further turns out that, since the dissipation is nearly zero in the QHE states, even the shape of the sample and the precise location of the Hall voltage probes are almost completely irrelevant. (There is no voltage drop along the direction of the current so the sample edges are isopotentials; as a result mis-aligning the Hall voltage probes, which normally would pick up some *longitudinal* voltage drop if  $R$  were non-zero, has no consequence since  $R \rightarrow 0$ .) We will trace the lack of dissipation to a quantum effect that is also peculiar to 2D, namely the quenching of the kinetic energy by the strong magnetic field. Taken together, all of the factors we have mentioned above mean that there is essentially no need to distinguish between the Hall resistance and the Hall resistivity.

## 12.3 Why Disorder and Localization Are Important

Paradoxically, the extreme universality of the transport properties in the quantum Hall regime occurs because of, rather than in spite of, the random disorder and uncontrolled imperfections which the devices contain. Anderson localization in the presence of disorder plays an essential role in the quantization, but this localization is strongly modified from the usual case because of the strong magnetic field.

In our study of Anderson localization we found that in 2D all states are localized even for arbitrarily weak disorder. The essence of this weak localization effect was shown to be the logarithmic divergence of the time integral of the current “echo” associated with the quantum interference corrections to classical transport. We further saw that these quantum interference effects relied crucially on the existence of time-reversal symmetry. In the presence of a strong quantizing magnetic field (which breaks time-reversal symmetry), the localization properties of the disordered 2D electron gas are radically altered. We will shortly see that there exists a sequence of novel phase transitions, not between a metal and an insulator, but rather between two distinctly different integer quantum Hall states (here we view the insulating state as a special quantum Hall state with zero Hall conductance).

In the absence of any impurities, the 2DEG is translationally invariant and there does not exist any preferred frame of reference.<sup>7</sup> As a result, we can transform to a frame of reference moving with velocity  $-\vec{v}$  relative to the lab frame. In this frame the electrons appear to be moving at velocity  $+\vec{v}$  and carrying current density

$$\vec{J} = -ne\vec{v}, \quad (12.11)$$

<sup>6</sup> Practitioners in the field often describe the aspect ratio as the length of the sample in “squares,” that is, the number of squares of side  $W$  that fit in the length  $L$ .

<sup>7</sup> This assumes that we can ignore the periodic potential of the crystal, which is of course fixed in the lab frame. Within the effective-mass approximation this potential modifies the mass but does not destroy the Galilean invariance, since the energy is still quadratic in the momentum.

where  $n$  is the areal density. In the lab frame, the electromagnetic fields are

$$\vec{E} = \vec{0}, \quad (12.12)$$

$$\vec{B} = B\hat{z}. \quad (12.13)$$

In the moving frame they are Lorentz-transformed (to lowest order in  $v/c$ ) to

$$\vec{E} = -\frac{1}{c}\vec{v} \times \vec{B}, \quad (12.14)$$

$$\vec{B} = B\hat{z}. \quad (12.15)$$

This Lorentz transformation picture is precisely equivalent to the usual statement that an electric field must exist which just cancels out the Lorentz force  $-(e/c)\vec{v} \times \vec{B}$  in order for the device to carry the current straight through without deflection. Thus we have

$$\vec{E} = \frac{B}{nec} \vec{J} \times \hat{B}. \quad (12.16)$$

The resistivity tensor is defined by Eq. (12.2), and from this we can make the identification

$$\vec{\rho} = \frac{B}{nec} \begin{pmatrix} 0 & +1 \\ -1 & 0 \end{pmatrix}. \quad (12.17)$$

The conductivity tensor from Eq. (12.1) is the matrix inverse of this, so that

$$\vec{\sigma} = \frac{nec}{B} \begin{pmatrix} 0 & -1 \\ +1 & 0 \end{pmatrix}. \quad (12.18)$$

Notice that, paradoxically, the system looks insulating since  $\sigma_{xx} = 0$  and yet it also looks like a perfect conductor since  $\rho_{xx} = 0$ . In an ordinary insulator  $\sigma_{xy} = 0$ , so  $\sigma_{xx} = 0$  implies  $\rho_{xx} = \infty$ . Here  $\sigma_{xy} = -nec/B \neq 0$ , so the inverse of  $\vec{\sigma}$  always exists.

The argument given above relies only on Lorentz covariance. The only property of the 2DEG that entered was the density. The argument works equally well irrespective of whether the system is classical or quantum, and whether the electron state is liquid, vapor, or solid. It simply does not matter. Thus, in the absence of disorder, the Hall effect teaches us nothing about the system other than its density. We cannot even see that the kinetic energy is quantized into Landau levels. The Hall resistivity is simply a linear function of the magnetic field whose slope tells us about the electron density. In the quantum Hall regime we would therefore see none of the novel physics exhibited in Fig. 12.1 if we had a perfect sample, since disorder is *needed* in order to destroy translation invariance.

We have thus shown that the random impurity potential is a necessary condition for deviations from simple linear behavior to occur. Understanding how it leads to the IQHE is our task in the remainder of this chapter. A hint that Anderson localization must play a major role comes from the fact on the quantum Hall plateaux there is no dissipative response ( $\sigma_{xx} \rightarrow 0$ ), suggesting that states at the Fermi level are localized (in the bulk).

## 12.4 Classical and Semiclassical Dynamics

### 12.4.1 Classical Dynamics

The classical equations of motion for a particle of charge  $-e$  moving in 2D under the influence of the Lorentz force  $-(e/c)\vec{v} \times \vec{B}$  caused by a magnetic field  $\vec{B} = B\hat{z}$  are

$$\begin{aligned} m_e \ddot{x} &= -\frac{eB}{c} \dot{y}; \\ m_e \ddot{y} &= +\frac{eB}{c} \dot{x}. \end{aligned} \quad (12.19)$$

The general solution of these equations corresponds to motion in a circle of arbitrary radius  $R$ :

$$\vec{r} = R[\cos(\omega_c t + \delta), \sin(\omega_c t + \delta)]. \quad (12.20)$$

Here  $\delta$  is an arbitrary phase for the motion and

$$\omega_c \equiv \frac{eB}{m_e c} \quad (12.21)$$

is known as the classical **cyclotron frequency**. Notice that the period of the orbit is independent of the radius and that the tangential speed

$$v = R\omega_c \quad (12.22)$$

controls the radius. A fast particle travels in a large circle but returns to the starting point in the same length of time as a slow particle which (necessarily) travels in a small circle. The motion is thus *isochronous*, much like that of a harmonic oscillator, whose period is independent of the amplitude of the motion. This apparent analogy is not an accident, as we shall see when we study the Hamiltonian (which we will need for the full quantum solution).

Because of some subtleties involving distinctions between canonical and mechanical momentum in the presence of a magnetic field, it is worth reviewing the formal Lagrangian and Hamiltonian approaches to this problem. The above classical equations of motion follow from the Lagrangian

$$\mathcal{L} = \frac{1}{2}m_e \dot{x}^\mu \dot{x}^\mu - \frac{e}{c} \dot{x}^\mu A^\mu, \quad (12.23)$$

where  $\mu = 1, 2$  refers to  $x$  and  $y$ , respectively, and  $\vec{A}$  is the vector potential evaluated at the position of the particle. (We use the Einstein summation convention throughout this discussion.) Using

$$\frac{\delta \mathcal{L}}{\delta x^\nu} = -\frac{e}{c} \dot{x}^\mu \partial_\nu A^\mu \quad (12.24)$$

and

$$\frac{\delta \mathcal{L}}{\delta \dot{x}^\nu} = m_e \dot{x}^\nu - \frac{e}{c} A^\nu, \quad (12.25)$$

the Euler–Lagrange equation of motion becomes

$$m_e \ddot{x}^\nu = -\frac{e}{c} [\partial_\nu A^\mu - \partial_\mu A^\nu] \dot{x}^\mu. \quad (12.26)$$

Using

$$\vec{B} = \nabla \times \vec{A}, \quad (12.27)$$

or equivalently

$$B^\alpha = \epsilon^{\alpha\beta\gamma} \partial_\beta A^\gamma, \quad (12.28)$$

it is easy to show that Eq. (12.26) is equivalent to Eqs. (12.19).

Once we have the Lagrangian, we can deduce the canonical momentum

$$p^\mu \equiv \frac{\delta \mathcal{L}}{\delta \dot{x}^\mu} = m_e \dot{x}^\mu - \frac{e}{c} A^\mu, \quad (12.29)$$

and the Hamiltonian<sup>8</sup>

$$\begin{aligned} H[\vec{p}, \vec{x}] &\equiv \dot{x}^\mu p^\mu - \mathcal{L}(\dot{x}, \vec{x}) \\ &= \frac{1}{2m_e} \left( p^\mu + \frac{e}{c} A^\mu \right) \left( p^\mu + \frac{e}{c} A^\mu \right) = \frac{\Pi^\mu \Pi^\mu}{2m_e}. \end{aligned} \quad (12.30)$$

<sup>8</sup> Recall that the Lagrangian is canonically a function of the positions and velocities, while the Hamiltonian is canonically a function of the positions and momenta.

The quantity

$$\Pi^\mu \equiv p^\mu + \frac{e}{c} A^\mu \quad (12.31)$$

is known as the *mechanical* momentum. Hamilton's equations of motion

$$\dot{x}^\mu = \frac{\partial H}{\partial p^\mu} = \frac{\Pi^\mu}{m_e}, \quad (12.32)$$

$$\dot{p}^\mu = -\frac{\partial H}{\partial x^\mu} = -\frac{e}{m_e c} \left( p^\nu + \frac{e}{c} A^\nu \right) \partial_\mu A^\nu \quad (12.33)$$

show that it is the mechanical momentum, not the canonical momentum, that is equal to the usual expression related to the velocity,

$$\Pi^\mu = m_e \dot{x}^\mu. \quad (12.34)$$

Equation (12.33) can be recast as

$$\dot{\Pi}^\mu = -\frac{e}{m_e c} \Pi^\nu [\partial_\mu A^\nu - \partial_\nu A^\mu]. \quad (12.35)$$

Equivalently, using Hamilton's equations of motion we can recover Newton's law for the Lorentz force given in Eq. (12.26) by simply taking a time derivative of  $\dot{x}^\mu$  in Eq. (12.32).

The distinction between canonical and mechanical momentum can lead to confusion. For example, it is possible for the particle to have a finite velocity while having zero (canonical) momentum! Furthermore, the canonical momentum is dependent (as we will see later) on the choice of gauge for the vector potential and hence is not a physical observable. The mechanical momentum, being simply related to the velocity (and hence the current) is physically observable and gauge-invariant. The classical equations of motion involve only the curl of the vector potential, so the particular gauge choice is not very important at the classical level. We will therefore delay discussion of gauge choices until we study the full quantum solution, where the issue is unavoidable.

## 12.4.2 Semiclassical Approximation

Recall that, in the semiclassical approximation previously developed when we studied transport theory in Chapter 8, we considered wave packets  $\Psi_{\vec{R}(t), \vec{K}(t)}(\vec{r}, t)$  made up of a linear superposition of Bloch waves. These packets are large on the scale of the de Broglie wavelength so that they have a well-defined central wave vector  $\vec{K}(t)$ , but small on the scale of everything else (external potentials, etc.) so that they simultaneously can be considered to have well-defined mean position  $\vec{R}(t)$ . (Note that  $\vec{K}$  and  $\vec{R}$  are *parameters* labeling the wave packet, not arguments of the wave function.) We then argued that the solution of the Schrödinger equation in this semiclassical limit gives a wave packet whose parameters  $\vec{K}(t)$  and  $\vec{R}(t)$  obey the appropriate analog of the classical Hamilton equations of motion,

$$\dot{R}^\mu = \frac{\partial \langle \Psi_{\vec{R}, \vec{K}} | H | \Psi_{\vec{R}, \vec{K}} \rangle}{\partial (\hbar K^\mu)}; \quad (12.36)$$

$$\hbar \dot{K}^\mu = -\frac{\partial \langle \Psi_{\vec{R}, \vec{K}} | H | \Psi_{\vec{R}, \vec{K}} \rangle}{\partial R^\mu}. \quad (12.37)$$

Naturally this leads to the same circular motion of the wave packet at the classical cyclotron frequency discussed above. For weak fields and fast electrons, the radii of these circular orbits will be large compared with the size of the wave packets, and the semiclassical approximation will be valid. However, at strong fields, the approximation begins to break down because the orbits are too small and because  $\hbar\omega_c$  becomes a significant (large) energy. Thus we anticipate that the semiclassical regime requires that the cyclotron energy be much smaller than the Fermi energy,  $\hbar\omega_c \ll \epsilon_F$ .

We have already seen hints that the problem we are studying is really a harmonic oscillator problem. For the harmonic oscillator there is a characteristic energy scale  $\hbar\omega$  (in this case  $\hbar\omega_c$ ) and a characteristic length scale<sup>9</sup>  $\ell$  for the zero-point fluctuations of the position in the ground state. The analogous quantity in this problem is the so-called **magnetic length**

$$\ell \equiv \sqrt{\frac{\hbar c}{eB}} \approx \frac{257 \text{ \AA}}{\sqrt{B/\text{T}}}. \quad (12.38)$$

The physical interpretation of this length is that the area  $2\pi\ell^2$  contains one quantum of magnetic flux  $\Phi_0 = hc/e$ . That is to say, the density of magnetic flux is

$$B = \frac{\Phi_0}{2\pi\ell^2}. \quad (12.39)$$

For the system to be in the semiclassical limit one then requires that the de Broglie wavelength be small on the scale of the magnetic length, or  $k_F\ell \gg 1$ . This condition turns out to be equivalent to  $\hbar\omega_c \ll \epsilon_F$ , so they are not separate constraints.

**Exercise 12.1.** Use the Bohr–Sommerfeld quantization condition that the orbit have a circumference containing an integral number of de Broglie wavelengths to find the allowed orbits of a 2D electron moving in a uniform magnetic field. Show that each successive orbit encloses precisely one additional quantum of flux in its interior. Hint: it is important to make the distinction between the canonical momentum (which controls the de Broglie wavelength) and the mechanical momentum (which controls the velocity). The calculation is simplified if one uses the symmetric gauge  $\vec{A} = -\frac{1}{2}\vec{r} \times \vec{B}$ , in which the vector potential is purely azimuthal and independent of the azimuthal angle.

## 12.5 Quantum Dynamics in Strong $B$ Fields

Since we will be dealing with the Hamiltonian and the Schrödinger equation, our first order of business is to choose a gauge for the vector potential. One convenient choice is the so-called **Landau gauge**,

$$\vec{A}(\vec{r}) = -xB\hat{y}, \quad (12.40)$$

which obeys  $\nabla \times \vec{A} = -B\hat{z}$  (we change the sign or direction of the  $B$  field simply to avoid some awkward minus signs here as well as in Chapter 16). In this gauge, the vector potential points in the  $y$  direction but varies only with the  $x$  position. Hence the system still has translation invariance in the  $y$  direction. Notice that the magnetic field is translationally invariant, but the Hamiltonian is not! This is one of many peculiarities of dealing with vector potentials. Translation leaves the curl of  $\vec{A}$  invariant but not  $\vec{A}$  itself. That is, a gauge change has occurred. The eigenvalues of the Hamiltonian are invariant under translation but the eigenfunctions are subject to a unitary transformation which depends on the gauge choice.

**Exercise 12.2.** Show for the Landau gauge that, even though the Hamiltonian is not invariant for translations in the  $x$  direction, the physics is still invariant, since the change in the Hamiltonian that occurs under translation is simply equivalent to a gauge change. Prove this for any arbitrary gauge, assuming only that the magnetic field is uniform.

<sup>9</sup> Not to be confused with the mean free path, for which we used the same symbol in previous chapters!

We will also ignore the periodic potential and associated band structure due to the lattice for the moment. This is justifiable as long as the effective-mass approximation is valid.<sup>10</sup> Possible band structure effects at very high magnetic fields will be discussed in Section 12.9. The Hamiltonian can thus be written in the Landau gauge as

$$H = \frac{1}{2m_e} \left[ p_x^2 + \left( p_y - \frac{eB}{c}x \right)^2 \right]. \quad (12.41)$$

Taking advantage of the translation symmetry in the  $y$  direction, let us attempt a separation of variables by writing the wave function in the form

$$\psi_k(x, y) = e^{iky} f_k(x). \quad (12.42)$$

This has the advantage that it is an eigenstate of  $p_y$  and hence we can make the replacement  $p_y \rightarrow \hbar k$  in the Hamiltonian. After separating variables, we have the effective 1D Schrödinger equation

$$h_k f_k(x) = \epsilon_k f_k(x), \quad (12.43)$$

where (note the similarity to Eq. (7.65))

$$h_k \equiv \frac{p_x^2}{2m_e} + \frac{1}{2m_e} \left( \hbar k - \frac{eB}{c}x \right)^2. \quad (12.44)$$

This is simply a 1D displaced harmonic oscillator

$$h_k = \frac{p_x^2}{2m_e} + \frac{1}{2} m_e \omega_c^2 (x - k\ell^2)^2, \quad (12.45)$$

whose frequency is the classical cyclotron frequency and whose central position (also known as the **guiding center**<sup>11</sup> position)

$$X_k = k\ell^2 \quad (12.46)$$

is (somewhat paradoxically) determined by the  $y$  momentum quantum number. Thus for each plane wave chosen for the  $y$  direction there will be an entire family of energy eigenvalues

$$\epsilon_{kn} = \left( n + \frac{1}{2} \right) \hbar \omega_c, \quad (12.47)$$

which depend only on  $n$  but are completely independent of the  $y$  momentum  $\hbar k$ . The corresponding (unnormalized) eigenfunctions are

$$\psi_{nk}(\vec{r}) = e^{iky} H_n(x/\ell - k\ell) e^{-\frac{1}{2\ell^2}(x - k\ell^2)^2}, \quad (12.48)$$

where  $H_n$  is (as usual for harmonic oscillators) the  $n$ th Hermite polynomial (in this case displaced to the new central position  $X_k$ ).

**Exercise 12.3.** Verify that Eq. (12.48) is in fact a solution of the Schrödinger equation as claimed.

<sup>10</sup> Provided that the lattice constant is much smaller than the magnetic length  $\ell$ , it is a good approximation to simply use the band mass instead of the bare mass in Schrödinger's equation.

<sup>11</sup> Classically the guiding center is the center of the cyclotron motion of a charged particle in a magnetic field. What we have here is its quantum analog. In the Landau gauge used here, it is simply the  $X$  coordinate of the center of motion, the  $Y$  position being uncertain. We will later encounter in Eq. (16.8) a gauge-invariant formulation of the guiding center as a 2D vector position operator.

These harmonic oscillator levels are called **Landau levels**. Owing to the lack of dependence of the energy on  $k$ , the degeneracy of each level is enormous, as we will now show. We assume periodic boundary conditions in the  $y$  direction. Because of the vector potential, it is *impossible* to simultaneously have periodic boundary conditions in the  $x$  direction. However, since the basis wave functions are harmonic oscillator polynomials multiplied by strongly converging Gaussians, they rapidly vanish for positions away from the center position  $X_k$ . Let us suppose that the sample is rectangular with dimensions  $L_x, L_y$  and that the left-hand edge is at  $x = 0$  and the right-hand edge is at  $x = L_x$ . Then the values of the wave vector  $k$  for which the basis state is substantially inside the sample run from  $k = 0$  to  $k = L_x/\ell^2$ . It is clear that the states at the left edge and the right edge differ strongly in their  $k$  values, and hence periodic boundary conditions are impossible.<sup>12</sup>

The total number of states in *each* Landau level is then

$$N = \sum_k 1 = \frac{L_y}{2\pi} \int_0^{L_x/\ell^2} dk = \frac{L_x L_y}{2\pi \ell^2} = N_\Phi, \quad (12.49)$$

where

$$N_\Phi \equiv \frac{BL_x L_y}{\Phi_0} \quad (12.50)$$

is the number of flux quanta penetrating the sample. Thus there is one state per Landau level per flux quantum. As a result, when the filling factor  $\nu$  of Eq. (12.7) takes an integer value, we have in the ground state exactly  $\nu$  (lowest-energy) Landau levels completely filled, and all higher Landau levels empty.<sup>13</sup> Such a state is clearly special, as there is a gap for excitations *in the bulk*, and the 2DEG is *incompressible*, just like in band insulators.<sup>14</sup>

Notice that, even though the family of allowed wave vectors is only 1D, we find that the degeneracy of each Landau level is extensive in the 2D area. The reason for this is that the spacing between wave vectors allowed by the periodic boundary conditions  $\Delta_k = 2\pi/L_y$  *decreases* with increasing  $L_y$ , while the range of allowed wave vectors  $[0, L_x/\ell^2]$  *increases* with increasing  $L_x$ . The reader may also worry that for very large samples, the range of allowed values of  $k$  will be so large that it will fall outside the first Brillouin zone, forcing us to include band mixing and the periodic lattice potential beyond the effective-mass approximation. This is not true, however, since the canonical momentum is a gauge-dependent quantity. The value of  $k$  in any particular region of the sample can be made small by shifting the origin of the coordinate system to that region (thereby making a gauge transformation).

The width of the harmonic oscillator wave functions in the  $n$ th Landau level is of order  $\sqrt{n}\ell$ . This is microscopic compared with the system size, but note that the spacing between the centers

$$\Delta = \Delta_k \ell^2 = \frac{2\pi \ell^2}{L_y} \quad (12.51)$$

is vastly smaller (assuming that  $L_y \gg \ell$ ). Thus the supports of the different basis states are strongly overlapping (but they are still orthogonal due to their different  $k$  quantum numbers).

<sup>12</sup> The best one can achieve is so-called quasi-periodic boundary conditions in which the phase difference between the left and right edges is zero at the bottom and rises linearly with height, reaching  $2\pi N_\Phi \equiv L_x L_y/\ell^2$  at the top. The eigenfunctions with these boundary conditions are elliptic theta functions, which are linear combinations of the Gaussians discussed here. These issues will be discussed more generally and in a gauge-independent manner in Section 12.9.

<sup>13</sup> Here we neglected the fact that electrons carry spin, and each Landau-level orbital state can accommodate two electrons. This is because the magnetic field also lifts the spin degeneracy, and we choose to view different spin states with the same orbital Landau-level wave functions as *different* Landau levels.

<sup>14</sup> We will see very soon that the situation is quite different at the edge of the sample.

**Exercise 12.4.** Using the fact that the energy for the  $n$ th harmonic oscillator state is  $(n + \frac{1}{2})\hbar\omega_c$ , present a semiclassical argument explaining the result claimed above, namely that the width of the support of the wave function scales as  $\sqrt{n}\ell$ .

**Exercise 12.5.** Using the Landau gauge, construct a Gaussian wave packet in the lowest Landau level of the form

$$\Psi(x, y) = \int_{-\infty}^{+\infty} a_k e^{iky} e^{-\frac{1}{2\ell^2}(x-k\ell^2)^2} dk,$$

choosing  $a_k$  in such a way that the wave packet is localized as closely as possible around some point  $\vec{R}$ . What is the smallest-size wave packet that can be constructed without mixing in higher Landau levels?

Having now found the eigenfunctions for an electron in a strong magnetic field, we can relate them back to the semiclassical picture of wave packets undergoing circular cyclotron motion. Consider an initial semiclassical wave packet located at some position and having some specified *mechanical* momentum  $\hbar\vec{K}$ . In the semiclassical limit the mean energy of this packet will greatly exceed the cyclotron energy ( $\hbar^2 K^2/2m \gg \hbar\omega_c$ ) and hence it will be made up of a linear combination of a large number of different Landau-level states centered around  $\bar{n} = \hbar^2 K^2/2m\hbar\omega_c$ :

$$\Psi(\vec{r}, t) = \frac{L_y}{2\pi} \sum_n \int dk a_n(k) \psi_{nk}(\vec{r}) e^{-i(n+\frac{1}{2})\omega_c t}. \quad (12.52)$$

Notice that, in an ordinary 2D problem at zero field, the complete set of plane-wave states would be labeled by a 2D continuous momentum label. Here we have one discrete label (the Landau-level index) and a 1D continuous label (the  $y$  wave vector). Thus the “sum” over the complete set of states is actually a combination of a summation and an integration.

The details of the initial position and momentum are controlled by the amplitudes  $a_n(k)$ . We can immediately see, however, that, since the energy levels are exactly evenly spaced, the motion is exactly periodic:

$$\Psi\left(\vec{r}, t + \frac{2\pi}{\omega_c}\right) = -\Psi(\vec{r}, t). \quad (12.53)$$

If one works through the details, one finds that the motion is indeed circular and corresponds to the expected semiclassical cyclotron orbit.

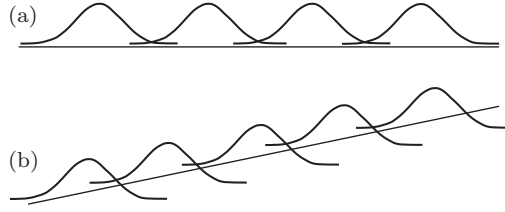
**Exercise 12.6.** Start with a Gaussian wave packet (not projected into any particular Landau level) with an initial position and velocity. Express it in terms of Landau-level eigenfunctions and show that its time evolution corresponds to a semiclassical circular cyclotron orbit.

For simplicity we will restrict the remainder of our discussion mostly to the lowest Landau level, where the (correctly normalized) eigenfunctions in the Landau gauge are (dropping the index  $n = 0$  from now on)

$$\psi_k(\vec{r}) = \frac{1}{\sqrt{\pi^{1/2} L_y \ell}} e^{iky} e^{-\frac{1}{2\ell^2}(x-k\ell^2)^2} \quad (12.54)$$

and every state has the same energy eigenvalue  $\epsilon_k = \frac{1}{2}\hbar\omega_c$ ; see Fig. 12.3(a). Many of the following results will be applicable to higher Landau levels as well, and we will point them out when appropriate.

We imagine that the magnetic field is high and hence the Landau-level splitting is very large compared with the scale of the disorder, so that we can ignore higher Landau levels. (There are some



**Figure 12.3** Illustration of lowest-Landau-level wave functions in the Landau gauge, which are plane waves along the  $\hat{y}$  direction (into the paper) and Gaussian along the  $\hat{x}$  direction (left to right), with different guiding center locations. (a) Without the electric field, all states are degenerate. (b) With the electric field along the  $\hat{x}$  direction, the degeneracy is lifted.

subtleties here, to which we will return.) Because the states are all degenerate, any wave packet made up of any combination of the basis states will be a stationary state. The total current will be zero. We anticipate, however, from semiclassical considerations that there should be some remnant of the classical circular motion visible in the local current density. To see this, note that the current density in the  $k$ th basis state is

$$\langle \vec{J}(x, y) \rangle = -\frac{e}{m_e} \psi_k^*(x, y) \left[ -i\hbar \nabla + \frac{e}{c} \vec{A}(x, y) \right] \psi_k(x, y), \quad (12.55)$$

and the  $y$  component of the current is

$$\begin{aligned} \langle I_y \rangle &= -\frac{e}{m_e \pi^{1/2} \ell} \frac{1}{L_y} \int_{-\infty}^{\infty} dx e^{-\frac{1}{2\ell^2}(x-k\ell^2)^2} \left( \hbar k - \frac{eB}{c} x \right) e^{-\frac{1}{2\ell^2}(x-k\ell^2)^2} \\ &= \frac{e\omega_c}{\pi^{1/2} \ell} \frac{1}{L_y} \int_{-\infty}^{\infty} dx e^{-\frac{1}{\ell^2}(x-k\ell^2)^2} (x - k\ell^2). \end{aligned} \quad (12.56)$$

We see from the integrand that the current density is anti-symmetric about the peak of the Gaussian at  $X_k = k\ell^2$  and hence the total current vanishes. This anti-symmetry is the remnant of the semiclassical circular motion, as the *local* current density points in opposite directions on the two sides of  $X_k$ . The reader should verify that exactly the same happens in all Landau levels.

One might be surprised by the vanishing current we found in Eq. (12.56), given the fact we are dealing with plane waves (along the  $y$  direction), which normally describe propagating states. This is one manifestation of the subtlety associated with the difference between canonical and mechanical momenta mentioned earlier. In Eq. (12.56) the contribution of the canonical momentum to the total current is canceled out *exactly* by the vector potential contribution to the mechanical momentum, yielding a null result. This could perhaps be anticipated from the observation that the energy of the Landau-level states is *independent* of the wave vector  $k$ , thus the group velocity is given by

$$v_y = \frac{1}{\hbar} \frac{\partial \epsilon_{kn}}{\partial k} = 0. \quad (12.57)$$

The lack of dependence of the energy on  $k$  can in turn be traced to the fact that  $k$  determines the guiding center position  $X_k$ , but, in the absence of a potential-energy term in the Hamiltonian (12.41), the energy is independent of the guiding center position. As we will see below, the situation changes completely when we add such a potential energy.

Let us now consider the case of a uniform electric field pointing in the  $x$  direction and giving rise to the potential energy

$$V(\vec{r}) = +eEx. \quad (12.58)$$

This still has translation symmetry in the  $y$  direction, so our Landau gauge choice is still the most convenient. Again, on separating variables we see that the solution is nearly the same as before, except

that the displacement of the harmonic oscillator is slightly different. The Hamiltonian in Eq. (12.44) becomes

$$h_k = \frac{p_x^2}{2m_e} + \frac{1}{2}m_e\omega_c^2 (x - k\ell^2)^2 + eEx. \quad (12.59)$$

Completing the square, we see that the oscillator is now centered at the new (and slightly shifted) position

$$X'_k = k\ell^2 - \frac{eE}{m_e\omega_c^2} = X_k - \frac{eE}{m_e\omega_c^2} \quad (12.60)$$

and the energy eigenvalue is now linearly dependent on the particle's peak position  $X'_k \approx X_k$  (and therefore linear in the  $y$  momentum):

$$\epsilon_{kn} = \left(n + \frac{1}{2}\right)\hbar\omega_c + eEX'_k + \frac{1}{2}m_e\bar{v}^2, \quad (12.61)$$

where, as we will discuss further below,

$$\bar{v} \equiv c \frac{E}{B} \quad (12.62)$$

is the drift velocity. Because of the shift in the peak position of the wave function, the perfect anti-symmetry of the current distribution is destroyed and there is a net current

$$\langle I_y \rangle = -e \frac{\bar{v}}{L_y}, \quad (12.63)$$

showing that  $\bar{v}$  is simply the usual  $c\vec{E} \times \vec{B}/B^2$  drift velocity. Thus the time it takes the electron to travel through the system is  $L_y/\bar{v}$ . This result can be derived either by explicitly doing the integral for the current or by noting that the wave-packet group velocity is

$$\frac{1}{\hbar} \frac{\partial \epsilon_{kn}}{\partial k} = \frac{eE}{\hbar} \frac{\partial X'_k}{\partial k} = \frac{eE}{\hbar} \frac{\partial X_k}{\partial k} = \bar{v}, \quad (12.64)$$

independently of the values of  $k$  and the Landau-level index  $n$  (since the electric field is a constant in this case, giving rise to a strictly linear potential). Thus we have recovered the correct kinematics from our quantum solution.

We now have a clear understanding of the origin of the three terms on the RHS of Eq. (12.61). The first term is of course the original Landau level energy. The second term is the *potential* energy, which is determined by the peak (or guiding center) position. The last term is the *additional* kinetic energy associated with the non-zero drift velocity. Owing to the second term, the applied electric field “tilts” the Landau levels in the sense that their energy is now linear in position as illustrated in Fig. 12.3(b). This means that the degeneracy between states in the same Landau level gets lifted, but there are now degeneracies between states in *different* Landau levels, because the difference in Landau-level energy (the first term of Eq. (12.61)) can be compensated for by the difference in potential energy (the second term of Eq. (12.61)). Nevertheless, what we have found here are the exact eigenstates (i.e. the stationary states). It is not possible for an electron to decay into one of the other degenerate states, because they have different canonical momenta. If, however, disorder or phonons are available to break translation symmetry, then these decays become allowed, and dissipation can appear. The matrix elements for such processes are small if the electric field is weak, because the degenerate states are widely separated spatially due to the small tilt of the Landau levels.

**Exercise 12.7.** It is interesting to note that the exact eigenstates in the presence of the electric field can be viewed as displaced oscillator states in the original (zero  $E$  field) basis. In this basis the displaced states are linear combinations of all the Landau-level excited states of the

same  $k$ . Use first-order perturbation theory to find the amount by which the  $n = 1$  Landau level is mixed into the  $n = 0$  state. Compare this with the exact amount of mixing computed using the exact displaced oscillator state. Show that the two results agree to first order in  $E$ . Because the displaced state is a linear combination of more than one Landau level, it can carry a finite current. Give an argument, derived from perturbation theory, for why the amount of this current is inversely proportional to the  $B$  field, but is independent of the mass of the particle. Note that this quantum perturbation-theory result is in agreement with Eq. (12.62). Hint: how does the mass affect the Landau-level energy spacing and the current operator?

## 12.6 IQHE Edge States

Now that we understand drift in a uniform electric field, we can consider the problem of electrons confined in a Hall bar of finite width by a non-uniform electric field. For simplicity, we will consider the situation where the potential  $V(x)$  is smooth on the scale of the magnetic length, but this is not central to the discussion. If we assume that the system still has translation symmetry in the  $y$  direction, the solution to the Schrödinger equation must still be of the form

$$\psi(x, y) = \frac{1}{\sqrt{L_y}} e^{iky} f_k(x). \quad (12.65)$$

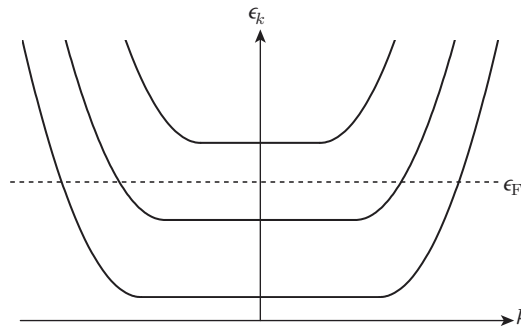
The function  $f_k$  will no longer be a simple harmonic wave function as we found in the case of the uniform electric field. However, we can anticipate that  $f_k$  will still be peaked near (but in general not precisely at) the point  $X_k \equiv k\ell^2$ . The eigenvalues  $\epsilon_{kn}$  will no longer be linear in  $k$  but will still reflect the kinetic energy of the cyclotron motion plus the local potential energy  $V(X_k)$  (plus small corrections analogous to the one in Eq. (12.61)). This is illustrated in Fig. 12.4.

We see that the group velocity

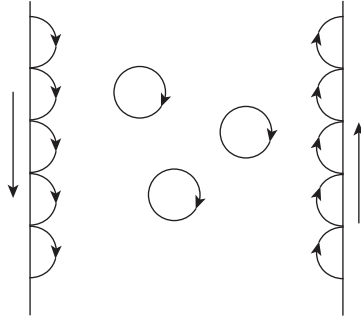
$$\vec{v}_k = \frac{1}{\hbar} \frac{\partial \epsilon_{kn}}{\partial k} \hat{y} \approx \frac{\ell^2}{\hbar} \frac{dV(x)}{dx} \bigg|_{x=X_k} \hat{y} \quad (12.66)$$

has the opposite sign on the two edges of the sample. This means that in the ground state there are edge currents of opposite sign flowing in the sample. The semiclassical interpretation of these currents is that they represent “skipping orbits” in which the circular cyclotron motion is interrupted by collisions with the walls at the edges as illustrated in Fig. 12.5.

One way to analyze the Hall effect in this system is quite analogous to the Landauer picture of transport in narrow wires that we discussed in Chapter 10. The edge states play the role of the left-



**Figure 12.4** Landau-level dispersions in the presence of a confining potential, for a sample with Hall bar geometry.



**Figure 12.5** Cyclotron orbits of electrons, in the presence of a magnetic field pointing into the paper. These orbits are circles in the bulk; however, near the sample edges they cannot be completed due to collisions with the boundaries, thus the electron trajectories are a sequence of “skipping” orbits, resulting in non-zero (average) drift velocities along the edges. The drift velocities point in opposite directions in the opposite edges.

and right-moving states at the two Fermi points. Because momentum in a magnetic field corresponds to position, the edges are essentially real-space realizations of the Fermi surface. A Hall voltage drop across the sample in the  $x$  direction corresponds to a difference in electrochemical potential between the two edges. As in our analysis of the Landauer formulation of transport, we will choose to apply this in the form of a chemical-potential difference and ignore any changes in electrostatic potential.<sup>15</sup> This has the effect of increasing the number of electrons in skipping orbits on one edge of the sample and/or decreasing the number on the other edge. Previously the net current due to the two edges was zero, but now there is a net Hall current. To calculate this current we have to add up the group velocities of all the occupied states,

$$I = -\frac{e}{L_y} \int_{-\infty}^{+\infty} dk \frac{L_y}{2\pi} \frac{1}{\hbar} \frac{\partial \epsilon_k}{\partial k} n_k, \quad (12.67)$$

where for the moment we assume that, in the bulk, only a single Landau level is occupied and  $n_k$  is the probability that state  $k$  in that Landau level is occupied. Assuming zero temperature and noting that the integrand is a perfect derivative, we have

$$I = -\frac{e}{h} \int_{\mu_L}^{\mu_R} d\epsilon = -\frac{e}{h} [\mu_R - \mu_L]. \quad (12.68)$$

The definition of the Hall voltage drop is

$$(-e)V_H \equiv (-e)[V_R - V_L] = [\mu_R - \mu_L]. \quad (12.69)$$

Hence

$$I = \nu \frac{e^2}{h} V_H, \quad (12.70)$$

where we have now allowed for the possibility that  $\nu$  different Landau levels are occupied in the bulk and hence there are  $\nu$  separate edge channels contributing to the current. This is the analog of having  $\nu$  “open” channels in the Landauer transport picture. In the Landauer picture for an ordinary wire, we are considering the longitudinal voltage drop (and computing  $G_{xx}$ ), while here we have the Hall

<sup>15</sup> This has led to various confusions in the literature. If there is an electrostatic potential gradient then some of the net Hall current may be carried in the bulk rather than at the edges, but the final answer is the same. In any case, the essential part of the physics is that the only places where there are low-lying excitations are the edges.

voltage drop (and are computing  $G_{xy} = \sigma_{xy}$ ). The analogy is quite precise, however, because we view the right and left movers as having distributions controlled by separate chemical potentials. It just happens in the QHE case that the right and left movers are physically separated in such a way that the voltage drop is transverse to the current. Using the above result and the fact that the current flows at right angles to the voltage drop, we have the desired results

$$\sigma_{xx} = 0, \quad (12.71)$$

$$\sigma_{xy} = \nu \frac{e^2}{h}, \quad (12.72)$$

with the quantum number  $\nu$  being an integer.

So far we have been ignoring the possible effects of disorder. Recall that, for a single-channel 1D wire in the Landauer picture, a disordered region in the middle of the wire will reduce the conductivity to

$$I = \frac{e^2}{h} T, \quad (12.73)$$

where  $T$  is the probability that an electron will be transmitted through the disordered region. The reduction in transmitted current is due to *backscattering*. Remarkably, in the QHE case, the backscattering is essentially zero in very wide samples. To see this, note that, in the case of the Hall bar, scattering into a backward-moving state would require transfer of the electron from one edge of the sample to the other, since the edge states are spatially separated. For samples which are very wide compared with the magnetic length (more precisely, compared with the Anderson localization length), the matrix element for this is exponentially small (see Exercise 12.9). In short, there can be nothing but forward scattering. An incoming wave given by Eq. (12.65) can only be transmitted in the forward direction, with the outgoing wave at most suffering a simple phase shift  $\delta_k$  as it continues forward:

$$\psi(x, y) = \frac{1}{\sqrt{L_y}} e^{i\delta_k} e^{iky} f_k(x). \quad (12.74)$$

This is because no other states of the same energy are available. If the disorder causes Landau-level mixing at the edges to occur (because the confining potential is relatively steep) then it is possible for an electron in one edge channel to scatter into another, but the current is still going in the same direction so that there is no reduction in the overall transmission probability. It is this *chiral* (unidirectional) nature of the edge states which is responsible for the fact that the Hall conductance is correctly quantized independently of the disorder.<sup>16</sup> This physics will be further illustrated in Exercise 12.9.

In fact, when the Fermi energy  $\epsilon_F$  is between Landau levels, disorder not only does *not* cause backscattering that would destroy the quantization of  $\sigma_{xy}$ , but is actually responsible for the visibility of the quantized value over a finite range of magnetic field or electron density. That is, disorder is responsible for the finite width of the quantum Hall plateaux. This is because, without disorder, all states in the bulk have energies precisely at one of the Landau-level energies, and so does  $\epsilon_F$ , unless the filling factor  $\nu$  is *exactly* an integer, which requires fine-tuning. These are the special points

<sup>16</sup> Our discussion on the absence of backscattering here has been quantum mechanical. In fact, this conclusion holds for classical particles as well. Imagine distorting the shape of the boundary in Fig. 12.5, or adding some random potential near the boundary. This will distort the shape of the skipping orbitals, but will not change the fact that electrons near the right edge can only have overall upward motion, and electrons near the left edge can only have overall downward motion.

when  $\epsilon_F$  jumps from one Landau level to the next one. Having edge states whose energies deviate away from the quantized Landau-level energies does not solve this problem (see Exercise 12.8). Disorder will broaden the Landau levels in the bulk and provide a reservoir of (localized) states which will allow the chemical potential to vary smoothly with density. These localized states will not contribute to the transport, so the Hall conductance will be quantized over a plateau of finite width in  $B$  (or density) as seen in Fig. 12.1. Thus obtaining the universal value of the quantized Hall conductance to a precision of  $10^{-10}$  does not require fine-tuning the applied  $B$  field to a similar precision.

The localization of states in the bulk by disorder is an essential part of the physics of the quantum Hall effect, as we saw when we studied the role of translation invariance. We learned previously that, in zero magnetic field, all states are (weakly) localized in 2D. In the presence of a quantizing magnetic field, most states are actually strongly localized (as discussed in the following section). However, if all states were localized, then it would be impossible to have a quantum phase transition from one QHE plateau to the next. To understand how this works, it is convenient to work in a semiclassical percolation picture to be described below.

**Exercise 12.8.** Show that the number of edge states whose energies lie in the gap between two Landau levels scales with the length  $L$  of the sample, while the number of bulk states scales with the area. Use these facts to show that the range of magnetic field in which the chemical potential lies in between two Landau levels scales to zero in the thermodynamic limit. Hence, finite-width quantized Hall plateaux cannot occur in the absence of disorder that produces a reservoir of localized states in the bulk whose number is proportional to the area.

**Exercise 12.9.** We discussed the similarity between transport via edge states in the integer quantum Hall effect and the Landauer transport in 1D quantum wires. What is very different is the effect of impurity scattering. (i) First consider a 1D wire of length  $L$ , in the presence of a delta-function impurity potential:  $V(x) = V_0\delta(x)$ . Show that the matrix element for backscattering is proportional to  $1/L$ . (ii) Now consider a Hall bar of length  $L_y$  and width  $L_x$ , in the presence of a single impurity:  $V(x, y) = V_0\delta(x - x_0)\delta(y - y_0)$ . Show that the matrix element for backscattering is exponentially small as a function of  $L_x/\ell$  for all edge channels (where  $\ell$  is the magnetic length), regardless of the position of the impurity  $(x_0, y_0)$ . Here we assume that the electron eigenstates are plane waves along the  $\hat{y}$  direction and there is a confinement potential along the  $\hat{x}$  direction.

## 12.7 Semiclassical Percolation Picture of the IQHE

Let us consider a smooth random potential caused, say, by ionized silicon donors remotely located away from the 2DEG in the GaAs semiconductor host. (See Fig. 9.15 and Exercise 9.7.) We take the magnetic field to be very large so that the magnetic length is small on the scale over which the potential varies. In addition, we ignore the Coulomb interactions among the electrons.

What is the nature of the eigenfunctions in this random potential? We have learned how to solve the problem exactly for the case of a constant electric field and know the general form of the solution when there is translation invariance in one direction; see Eq. (12.65). We found that the wave functions were plane waves running along lines of constant potential energy and having a width perpendicular to them which is very small and on the order of the magnetic length. The reason for this is the discreteness of the kinetic energy in a strong magnetic field. It is impossible for an electron stuck in a given

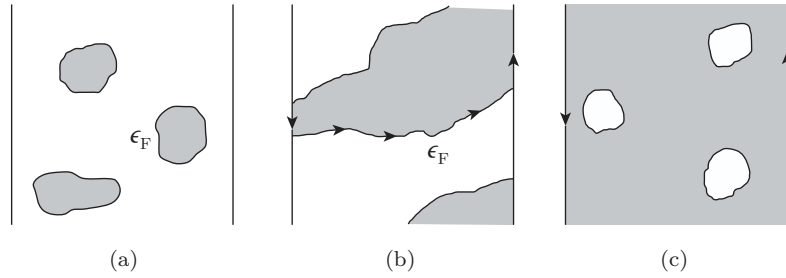
Landau level to continuously vary its kinetic energy. Hence energy conservation restricts its motion to regions of constant potential energy. In the limit of infinite magnetic field where Landau-level mixing is completely negligible, this confinement to lines of constant potential becomes exact.<sup>17</sup>

From this discussion it then seems very reasonable that, in the presence of a smooth random potential, with no particular translation symmetry, the eigenfunctions will have support lying close to contour lines of constant potential energy on the random potential surface. Thus, low-energy states will be found lying along contours in deep valleys in the potential landscape while high-energy states will be found encircling “mountain tops” in the landscape. Naturally these extreme states will be strongly localized about these extrema in the potential.

To understand the nature of states at intermediate energies, it is useful to imagine gradually filling a random landscape with water as illustrated in Fig. 12.6. In this analogy, sea level represents the chemical potential for the electrons. When only a small amount of water has been added, the water will fill the deepest valleys and form small lakes. As the sea level is increased, the lakes will grow larger and their shorelines will begin to take more complex shapes. At a certain critical value of the sea level, a phase transition will occur in which the shoreline percolates from one side of the system to the other. As the sea level is raised still further, the ocean will cover the majority of the land and only a few mountain tops will stick out above the water. The shoreline will no longer percolate but only surround the mountain tops.

As the sea level is raised still higher, a sequence of additional percolation transitions will occur as each successive Landau level passes under water. If Landau-level mixing is small and the disorder potential is symmetrically distributed about zero, then the critical value of the chemical potential for the  $n$ th percolation transition will occur near the center of the  $n$ th Landau level:

$$\mu_n^* = \left(n + \frac{1}{2}\right) \hbar\omega_c. \quad (12.75)$$



**Figure 12.6** Illustration of filling a Landau level with electron liquid, as the (average) electron density or Fermi energy  $\epsilon_F$  increases. (a) Low density, below the percolation threshold. (b) At the percolation threshold. (c) Above the percolation threshold.

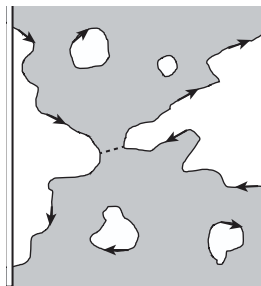
<sup>17</sup> As the magnetic length goes to zero, all (reasonable) potentials begin to look smooth and locally linear, thus producing semiclassical  $\vec{E} \times \vec{B}$  drift. We are led to the following somewhat paradoxical picture. The strong magnetic field should be viewed as putting the system in the quantum limit, in the sense that  $\hbar\omega_c$  is a very large energy (comparable to the zero-field Fermi energy,  $\epsilon_F$ ). At the same time (if one assumes the potential is smooth), one can argue that, since the magnetic length is small compared with the scale over which the random potential varies, the random potential begins to look locally linear and the system is in a semiclassical limit where small wave packets (on the scale of  $\ell$ ) follow classical  $\vec{E} \times \vec{B}$  drift trajectories. The way to reconcile these statements is the following. In the strong-field limit, the electron motion can be decomposed into two parts: the very fast (frequency  $\omega_c$ ) cyclotron motion (with radius of order  $\ell$ ) around the guiding center (the center of the circles in Fig. 12.5), which is going to the extreme quantum limit, and the very slow  $\vec{E} \times \vec{B}$  drift motion of the guiding center itself, which is semiclassical.

This percolation transition corresponds to the transition between quantized Hall plateaux. To see why, note that, when the sea level is below the percolation point, most of the sample is dry land. The electron liquid is therefore insulating. When the sea level is above the percolation point, most of the sample is covered with water. The electron liquid is therefore connected throughout the majority of the sample, and a quantized Hall current can be carried. Another way to see this is to note that, when the sea level is above the percolation point, the confining potential will make a shoreline along the full length of each edge of the sample. The edge states will then be populated in this case, and they carry current from one end of the sample to the other as we discussed in the previous section. In other words, we get a (new) open conduction channel.

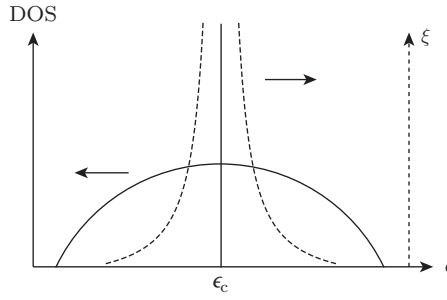
We can also understand from this picture why the dissipative conductivity  $\sigma_{xx}$  has a sharp peak just as the plateau transition occurs. (Recall the data in Fig. 12.1). Away from the critical point the circumference of any particular patch of shoreline is finite. The period of the semiclassical orbit around this is finite and hence so is the quantum level spacing. Thus there are small energy gaps for excitation of states across these real-space Fermi levels. Adding an infinitesimal electric field will only weakly perturb these states due to the gap and the finiteness of the perturbing matrix element, which will be limited to values on the order of  $\sim eED$ , where  $D$  is the diameter of the orbit. If, however, the shoreline percolates from one end of the sample to the other, then the orbital period diverges and the gap vanishes. An infinitesimal electric field can then cause dissipation of energy.

Another way to see this is that, as the percolation transition is approached from above, the edge states on the two sides will begin taking detours deeper and deeper into the bulk. They will thus begin communicating with each other as the localization length diverges and the zig-zags of the shorelines bring the two edges close to each other. Thus, electrons in one edge state can be backscattered into the other edge states and ultimately reflected from the sample as illustrated in Fig. 12.7. Precisely at the percolation level, we can even have shorelines directly connecting the two edges through the bulk, as seen in Fig. 12.6(b). Below the percolation threshold the incoming edge mode is *always* connected to the mode on the other side, and the transmission coefficient through the sample is zero.

Because the random potential broadens out the Landau levels (spreading the density of states across some finite width), the quantized Hall plateaux will have finite width. As the chemical potential is varied in the regime of localized states in between the Landau-level peaks, only the occupancy of localized states is changing. Hence the transport properties remain constant until the next percolation transition occurs. It is important to have the disorder present in order to produce this finite density of states and to localize those states.



**Figure 12.7** The electron distribution when the Fermi energy is above but very close to the percolation threshold. The grey area indicates the region occupied by electrons in the random potential. Backscattering between counter-propagating edge states becomes possible as they come close together in space, as indicated by the dashed line representing a quantum tunneling path connecting the two edges where they approach each other. The white regions in the interior are “mountain peaks” which are above “sea level” and therefore unoccupied by electrons. Their shorelines have finite length, and so local excitations have at least a small gap and cannot cause dissipation for weak electric fields at low frequencies.



**Figure 12.8** Schematic density of states (DOS; solid line) and localization length ( $\xi$ , dashed line) of a disorder-broadened Landau level, as a function of energy.  $\xi$  diverges at a single energy,  $\epsilon_c$ .

It is known that, as the percolation point is approached in 2D, the characteristic size (diameter) of the shoreline orbits diverges like

$$\xi \sim |\delta|^{-4/3}, \quad (12.76)$$

where  $\delta$  measures the deviation of the sea level from its critical value. The shoreline structure is not smooth, and in fact its circumference diverges with a larger exponent  $C \sim |\delta|^{-7/3} \sim \xi^{7/4}$ , showing that these are highly ramified fractal objects whose circumference scales as the 7/4th power of the diameter.

So far we have assumed that the magnetic length is essentially zero. That is, we have ignored the fact that the wave-function support extends a small distance transverse to the isopotential lines. If two different orbits with the same energy pass near each other but are classically disconnected, the particle can still tunnel between them if the magnetic length is finite. This quantum tunneling causes the localization length to diverge faster than the classical percolation model predicts. Numerical simulations find that the localization length diverges like [84, 85]

$$\xi \sim |\delta|^{-\nu}, \quad (12.77)$$

where the exponent  $\nu$  (not to be confused with the Landau-level filling factor!) has a value close (but probably not exactly equal) to 7/3 rather than the 4/3 described above for classical percolation. It is believed that this exponent is the same, irrespective of the Landau-level index. The energy dependence of the density of states as well as the localization length is illustrated in Fig. 12.8. It is interesting to compare that with Fig. 11.15; loosely speaking, in Fig. 12.8 the finite range of extended states (in energy) of Fig. 11.15 shrinks to a single critical energy.

Experiments on the quantum critical behavior are quite difficult, but there is evidence, at least in selected samples which show good scaling, that  $\nu$  is indeed close to 7/3. Why Coulomb interactions that are present in real samples do not spoil the agreement with the numerical simulations which is based on non-interacting electron models (as in our discussions in this chapter) is something of a mystery at the time of writing. For a discussion of these issues see Ref. [86].

## 12.8 Anomalous Integer Quantum Hall Sequence in Graphene

Thus far our discussion of the IQHE has mostly focused on the Landau levels of electrons whose wave functions satisfy Schrödinger's equation in the presence of a uniform magnetic field. On the other hand, as we learned in Section 7.5, the low-energy electronic structure of graphene is described by the Dirac equation. In this section we discuss the electronic states of Dirac electrons in the presence

of a uniform magnetic field, and the corresponding integer quantum Hall sequence that turns out to be quite different from that for Schrödinger electrons.

The low-energy states of graphene live near the two Dirac points  $K$  and  $K'$ , each described by a corresponding two-component wave function that satisfies the Dirac equation:<sup>18</sup>

$$H_{K,K'} = v_F(\pm\sigma_x p_x + \sigma_y p_y). \quad (12.78)$$

In the presence of a magnetic field we need to perform the standard minimum substitution of

$$\vec{p} \rightarrow \vec{\Pi} = \vec{p} + \frac{e}{c} \vec{A}(\vec{r}), \quad (12.79)$$

where  $\vec{\Pi}$  is the mechanical momentum. In the following we focus on the  $K$  point; solutions near the  $K'$  point can be obtained in a similar manner.

In Section 12.5 we proceeded by choosing the Landau gauge to fix the explicit form of the vector potential  $\vec{A}$ . We can proceed in a similar manner here. But the physics (including the energy spectrum) is gauge-independent, and in the following we warm up by obtaining the Schrödinger-electron Landau-level spectrum in a gauge-independent manner. Before considering the Dirac Hamiltonian, let us re-examine the Schrödinger-electron Hamiltonian

$$H = \frac{1}{2m_e} \left( \vec{p} + \frac{e}{c} \vec{A} \right)^2 = \frac{1}{2m_e} \vec{\Pi}^2 = \frac{1}{2m_e} (\Pi_x^2 + \Pi_y^2). \quad (12.80)$$

Unlike the canonical momentum  $\vec{p}$ , the different components of  $\vec{\Pi}$  do *not* commute:

$$[\Pi_x, \Pi_y] = -\frac{i\hbar e}{c} (\partial_x A_y - \partial_y A_x) = \frac{i\hbar e B}{c} = \frac{i\hbar^2}{\ell^2}, \quad (12.81)$$

where for convenience we again choose  $\vec{B} = -B\hat{z}$ . We see that the commutation relation between  $\Pi_x$  and  $\Pi_y$  is similar to that between  $x$  and  $p_x$ ; thus Eq. (12.80) takes the form of a 1D harmonic oscillator Hamiltonian. Introducing ladder operators

$$a = \frac{\ell}{\sqrt{2}\hbar} (\Pi_x + i\Pi_y), \quad a^\dagger = \frac{\ell}{\sqrt{2}\hbar} (\Pi_x - i\Pi_y), \quad (12.82)$$

we have  $[a, a^\dagger] = 1$  and

$$H = \hbar\omega_c (a^\dagger a + 1/2). \quad (12.83)$$

We thus immediately obtain the Landau-level spectrum

$$\epsilon_n = \hbar\omega_c (n + 1/2), \quad (12.84)$$

without specifying the gauge.

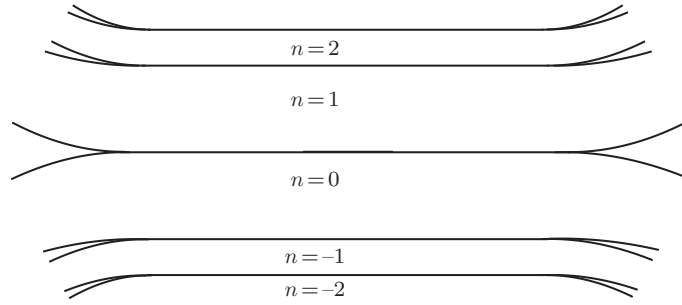
Now consider the corresponding Dirac equation for point  $K$ :

$$H_K = v_F \vec{\sigma} \cdot \vec{\Pi} = v_F \begin{pmatrix} 0 & \Pi_x - i\Pi_y \\ \Pi_x + i\Pi_y & 0 \end{pmatrix} = \epsilon_D \begin{pmatrix} 0 & a^\dagger \\ a & 0 \end{pmatrix}, \quad (12.85)$$

where  $\epsilon_D = \sqrt{2}\hbar v_F/\ell$  is an energy scale playing a role similar to  $\hbar\omega_c$ . To obtain the spectrum of (12.85), we employ the usual trick of solving the Dirac equation by considering the *square* of the Dirac Hamiltonian:

$$H_K^2 = \epsilon_D^2 \begin{pmatrix} a^\dagger a & 0 \\ 0 & a a^\dagger \end{pmatrix} = \epsilon_D^2 \begin{pmatrix} a^\dagger a & 0 \\ 0 & a^\dagger a + 1 \end{pmatrix}, \quad (12.86)$$

<sup>18</sup> Recall that the two components represent the amplitudes on the two sublattices. The two-component sublattice degree of freedom is often referred to as the pseudospin. The physical electron spin contributes a trivial factor of two; it will be ignored until the end of the discussion.



**Figure 12.9** The Landau-level spectrum of graphene, including edge states.

which is of diagonal form, and the eigenvalues are  $\epsilon_D^2 |n|$  with  $|n| = 0, 1, 2, \dots$ . Thus the spectrum of  $H_K$  is

$$\epsilon_n = \pm \epsilon_D \sqrt{|n|} = \text{sgn}(n) \epsilon_D \sqrt{|n|}, \quad (12.87)$$

with  $n = 0, \pm 1, \pm 2, \dots$ , and  $\text{sgn}(n)$  is the sign of  $n$ ; see Fig. 12.9 for illustration.

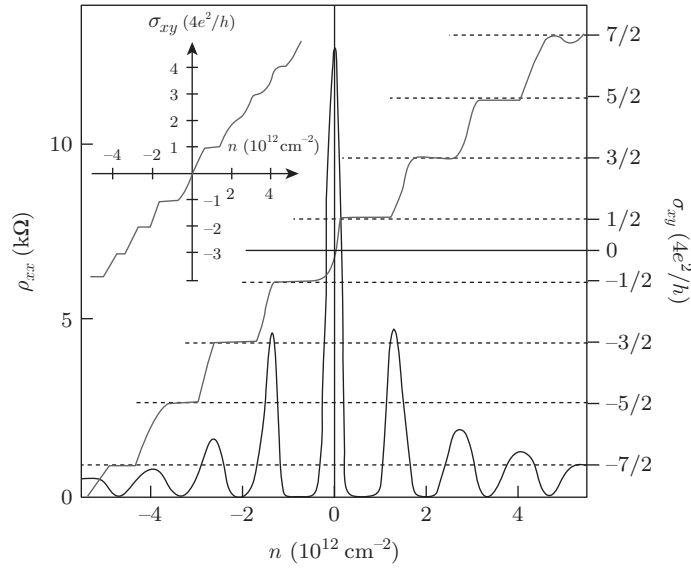
We thus find that Dirac electrons in a uniform magnetic field also form Landau levels, but with some key differences from the Schrödinger Landau levels. (i) Unlike Schrödinger Landau levels, Dirac Landau levels are *not* equally spaced. The equal spacing in the Schrödinger case is simply a reflection of the constant density of states for  $B = 0$  in 2D for parabolic dispersion. The Dirac dispersion gives a linearly rising density of states and hence the Landau-level spacing must decrease with energy. (ii) Dirac Landau levels come with both positive and negative energies. The former correspond to conduction band (or electron-like) states, while the latter correspond to valence band (or hole-like) states. (iii) Perhaps the most remarkable feature is the existence of a Landau level right at zero energy,  $n = 0$ , which is formed by conduction and valence band states with equal weight (as guaranteed by particle–hole symmetry of the underlying Hamiltonian). As we now discuss, the existence of the  $n = 0$  Landau level, which is the hallmark of massless Dirac particles, gives rise to an “anomalous” sequence of integer quantum Hall states.

As we learned earlier, each electron Landau level, if fully occupied, contributes 1 (in units of  $e^2/h$ ) to the Hall conductance  $\sigma_{xy}$ . Similarly, each hole Landau level, if fully occupied (by holes), contributes  $-1$ . At the neutral point we expect  $\sigma_{xy} = 0$ . On the other hand, this corresponds to a *half-filled*  $n = 0$  Landau level! As a result, when the zeroth Landau level is fully occupied, if only the  $K$  point were present, we would expect  $\sigma_{xy} = \frac{1}{2} e^2/h$ , namely we have quantization at half integers (instead of integers) for  $\sigma_{xy}$ . The remedy, of course, is provided by the existence of the other Dirac point  $K'$ , which contributes an identical set of Landau levels. Including contributions from both  $K$  and  $K'$  points, as well as the two-fold spin degeneracy, we expect the IQHE sequence in graphene to be<sup>19</sup>

$$\sigma_{xy} = 4 \left( n + \frac{1}{2} \right) \frac{e^2}{h} = (\pm 2, \pm 6, \pm 10, \dots) \frac{e^2}{h}. \quad (12.88)$$

The experimental observation (see Fig. 12.10) of this sequence in 2005 clearly demonstrated the Dirac nature of electrons in graphene, and led to the 2010 Nobel prize in physics for graphene work. It is worth noting that the edge-state spectra of Dirac Landau levels are also different from

<sup>19</sup> Here, for simplicity, we have neglected the Zeeman splitting of electron spin states, since these splittings are nominally much smaller than the Landau-level spacing, and unimportant except in very-high-quality graphene samples. Interactions can in some cases lead to exchange enhancement of the spin splitting and the existence of quantum Hall ferromagnetism.



**Figure 12.10** The Hall conductivity  $\sigma_{xy}$  and the longitudinal resistivity  $\rho_{xx}$  of graphene as a function of the carrier density at  $B = 14$  T and  $T = 4$  K. The inset shows the transport data of bilayer graphene, another fascinating system that we will not discuss here. Figure adapted from [87] by permission from Springer Nature.

their Schrödinger counterparts: For each Landau level with  $n \neq 0$ , and for each spin, there are two branches of edge modes at each edge (one from each of the two Dirac points), and dispersing in opposite ways for  $n > 0$  (electron-like) and  $n < 0$  (hole-like). The  $n = 0$  Landau level also has two branches, one electron-like and one hole-like, reflecting the fact it is an equal superposition of conduction and valence band states. These are illustrated in Fig. 12.9. The integer quantum Hall sequence of Eq. (12.88) can also be understood using edge transport in terms of the Dirac Landau-level spectrum.

**Exercise 12.10.** Use the Landau gauge to obtain the eigenstate wave functions of the Hamiltonian in Eq. (12.85), and do the same for the corresponding Hamiltonian of the  $K'$  point.

## 12.9 Magnetic Translation Invariance and Magnetic Bloch Bands

By now the reader should have become quite familiar with Landau-level states and wave functions, at least in the Landau gauge. It appears that these states behave (somewhat) like Bloch states in a Bloch band, as they all carry a band-like index  $n$  (the Landau-level index), and a momentum-like quantum number  $k$ ; also we can use the band dispersion to calculate the electron velocity, as we did in Eq. (12.66), in a manner similar to Eq. (8.9) in Chapter 8. But the differences are also quite obvious: in Landau gauge we have a 1D  $k$ -space, whereas in a 2D system the Bloch wave vector should be a 2D vector defined in the first Brillouin zone. Most importantly, here  $k$  is a *gauge-dependent* quantity; its value and even *direction* depend on the choice of gauge.<sup>20</sup> Physics, on the other hand,

<sup>20</sup> In our earlier discussion we chose a specific Landau gauge in which  $\vec{A}$  points in the  $\hat{y}$  direction so  $k$  is the  $\hat{y}$ -direction momentum; but this choice is arbitrary in the absence of any potential, and we could have chosen  $\vec{A}$  to be in any direction and the direction of  $k$  would change accordingly.

is gauge-independent. In the following we generalize Bloch's theorem in the presence of a uniform magnetic field (and eventually to include the possibility of non-uniform but periodic magnetic field), in a gauge-independent manner.

We start by reminding ourselves about the origin of Bloch's theorem. In the case of a periodic potential without a magnetic field, the Hamiltonian is invariant under a set of discrete lattice translations  $T_{\vec{a}}$ ,

$$T_{\vec{a}} H T_{\vec{a}}^{\dagger} = H, \quad (12.89)$$

where  $T_{\vec{a}} = e^{i\vec{p} \cdot \vec{a}}$  and  $\vec{a}$  is a lattice vector. Now, for an eigenstate of  $H$  with  $H\psi = \epsilon\psi$ , we have

$$T_{\vec{a}} H T_{\vec{a}}^{\dagger} T_{\vec{a}} \psi = H T_{\vec{a}} \psi = \epsilon T_{\vec{a}} \psi, \quad (12.90)$$

namely  $T_{\vec{a}} \psi$  is also an eigenstate of  $H$  with the *same* eigenvalue. We thus expect that, under a translation which leaves the Hamiltonian invariant, the eigenfunction remains unchanged up to an overall phase factor<sup>21</sup>

$$T_{\vec{a}} \psi = \lambda_{\vec{a}} \psi = e^{i\vec{k} \cdot \vec{a}} \psi, \quad (12.91)$$

which *defines* the Bloch wave vector  $\vec{k}$ .

Now consider

$$H = \frac{1}{2m_e} \left[ \vec{p} + \frac{e}{c} \vec{A}(\vec{r}) \right]^2, \quad (12.92)$$

with  $\nabla \times \vec{A}(\vec{r}) = \vec{B}$ . Performing the same manipulation on the eigenstate  $\psi$  as in Eq. (12.90), we obtain

$$T_{\vec{a}} H T_{\vec{a}}^{\dagger} T_{\vec{a}} \psi = \frac{1}{2m_e} \left[ \vec{p} + \frac{e}{c} \vec{A}(\vec{r} + \vec{a}) \right]^2 T_{\vec{a}} \psi = \epsilon T_{\vec{a}} \psi. \quad (12.93)$$

We find  $T_{\vec{a}} \psi$  is an eigenstate of a *different* Hamiltonian,

$$H' = \frac{1}{2m_e} \left[ \vec{p} + \frac{e}{c} \vec{A}(\vec{r} + \vec{a}) \right]^2, \quad (12.94)$$

that in general is *not* the same as the original Hamiltonian  $H$ . As a result, Bloch's theorem does *not* follow.

The origin of this is of course the  $\vec{r}$  dependence of  $\vec{A}$ , which ruins the translation symmetry of  $H$ . On the other hand, for a uniform magnetic field,  $\vec{A}(\vec{r})$  and its translated version,  $\vec{A}(\vec{r} + \vec{a})$ , correspond to the *same* magnetic field:

$$\nabla \times \vec{A}(\vec{r}) = \nabla \times \vec{A}(\vec{r} + \vec{a}) = \vec{B}. \quad (12.95)$$

This means that  $\vec{A}(\vec{r})$  and  $\vec{A}(\vec{r} + \vec{a})$  are related by a gauge transformation:

$$\vec{A}(\vec{r} + \vec{a}) = \vec{A}(\vec{r}) + \nabla f_{\vec{a}}(\vec{r}). \quad (12.96)$$

We can thus perform a *gauge transformation* to bring  $H'$  back to  $H$ :

$$e^{i\phi_{\vec{a}}(\vec{r})} H' e^{-i\phi_{\vec{a}}(\vec{r})} = H, \quad (12.97)$$

where

$$\phi_{\vec{a}}(\vec{r}) = \frac{ef_{\vec{a}}(\vec{r})}{\hbar c}. \quad (12.98)$$

<sup>21</sup> Here we assume for simplicity that the eigenstate is non-degenerate. More care has to be taken in the degenerate case.

This immediately suggests that we can combine translation and the corresponding gauge transformations to form a **magnetic translation operator**,

$$\tilde{T}_{\vec{a}} = e^{i\phi_{\vec{a}}(\vec{r})} T_{\vec{a}}, \quad (12.99)$$

which *commutes* with  $H$ :

$$[\tilde{T}_{\vec{a}}, H] = 0. \quad (12.100)$$

We have thus found the equivalent of translation symmetry, in the case of a uniform magnetic field.

There is, however, a fundamental difference between ordinary translation and magnetic translation. We know that  $[T_{\vec{a}}, T_{\vec{b}}] = 0$  for arbitrary  $\vec{a}$  and  $\vec{b}$ . This is *not* true for  $\tilde{T}$ , because it involves both position and momentum operators. As a result, it is impossible to have eigenstates of  $H$  that are also *simultaneous* eigenstates of all magnetic translation operators. On the other hand, for carefully chosen *different* vectors  $\vec{a}_1$  and  $\vec{a}_2$  that satisfy appropriate conditions, we can have

$$[\tilde{T}_{\vec{a}_1}, \tilde{T}_{\vec{a}_2}] = 0. \quad (12.101)$$

Let us find out what these conditions are. Equation (12.101) is equivalent to

$$\begin{aligned} \tilde{T}_{\vec{a}_2} &= \tilde{T}_{\vec{a}_1} \tilde{T}_{\vec{a}_2} \tilde{T}_{\vec{a}_1}^\dagger = e^{i\phi_{\vec{a}_1}(\vec{r})} T_{\vec{a}_1} e^{i\phi_{\vec{a}_2}(\vec{r})} T_{\vec{a}_2} T_{\vec{a}_1}^\dagger e^{-i\phi_{\vec{a}_1}(\vec{r})} \\ &= e^{i\phi_{\vec{a}_1}(\vec{r}) + i\phi_{\vec{a}_2}(\vec{r} + \vec{a}_1) - i\phi_{\vec{a}_1}(\vec{r} + \vec{a}_2)} T_{\vec{a}_2} \\ &= e^{i\phi_{\vec{a}_1}(\vec{r}) + i\phi_{\vec{a}_2}(\vec{r} + \vec{a}_1) - i\phi_{\vec{a}_1}(\vec{r} + \vec{a}_2) - i\phi_{\vec{a}_2}(\vec{r})} \tilde{T}_{\vec{a}_2}. \end{aligned} \quad (12.102)$$

Thus the condition we need  $\vec{a}_1$  and  $\vec{a}_2$  to satisfy is

$$\begin{aligned} [\phi_{\vec{a}_1}(\vec{r}) - \phi_{\vec{a}_1}(\vec{r} + \vec{a}_2)] - [\phi_{\vec{a}_2}(\vec{r}) - \phi_{\vec{a}_2}(\vec{r} + \vec{a}_1)] &= 2\pi N \\ &= \frac{e}{\hbar c} \oint_{\diamond} \vec{A}(\vec{r}) \cdot d\vec{r} \\ &= \frac{e\Phi_{\diamond}}{\hbar c} \\ &= \frac{2\pi\Phi_{\diamond}}{\Phi_0}. \end{aligned} \quad (12.104)$$

Here  $N$  is an integer,  $\diamond$  stands for the parallelogram spanned by  $\vec{a}_1$  and  $\vec{a}_2$ , and  $\Phi_{\diamond} = \vec{B} \cdot (\vec{a}_1 \times \vec{a}_2)$  is the flux it encloses. This phase is nothing more than the Aharonov–Bohm phase acquired in moving around the perimeter of the parallelogram. To arrive at Eq. (12.104) we combined Eqs. (12.96) and (12.98). Thus the condition for Eq. (12.101) is simply

$$\frac{\Phi_{\diamond}}{\Phi_0} = N, \quad (12.105)$$

namely the parallelogram spanned by  $\vec{a}_1$  and  $\vec{a}_2$  encloses an integer number of flux quanta  $N$ . For parallel  $\vec{a}_1$  and  $\vec{a}_2$ ,  $N = 0$  and Eq. (12.105) is trivially satisfied. In this case  $\tilde{T}_{\vec{a}_1}$  and  $\tilde{T}_{\vec{a}_2}$  are *not* independent, and we only get a 1D  $k$  quantum number for  $\psi$  from the magnetic translation symmetry of  $H$ , as in the case we studied earlier using the Landau gauge. To obtain a 2D vector quantum number  $\vec{k}$  for  $\psi$ , the minimum condition is  $N = 1$ , or the minimum *magnetic* unit cell must enclose one flux quantum. We can then combine Eqs. (12.100) and (12.101) to generalize Bloch's theorem and define *magnetic* Bloch bands and states that are simultaneous eigenstates of  $H$ ,  $\tilde{T}_{\vec{a}_1}$ , and  $\tilde{T}_{\vec{a}_2}$ ; clearly, in the present case they correspond to Landau levels and corresponding states.

Using the magnetic translation symmetry, we can obtain some of the key properties of Landau-level states, without actually solving  $H$ . One very special feature of the magnetic unit cell in this case is that the condition Eq. (12.105) determines only its area, *not* its shape or the actual lattice structure. Correspondingly, the reciprocal lattice structure is not determined either, other than the area

of the Brillouin zone. On the other hand, we know that the band dispersion is a periodic function in the reciprocal lattice; the only function that is periodic in such a *continuously* varying family of lattice structures is a constant. We thus conclude that each of the magnetic Bloch bands must have a constant energy, namely they form degenerate Landau levels! Furthermore, since each magnetic unit cell contains one flux quantum, and the number of states in each band is equal to the number of unit cells, we come to the conclusion that the Landau-level degeneracy must be equal to the number of flux quanta enclosed in the system, as we found earlier.

The discussions above can be easily generalized to the case where a magnetic field coexists with a periodic potential,  $V(\vec{r})$ . In this case, in order to find  $\tilde{T}_{\vec{a}_1}$  and  $\tilde{T}_{\vec{a}_2}$  that commute with each other and both commute with

$$H = \frac{1}{2m_e} \left[ \vec{p} + \frac{e}{c} \vec{A}(\vec{r}) \right]^2 + V(\vec{r}), \quad (12.106)$$

we need  $\vec{a}_1$  and  $\vec{a}_2$  to be lattice vectors of  $V(\vec{r})$  (but not necessarily primitive ones), in addition to satisfying Eq. (12.105). Thus, if the original unit cell of  $V(\vec{r})$  contains a rational number  $p/q$  of flux quanta, with  $p$  and  $q$  mutually prime integers, we need to repeat the original unit cell  $q$  times to form a **magnetic unit cell**, which contains an integer number  $p$  of flux quanta. In this case each original band (in the absence of  $\vec{B}$ ) is split into  $q$  magnetic subbands.<sup>22</sup> Equivalently, one can also say that each Landau level (in the absence of  $V(\vec{r})$ ) is split into  $p$  magnetic subbands. The area of the (minimal) magnetic unit cell is determined by  $\vec{B}$ . However, in this case the lattice vectors are not arbitrary, since they must be consistent with the translation symmetry of  $V(\vec{r})$ . Hence the argument above showing that the Landau-level states are degenerate does not apply and, in general, these magnetic Bloch bands are *not* degenerate. It should also be clear that  $\vec{B}(\vec{r})$  does *not* need to be a constant, but need only be periodic from one magnetic unit cell to another, in order for these conclusions to hold.

Owing to the (magnetic) translation symmetry, an eigenfunction of the Hamiltonian in Eq. (12.106) may be written as

$$\psi_{n\vec{k}}(\vec{r}) = e^{i\vec{k}\cdot\vec{r}} u_{n\vec{k}}(\vec{r}). \quad (12.107)$$

In the absence of a magnetic field (where we can choose the gauge  $\vec{A} = \vec{0}$ ), Eq. (12.91) tells us that the Bloch function satisfies periodic boundary conditions

$$u_{n\vec{k}}(\vec{r} + \vec{a}_j) = u_{n\vec{k}}(\vec{r}), \quad (12.108)$$

for any lattice translation vector  $\vec{a}_j$ . In the presence of a magnetic field, the magnetic translation operator is changed and Eq. (12.99) tells us that the Bloch function satisfies *generalized* periodic boundary conditions:

$$\tilde{T}_{\vec{a}_j} u_{n\vec{k}}(\vec{r}) = e^{i\phi_{\vec{a}_j}(\vec{r})} u_{n\vec{k}}(\vec{r} + \vec{a}_j) = u_{n\vec{k}}(\vec{r}). \quad (12.109)$$

### 12.9.1 Simple Landau Gauge Example

To better understand all these somewhat formal mathematical results, it is useful to consider the specific case of a uniform magnetic field treated in the Landau gauge. Suppose that the periodic potential yields a rectangular unit cell of width  $a_x$  and height  $a_y$ . We will take the magnetic field strength to be such that the unit cell contains a single flux quantum. Thus

$$a_x a_y = 2\pi \ell^2. \quad (12.110)$$

<sup>22</sup> For an arbitrary magnetic flux through the original unit cell, it may be necessary in some cases to choose extremely large  $p$  and  $q$  in order to find an accurate rational approximation to the flux density. The mathematical details needed to treat the case of irrational flux density are ignored in the present discussion.

The corresponding reciprocal lattice unit cell is also rectangular, with sides

$$G_x = \frac{2\pi}{a_y} = \frac{a_x}{\ell^2}, \quad (12.111)$$

$$G_y = \frac{2\pi}{a_x} = \frac{a_y}{\ell^2}. \quad (12.112)$$

To make the example particularly easy, we will assume that the periodic potential simply vanishes. The Landau gauge solutions of the Schrödinger equation are therefore those given in Eq. (12.48). It is very hard to see how to reconcile this expression with the standard form of the Bloch-wave solution given in Eq. (12.107). First, the solutions in Eq. (12.48) are parameterized by a 1D wave vector  $k$ , whereas those in Eq. (12.107) involve a 2D wave vector  $\vec{k}$ . Secondly, as was noted earlier in the discussion below Eq. (12.48), those solutions are localized Gaussians in the  $x$  direction and do not at all obey the generalized periodic boundary condition given in Eq. (12.109).

We escape from this quandary by noting that the discrepancies between Eqs. (12.48) and (12.107) are simply an accident associated with the complete degeneracy of the Landau level when the periodic potential vanishes. Because of the degeneracy, linear combinations of different solutions in Eq. (12.48) are also solutions. By choosing the linear combinations correctly, we can put the solutions into the standard Bloch-wave form.

Consider the following (unnormalized) linear combination of Landau wave functions centered on different  $x$  positions:

$$\begin{aligned} \Psi_{nkq}(\vec{r}) &= \sum_{m=-\infty}^{+\infty} e^{iqmG_x\ell^2} \psi_{n(k+mG_x)}(\vec{r}) \\ &= \sum_{m=-\infty}^{+\infty} e^{iqmG_x\ell^2} e^{i(k+mG_x)y} F_n(x - [k + mG_x]\ell^2), \end{aligned} \quad (12.113)$$

where

$$F_n(z) \equiv H_n(z/\ell) e^{-z^2/2\ell^2}. \quad (12.114)$$

These new basis states (which are essentially Jacobi elliptic theta functions) depend on  $k$  only through  $k \bmod G_x$  and on  $q$  only through  $q \bmod G_y$ , so we can without loss of generality employ the restrictions  $-G_x/2 \leq k < +G_x/2$  and  $-G_y/2 \leq q < +G_y/2$ . The presence of the new parameter  $q$  compensates for the restriction on  $k$  and will be the route to obtaining a 2D wave vector.

Under the translation  $x \rightarrow x + a_x$ , the basis states transform according to

$$\Psi_{nkq} \rightarrow e^{iqax} e^{iG_x y} \Psi_{nkq}, \quad (12.115)$$

while, under the translation  $y \rightarrow y + a_y$ , the basis states transform according to

$$\Psi_{nkq} \rightarrow e^{ikay} \Psi_{nkq}. \quad (12.116)$$

If we define a 2D wave vector  $\vec{K} = (q, k)$ , we can write this in the Bloch-wave form of Eq. (12.107),

$$\Psi_{nkq} = e^{i\vec{K} \cdot \vec{r}} u_{n\vec{K}}(\vec{r}), \quad (12.117)$$

where

$$u_{n\vec{K}}(\vec{r}) \equiv e^{-i\vec{K} \cdot \vec{r}} \Psi_{nkq} \quad (12.118)$$

obeys the generalized periodic boundary conditions

$$u_{n\vec{K}}(x, y + a_y) = u_{n\vec{K}}(x, y), \quad (12.119)$$

$$u_{n\vec{K}}(x + a_x, y) = e^{2\pi i y/a_y} u_{n\vec{K}}(x, y). \quad (12.120)$$

The final step in the derivation is to note that the Landau gauge vector potential given in Eq. (12.40) transforms under translation by one of the lattice vectors  $\vec{a}_j$  according to

$$\vec{A}(\vec{r} + \vec{a}_j) = \vec{A}(x, y) + \nabla f_{\vec{a}_j}(x, y), \quad (12.121)$$

where

$$f_{\vec{a}_j}(x, y) = -(a_{jx}B)y, \quad (12.122)$$

and hence

$$\phi_{\vec{a}}(\vec{r}) = -\frac{a_{jx}y}{\ell^2}, \quad (12.123)$$

which, for the case  $a_{jx} = a_x$ , reduces to

$$\phi_{\vec{a}}(\vec{r}) = -2\pi \frac{y}{a_y}. \quad (12.124)$$

Thus Eq. (12.120) is equivalent to Eq. (12.109) as required.

**Exercise 12.11.** Suppose that the sample has dimensions  $N_x a_x$  by  $N_y a_y$ . Count the number of allowed 2D wave vectors  $\vec{K}$  for the Bloch waves and show that this matches the number of allowed 1D wave-vector states  $k$ ,  $N_\Phi = N_x N_y$ , in agreement with Eq. (12.49).

**Exercise 12.12.** Consider an electron moving in the  $xy$  plane with a non-uniform but periodic magnetic field along the  $\hat{z}$  direction:  $\vec{B}(\vec{r}) = \hat{z}B(\vec{r})$ , with  $\vec{r}$  being a 2D vector. Its periodicity is specified by  $B(\vec{r} + \vec{a}_j) = B(\vec{r})$ , with  $\{\vec{a}_j\}$  forming a Bravais lattice with primitive unit cell area  $A$ . The magnetic flux enclosed in such a unit cell is  $p\Phi_0/q$ , where  $\Phi_0$  is the flux quantum and  $p$  and  $q$  are mutually prime integers. Find the size of the magnetic unit cell, and corresponding magnetic translation operators similar to that of Eq. (12.99).

## 12.10 Quantization of the Hall Conductance in Magnetic Bloch Bands

In the previous section we learned that Landau levels are nothing but very special cases of magnetic Bloch bands, which in turn are generalizations of Bloch bands in the presence of magnetic fields. This naturally leads to the following conundrum. In Chapter 7 we were told that completely filled bands do *not* carry current, and systems with Fermi energies inside a band gap are insulators. Yet we find that completely filled Landau levels give rise to quantized contributions to the Hall conductance, and the system can carry a *dissipationless* Hall current (and behave almost like a superconductor!) when its Fermi energy lies in between Landau levels. So something must be missing when we simply identified systems with completely filled and completely empty bands as insulators. We learned in this chapter that the dissipationless Hall current is carried by edge states, but need to understand better why we have these edge states.

We would also like to ask whether the quantization of the Hall conductance (that we demonstrated for Landau levels) is robust for magnetic Bloch bands. The answer is yes, and, like the answers to the questions posed in the previous paragraph, has a deep topological origin which we will explore in Chapter 13. In a few words, the Hall conductance is a *topological* invariant of a Bloch band, which is *insensitive* to continuous changes of the band dispersion (as long as no band crossing is involved). The contribution of individual states in a band to the Hall conductance can be viewed (along with

other characteristics) as a *geometric* property of the band. The overall Hall conductance of an isolated band (i.e. a band separated by an energy gap from other bands) is a topological invariant that does not depend on the particular geometry of the band. Furthermore, it is possible to show from purely topological arguments when gapless edge modes (such as the quantum Hall edge currents) will appear at the boundary between two different (topologically distinct) gapped regions. To fully appreciate these words, however, we need to take a detour and study some geometrical and topological features of quantum-mechanical systems in general, and then apply them to Bloch bands. This will be our task in Chapter 13, aided by the additional pedagogical material in Appendix D. In Chapter 14 we will extend these ideas to study the remarkable concepts of the topological insulator and topological semimetal.

Geometry and topology have begun to play an increasingly important role in modern quantum physics. Geometry was invented by the ancients partly for the purpose of measuring distances and surveying the areas of farm fields, etc. Indeed, the very word means “measuring the earth.” However, the formal Euclidean plane geometry that we first learn in high school explicitly eschews discussion of absolute distance. We are allowed to use a compass and a straight edge to construct geometric figures, but we are not allowed to use a ruler. In this sense, Euclidean plane geometry can thus be characterized as the study of invariants under linear scale changes and rotations. (An isosceles triangle made twice as large and then rotated through some angle is still an isosceles triangle with the same three internal angles, and two of the sides, though now twice as long, are still equal in length.) Topology is the study of invariants under more general arbitrary continuous transformations. (A coffee cup with one handle can be continuously deformed into a donut. An isosceles triangle can be continuously transformed into a circle.) There are far fewer invariants under arbitrary distortions than there are under linear scale changes, but, by the same token, those topological invariants are very robust and therefore quite special and fundamental. See Appendix D for a brief primer on the basic concepts of topology.

In recent years it has come to be recognized that the values of certain physical observables are universal and invariant under changes in the Hamiltonian. For example, in Chapter 12 we learned that the Hall conductance (or resistance) on a quantum Hall plateau is very precisely quantized, with a universal value completely independent of the details of the Hamiltonian (such as the particle mass, the strength of the magnetic field, and the particular realization of the random disorder potential). It is now understood [88] that this universality is the result of the fact that the physical observable is represented mathematically by a **topological invariant**. This so-called “topological protection” means that the quantity remains invariant even when the Hamiltonian undergoes changes (say, from one sample to the next, or the magnetic field is varied), unless the system goes through a quantum phase transition that results in a discontinuous change of the corresponding topological invariant.

We learn plane geometry in high school before we learn topology in college or graduate school. Hence we will start with a study of the geometry of Hilbert space in order to equip ourselves to move on to the study of topological invariants in quantum physics.

## 13.1 Adiabatic Evolution and the Geometry of Hilbert Space

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To explore the geometry of Hilbert space we need to move around inside Hilbert space. We already have some notion of length and distance because we have a well-defined inner product between

vectors. However, the fact that this is a complex vector space adds a new twist. Physical states are represented by rays,<sup>1</sup> not vectors in Hilbert space, and, as we shall see, this leads to interesting and novel features.

One nice way to move around in Hilbert space is to follow a particular eigenstate of the Hamiltonian as we slowly (adiabatically) vary the Hamiltonian. Such adiabatic processes can also be of great physical importance. For example, the naturally slow motion of the heavy nuclei in a diatomic molecule can be profitably viewed as a slow adiabatic variation of parameters in the Hamiltonian for the light (and speedy) electrons (see Exercise 5.6), which often have a significant excitation gap. It is this large excitation gap which prevents excitations of electrons into higher states as the nuclei slowly move.

Consider a Hamiltonian that depends on time through a set of parameters  $\vec{\mathcal{R}} = (\mathcal{R}_1, \mathcal{R}_2, \dots, \mathcal{R}_D)$ :

$$H(t) = H[\vec{\mathcal{R}}(t)]. \quad (13.1)$$

$\vec{\mathcal{R}}$  can be viewed as a point or a vector in a  $D$ -dimensional parameter space, where  $D$  is the number of independent parameters, and not necessarily related to the spatial dimension of the physical system  $d$  (or even the number of particles in the system). For every  $\vec{\mathcal{R}}$  we have an ( $\vec{\mathcal{R}}$ -dependent) set of orthonormal eigenstates of  $H(\vec{\mathcal{R}})$ :

$$H(\vec{\mathcal{R}})|n(\vec{\mathcal{R}})\rangle = \epsilon_n(\vec{\mathcal{R}})|n(\vec{\mathcal{R}})\rangle. \quad (13.2)$$

For simplicity, let us assume that the spectrum of  $H$  is discrete and non-degenerate everywhere in the  $\vec{\mathcal{R}}$  space. Then the **adiabatic theorem** tells us that, if the system is initially in the  $n$ th eigenstate of  $H$ ,

$$|\psi_n(t=0)\rangle = |n[\vec{\mathcal{R}}(t=0)]\rangle, \quad (13.3)$$

then it will evolve into the  $n$ th eigenstate of  $H$  at later times, as long as the time-variation of  $H$  is sufficiently slow:<sup>2</sup>

$$|\psi_n(t)\rangle = C_n(t)|n[\vec{\mathcal{R}}(t)]\rangle. \quad (13.4)$$

Thus a path in parameter space  $\vec{\mathcal{R}}(t)$  defines (maps onto) a path in Hilbert space  $|\psi_n(t)\rangle$ .

Since time-evolution under  $H$  is a unitary transformation, the proportionality constant  $C_n(t)$  has to be a pure phase.<sup>3</sup> For time-independent  $H$ , we simply have

$$C_n(t) = \exp\left[-\frac{i}{\hbar}\epsilon_n t\right]. \quad (13.5)$$

Generalizing to the case of  $H$  being time-dependent, we can write, without loss of generality,

$$C_n(t) = e^{i\gamma_n(t)} \exp\left[-\frac{i}{\hbar} \int_0^t dt' \epsilon_n(t')\right], \quad (13.6)$$

where the second factor above is the straightforward generalization of Eq. (13.5). We will see below that the phase  $\gamma_n(t)$  in the first factor is not zero in general, and contains much physics. It accounts for a special aspect of the state evolution not captured by the second factor. Combining Eqs. (13.4) and (13.2) and plugging into the Schrödinger equation

$$i\hbar \frac{\partial}{\partial t} |\psi_n(t)\rangle = H[\vec{\mathcal{R}}(t)] |\psi_n(t)\rangle, \quad (13.7)$$

<sup>1</sup> A ray is the equivalence class of vectors all related to a given vector by multiplication by an overall complex phase  $e^{i\theta}$  for any real  $\theta$ . More generally, since overall normalization does not matter, a ray is defined as the equivalence class of vectors all related to a given vector by multiplication by any complex number  $\lambda$ .

<sup>2</sup> The time scale that defines “slow” is set by the (inverse of the) excitation gap between the  $n$ th state and neighboring states.

<sup>3</sup> Since  $U^\dagger U = 1$ , unitary transformations preserve the inner product between two states and that of a state with itself.

and taking the inner product with  $\langle \psi_n(t) |$  on both sides, we obtain

$$\frac{\partial}{\partial t} \gamma_n(t) = i \langle n[\vec{\mathcal{R}}(t)] | \frac{\partial}{\partial t} | n[\vec{\mathcal{R}}(t)] \rangle. \quad (13.8)$$

Thus

$$\gamma_n(t) = i \int_0^t dt' \langle n[\vec{\mathcal{R}}(t')] | \frac{\partial}{\partial t'} | n[\vec{\mathcal{R}}(t')] \rangle \quad (13.9)$$

$$= \int_C \vec{\mathcal{A}}^n(\vec{\mathcal{R}}) \cdot d\vec{\mathcal{R}}, \quad (13.10)$$

where the so-called “**Berry connection**”<sup>4</sup>

$$\vec{\mathcal{A}}^n(\vec{\mathcal{R}}) = i \langle n(\vec{\mathcal{R}}) | \frac{\partial}{\partial \vec{\mathcal{R}}} | n(\vec{\mathcal{R}}) \rangle \quad (13.11)$$

is a (vector) function of  $\vec{\mathcal{R}}$  but *not* an explicit function of  $t$ . Thus the additional phase  $\gamma_n$  depends only on the path  $C$  that  $\vec{\mathcal{R}}$  traverses in the parameter space, *not* on its actual time-dependence. Thus  $\gamma_n$  is of *geometric* origin;<sup>5</sup> in contrast the phase  $-(1/\hbar) \int_0^t dt' \epsilon_n(t')$  is called a *dynamical* phase because it does depend on the time-dependence of the control path in parameter space.

**Exercise 13.1.** Let us explore the fact that the adiabatic phase depends only on the path  $\vec{\mathcal{R}}$  taken through parameter space and not on how that path is taken in time (as long as it is slow enough for the adiabatic approximation to be valid). Show that Eq. (13.9) exhibits “*time reparameterization invariance*.” That is, replace Eq. (13.9) by

$$\gamma_n(t) = i \int_0^t dt' \langle n[\vec{\mathcal{R}}(\tau)] | \frac{\partial}{\partial t'} | n[\vec{\mathcal{R}}(\tau)] \rangle, \quad (13.12)$$

where the modified time variable  $\tau(t')$  is an arbitrary (smooth) function of  $t'$ , subject only to the constraint that the starting and ending times (and therefore the end points of the trajectory) are left invariant,  $\tau(0) = 0$ ,  $\tau(t) = t$ , and show that the result is completely independent of the functional form of  $\tau(t')$ . See Appendix D for further discussion.

**Exercise 13.2.** Using the fact that  $\langle n(\vec{\mathcal{R}}) | n(\vec{\mathcal{R}}) \rangle = 1$  and hence  $(\partial/\partial \vec{\mathcal{R}}) \langle n(\vec{\mathcal{R}}) | n(\vec{\mathcal{R}}) \rangle = 0$ , show that  $\vec{\mathcal{A}}^n(\vec{\mathcal{R}})$  and  $\gamma_n$  are purely real. The interpretation of this mathematical result is the following. In computing the Berry connection, we are looking for the part of the change in  $|\psi\rangle$  with  $\vec{\mathcal{R}}$  that is parallel to the Hilbert space vector  $|\psi\rangle$  itself. Given that the length of the vector is fixed, the change must either be orthogonal to  $|\psi\rangle$  or represent a change in the overall phase of  $|\psi\rangle$ . Only the latter contributes to the Berry connection, and the contribution is such that  $\langle n(\vec{\mathcal{R}}) | (\partial/\partial \vec{\mathcal{R}}) | n(\vec{\mathcal{R}}) \rangle$  is purely imaginary.

<sup>4</sup> A word about our somewhat sloppy notation here. In the Dirac notation  $\langle \psi | \mathcal{O} | \psi \rangle$  represents the expectation value of the operator  $\mathcal{O}$ , or equivalently, the inner product between  $\mathcal{O}|\psi\rangle$  and  $|\psi\rangle$ .  $\partial/\partial \vec{\mathcal{R}}$  is not a physical observable (represented by an operator acting on states in the Hilbert space). Hence the proper interpretation of the notation is that it should be taken to mean the inner product between  $(\partial/\partial \vec{\mathcal{R}})|\psi\rangle$  and  $|\psi\rangle$ .

<sup>5</sup> For a useful analogy from daily life, consider you are driving home from your workplace and watching the odometer of your car to measure the distance between these two locations. The final result does *not* depend on how fast you drive (which is a piece of dynamical information!), but does depend on the route you choose, since distance is a purely geometric property of the route (or path).

It is important to emphasize, however, that in general  $\gamma_n$  is *gauge-dependent*, and thus *not* a physical observable. This is because Eq. (13.2) does not determine the overall phase of  $|n(\vec{\mathcal{R}})\rangle$ ; in particular, we can make an  $\vec{\mathcal{R}}$ -dependent phase change

$$|n(\vec{\mathcal{R}})\rangle \Rightarrow e^{i\zeta(\vec{\mathcal{R}})} |n(\vec{\mathcal{R}})\rangle, \quad (13.13)$$

which leads to

$$\vec{\mathcal{A}}^n(\vec{\mathcal{R}}) \Rightarrow \vec{\mathcal{A}}^n(\vec{\mathcal{R}}) - \partial_{\vec{\mathcal{R}}} \zeta(\vec{\mathcal{R}}) \quad (13.14)$$

and

$$\gamma_n \Rightarrow \gamma_n + \zeta[\vec{\mathcal{R}}(t=0)] - \zeta[\vec{\mathcal{R}}(t)]. \quad (13.15)$$

For this reason the potential physical effects of  $\gamma_n$  were ignored for many years. It was implicitly assumed that  $\gamma_n$  could always be “gauged away” and therefore had no physical content.

In 1984 Berry [89] (see also Refs. [90, 91] for more recent perspectives; parts of our discussions follow those of Ref. [91]) made the important discovery that, for adiabatic processes defined by motion over a closed loop in parameter space (i.e. ones for which  $\vec{\mathcal{R}}(t=0) = \vec{\mathcal{R}}(t_{\text{final}})$ , and thus  $H$  returns to its original value), the gauge-dependence of  $\gamma_n$  in Eq. (13.15) disappears. In these cases the line integral of the Berry connection around a closed loop in parameter space,

$$\gamma_n = \oint \vec{\mathcal{A}}^n(\vec{\mathcal{R}}) \cdot d\vec{\mathcal{R}}, \quad (13.16)$$

is *gauge-independent* and is known as the **Berry phase**. It is thus (in principle) observable and can have physical consequences if it turns out to be non-zero.

The reader may have recognized by now that the (gauge-dependent) **Berry connection**  $\vec{\mathcal{A}}^n(\vec{\mathcal{R}})$  is reminiscent of the familiar vector potential of a magnetic field, while the Berry phase, being a loop integral of  $\vec{\mathcal{A}}^n(\vec{\mathcal{R}})$ , is like the (gauge-independent) magnetic flux passing through the loop (in parameter space, not “real” space). This analogy with electromagnetism immediately suggests the existence of a gauge-independent local field known as the **Berry curvature**:

$$\omega_{\mu\nu}^n(\vec{\mathcal{R}}) = \partial_{\mathcal{R}_\mu} \mathcal{A}_\nu^n(\vec{\mathcal{R}}) - \partial_{\mathcal{R}_\nu} \mathcal{A}_\mu^n(\vec{\mathcal{R}}), \quad (13.17)$$

which is an anti-symmetric rank-2 tensor in the parameter space (similar to the field tensor  $F_{\mu\nu}$  in electromagnetism). Just as we do in electromagnetism, we can use Stokes’ theorem to relate the line integral of the connection around the boundary of a region to the surface integral of the curvature over that region:

$$\gamma_n = \oint_{\partial S} \vec{\mathcal{A}}^n(\vec{\mathcal{R}}) \cdot d\vec{\mathcal{R}} = \frac{1}{2} \int_S d\mathcal{R}_\mu \wedge d\mathcal{R}_\nu \omega_{\mu\nu}^n(\vec{\mathcal{R}}), \quad (13.18)$$

where  $S$  stands for the area in parameter space enclosed by the loop  $\partial S$ ,  $d\mathcal{R}_\mu \wedge d\mathcal{R}_\nu = -d\mathcal{R}_\nu \wedge d\mathcal{R}_\mu$  stands for an infinitesimal (oriented) element of the surface,<sup>6</sup> and again we use the Einstein summation convention. The factor of 1/2 compensates for the fact that each element is summed over twice due to the summation convention. If the parameter space happens to be 3D, we can express the Berry curvature  $\omega^n$  as a kind of “magnetic field” in terms of the curl of the connection vector,<sup>7</sup>

$$\vec{b}^n = \nabla_{\vec{\mathcal{R}}} \times \vec{\mathcal{A}}^n = i \langle \nabla_{\vec{\mathcal{R}}} n(\vec{\mathcal{R}}) | \times | \nabla_{\vec{\mathcal{R}}} n(\vec{\mathcal{R}}) \rangle, \quad (13.19)$$

<sup>6</sup> This “exterior” or “wedge” product notation is standard in differential geometry, see [35]. It represents the surface area of the parallelogram formed by the two vectors being multiplied. For the simplest case to visualize, namely a 2D surface embedded in 3D, it defines a surface normal vector given simply by the cross product of the two vectors.

<sup>7</sup> If  $\omega$  is a rank-2 anti-symmetric tensor in  $D$  dimensions, it contains  $D(D-1)/2$  independent components. For the special case of  $D=3$ , there are three independent components, the same number as in a 3D vector. As a result, it can be represented by a vector in 3D via  $b^\delta = \frac{1}{2} \epsilon^{\delta\mu\nu} \omega_{\mu\nu}$ , where  $\epsilon$  is the fully anti-symmetric tensor (Levi-Civita symbol). In this context  $\vec{b}$  is often called a pseudovector, reflecting the fact that it is really a rank-2 anti-symmetric tensor.

and the analogy with the vector potential/magnetic field is complete:

$$\gamma_n = \int_S \vec{b}^n \cdot d\vec{S}. \quad (13.20)$$

Motivated by this analogy, the Berry phase is also occasionally referred to as the **Berry flux**. In the following we use  $\omega_{\mu\nu}$  and  $\vec{b}$  interchangeably to represent the Berry curvature when  $D = 3$ . Indeed, we will see some examples in which  $\vec{b}$  actually is a physical magnetic field.

**Exercise 13.3.** Show from its definition in Eq. (13.17) that we can write the Berry curvature as

$$\omega_{\mu\nu}^n = i[\langle \partial_{\mathcal{R}_\mu} n(\vec{\mathcal{R}}) | \partial_{\mathcal{R}_\nu} n(\vec{\mathcal{R}}) \rangle - \langle \partial_{\mathcal{R}_\nu} n(\vec{\mathcal{R}}) | \partial_{\mathcal{R}_\mu} n(\vec{\mathcal{R}}) \rangle]. \quad (13.21)$$

In Eqs. (13.17) and (13.21), the Berry curvature  $\omega_{\mu\nu}^n$  is expressed in terms of the *variation* of the eigenstate  $|n(\vec{\mathcal{R}})\rangle$  with respect to the parameters  $\vec{\mathcal{R}}$ . In practice it is often convenient to express  $\omega_{\mu\nu}^n$  in terms of quantities at fixed  $\vec{\mathcal{R}}$ . To this end we will show below that

$$\omega_{\mu\nu}^n(\vec{\mathcal{R}}) = i \sum_{n' \neq n} \frac{\langle n(\vec{\mathcal{R}}) | (\partial H / \partial \mathcal{R}_\mu) | n'(\vec{\mathcal{R}}) \rangle \langle n'(\vec{\mathcal{R}}) | (\partial H / \partial \mathcal{R}_\nu) | n(\vec{\mathcal{R}}) \rangle - \text{c.c.}}{[\epsilon_n(\vec{\mathcal{R}}) - \epsilon_{n'}(\vec{\mathcal{R}})]^2}. \quad (13.22)$$

Equation (13.22) is explicitly gauge-invariant and often more useful in numerical calculations of  $\omega_{\mu\nu}^n$  because the numerically generated eigenvector  $|n(\vec{\mathcal{R}})\rangle$  is gauge-dependent and may carry an  $\vec{\mathcal{R}}$ -dependent phase factor that (accidentally<sup>8</sup>) does *not* vary smoothly with  $\vec{\mathcal{R}}$ ; this makes the derivatives in Eqs. (13.11) and (13.17) ill-defined, but does not affect the evaluation of Eq. (13.22).

To prove the equivalence of Eqs. (13.22) and (13.17), we take the derivative with respect to  $\mathcal{R}_\nu$  on both sides of Eq. (13.2), which yields

$$\left( \frac{\partial H}{\partial \mathcal{R}_\nu} \right) |n(\vec{\mathcal{R}})\rangle = \left( \frac{\partial \epsilon_n}{\partial \mathcal{R}_\nu} \right) |n(\vec{\mathcal{R}})\rangle + (\epsilon_n - H) | \partial_{\mathcal{R}_\nu} n(\vec{\mathcal{R}}) \rangle. \quad (13.23)$$

This leads to

$$\langle n'(\vec{\mathcal{R}}) | \left( \frac{\partial H}{\partial \mathcal{R}_\nu} \right) |n(\vec{\mathcal{R}})\rangle = (\epsilon_n - \epsilon_{n'}) \langle n'(\vec{\mathcal{R}}) | \partial_{\mathcal{R}_\nu} n(\vec{\mathcal{R}}) \rangle \quad (13.24)$$

for  $n \neq n'$ . Similarly,

$$\langle n(\vec{\mathcal{R}}) | \left( \frac{\partial H}{\partial \mathcal{R}_\mu} \right) |n'(\vec{\mathcal{R}})\rangle = (\epsilon_n - \epsilon_{n'}) \langle \partial_{\mathcal{R}_\mu} n(\vec{\mathcal{R}}) | n'(\vec{\mathcal{R}}) \rangle. \quad (13.25)$$

A good example of a pseudovector is the (real) magnetic field. One way to see that it is a pseudovector is to observe that, under spatial inversion that takes  $\vec{r} \rightarrow -\vec{r}$ , a true vector field transforms with odd parity as  $\vec{A}(\vec{r}) \rightarrow -\vec{A}(-\vec{r})$ , whereas a pseudovector field transforms with even parity as  $\vec{B}(\vec{r}) \rightarrow +\vec{B}(-\vec{r})$ . The reason why the magnetic field is even is that it is the curl of a true vector. The curl and the vector potential are both odd under spatial inversion, and their combination is even.

<sup>8</sup> For example, two different computer programs that find the eigenvectors of a given Hamiltonian matrix could produce results for (some of) the eigenvectors differing by a minus sign, or, in the case of complex eigenvectors, by an arbitrary overall phase that could be different for each eigenvector. Suppose, for instance, that the first program reports its eigenvectors using the arbitrary choice that the first entry in each eigenvector is real and positive. The second program might choose to make the last entry of the eigenvector real and positive. The two programs might produce identical results for some regions of parameter space and (suddenly) different results in another region (because in that region the eigenvector develops a sign difference between its first and last entries).

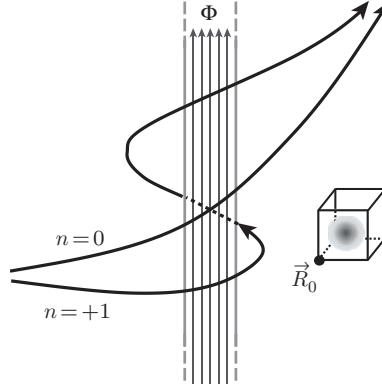
Combining these and using the resolution of the identity,  $\sum_{n'} |n'\rangle\langle n'|$ , shows that Eqs. (13.22) and (13.21) are equivalent. Equation (13.22) looks like a perturbation-theory result, but in fact it is formally exact.

## 13.2 Berry Phase and the Aharonov–Bohm Effect

A beautiful illustration of a non-trivial Berry connection can be found in the **Aharonov–Bohm effect**. Consider an infinitely long, very thin, tube of magnetic flux running along the  $z$  axis as shown in Fig. 13.1. Let there be a strong potential barrier which prevents charged particles from ever entering the thin region containing the magnetic field. Hence the particles never see the Lorentz force and classically the magnetic flux has no effect whatsoever on the motion. Quantum mechanically, however, there is an effect. Even though the magnetic field is zero outside the flux tube, the electromagnetic vector potential  $\vec{A}(\vec{r})$  is not. The vector potential obeys  $\nabla \times \vec{A} = \vec{0}$  everywhere the particle travels and hence, in any given local region, it can be gauged away. Nevertheless, there is a non-trivial global effect associated with the fact that, for any closed trajectory which winds around the flux tube, the line integral of the vector potential obeys

$$\oint d\vec{r} \cdot \vec{A}(\vec{r}) = n\Phi, \quad (13.26)$$

where  $\Phi$  is the total flux inside the flux tube and  $n$  is the integer winding number of the trajectory around the tube. The winding number is a well-defined topological invariant (see Appendix D),



**Figure 13.1** A tube of magnetic flux running along the  $z$  axis surrounded by a barrier which prevents charged particles from entering the region of the magnetic field. Because  $\vec{B} = \nabla \times \vec{A} = \vec{0}$  outside the barrier, the particles see no Lorentz force and hence the classical motion, though affected by the presence of the barrier, is independent of the value of the magnetic flux. Quantum mechanically, however, the flux has a non-trivial effect on the interference between the complex amplitudes associated with topologically distinct paths (which cannot be deformed into each other continuously). The two paths shown here are topologically distinct, as their difference (to take the difference, simply reverse the arrow on one of them) forms a closed loop with winding number  $n = +1$  (closed loops formed by pairs of topologically equivalent paths all have winding number  $n = 0$ ). This allows one to assign a winding number for each individual path, once a specific path has been *defined* to have zero winding number as shown in the figure. Because they are of different length, they will have different corresponding quantum amplitudes. The phase difference between the amplitudes will also depend on the total flux  $\Phi$  inside the flux tube, even though the Lorentz force is zero everywhere along both trajectories. The box in the lower right illustrates a method for adiabatically transporting a trapped electron along any desired trajectory. The orientation of the box is fixed, and the position  $\vec{R}_0$  of one corner is slowly varied. The finite size of the box provides the excitation gap necessary for adiabatic transport.

because the 3D space has been punctured by the excluded region of the flux tube. Continuous deformation of the trajectory cannot change the winding number, because passing through the punctured region is not allowed.<sup>9</sup>

Even though locally there appears to be no effect on the quantum mechanics, globally there is, and charged particles scatter from the flux tube in a way that varies with the product of the charge  $q$  and the total flux  $\Phi$ . One way to see this is to consider the Feynman path-integral approach, which assigns to any trajectory  $\vec{r}(t)$  an amplitude

$$e^{\frac{i}{\hbar} S[\vec{r}]}, \quad (13.27)$$

where  $S$  is the classical action for that trajectory,

$$S[\vec{r}] = \int_0^t d\tau \mathcal{L}[\dot{\vec{r}}, \vec{r}]. \quad (13.28)$$

Recall that in the presence of a vector potential the Lagrangian undergoes a modification (that leads to the Lorentz force in the classical equations of motion)

$$\mathcal{L} \rightarrow \mathcal{L} + (-e)\vec{A}(\vec{r}) \cdot \dot{\vec{r}}. \quad (13.29)$$

Thus we see that the amplitude associated with any given trajectory acquires a vector-potential-dependent phase

$$e^{i\theta} = e^{\frac{i(-e)}{\hbar} \int d\vec{r} \cdot \vec{A}(\vec{r})}, \quad (13.30)$$

which for closed trajectories is topological (and gauge invariant) – it depends only on the winding number of the trajectory around the Aharonov–Bohm flux tube. Rather than solving the scattering problem, we will here follow Berry’s example [89] and show that adiabatic transport of an electron around the flux tube yields a Berry phase equal to the Aharonov–Bohm phase,

$$\theta = \frac{i(-e)}{\hbar c} \oint d\vec{r} \cdot \vec{A}(\vec{r}) = -n2\pi \frac{\Phi}{\Phi_0}, \quad (13.31)$$

where  $\Phi_0$  is the flux quantum.

In order to do adiabatic transport of an electron (say) around the flux tube, we require that the Hamiltonian has an excitation gap. We can achieve this by trapping the electron inside a box with hard walls<sup>10</sup> as shown in Fig. 13.1. One corner of the box is located at position  $\vec{R}_0$ . The position of the electron is  $\vec{r}$  and the Hamiltonian is

$$H = \frac{1}{2m_e} \left[ \vec{p} + \frac{e}{c} \vec{A}(\vec{r}) \right]^2 + V(\vec{r} - \vec{R}_0). \quad (13.32)$$

<sup>9</sup> Using topology jargon, the puncture due to the flux tube has changed 3D space from being simply connected to being multiply connected, with closed trajectories or loops forming different topological classes, each of which is characterized by an integer-valued winding number. Formally, this statement can be expressed by noting that the “fundamental group” [35, 43, 44] of the punctured space is  $\pi_1 = \mathbb{Z}$ , where  $\mathbb{Z}$  is the group made of all integers with the group operation defined to be addition. The elements of the fundamental group are equivalence classes of trajectories labeled by their winding number and the group operation is composition of two trajectories. The equation  $\pi_1 = \mathbb{Z}$  simply states the fact that, if we “glue” two loops together to form a new loop, its winding number is the sum of the winding numbers of the two original loops. This explains why the group operation is addition; the fact that the group is  $\mathbb{Z}$  is explained (loosely speaking) by the fact that any loop with winding number  $n$  (necessarily an integer) can be smoothly deformed into any other loop with the same winding number, but not into any loop with a different winding number. See Appendix D for further details.

<sup>10</sup> Any other strongly confining potential such as a harmonic well will also work so long as there is a finite energy gap from the ground state to the first excited state. We also require that the confinement be strong enough that the wave function does not leak out into the vicinity of the flux tube.

We implicitly assume that the orientation of the box remains fixed as we transport it by varying  $\vec{R}_0$  so that we can neglect the dependence of the potential on the orientation.<sup>11</sup> We will further assume that the particle in the box remains completely outside the flux tube at all times so that we may take  $\nabla \times \vec{A} = \vec{0}$  and thus can choose an electromagnetic vector potential of the form

$$\vec{A}(\vec{r}) = \frac{\Phi_0}{2\pi} \nabla \chi(\vec{r}), \quad (13.33)$$

with<sup>12</sup>

$$\oint d\vec{r} \cdot \vec{A} = \frac{\Phi_0}{2\pi} \oint d\vec{r} \cdot \nabla \chi(\vec{r}) = n\Phi, \quad (13.34)$$

where  $n$  is the winding number of the closed loop around the flux tube containing flux  $\Phi$ . An obvious (possible) gauge choice is having  $\chi(\vec{r}) = (\Phi/\Phi_0)\varphi(\vec{r})$ , with  $\varphi(\vec{r})$  being the azimuthal angle of  $\vec{r}$ ; however, we will not specify a particular gauge at this stage.

If the quantum ground state wave function is  $\xi_0(\vec{r} - \vec{R}_0)$  when  $\vec{A}$  vanishes everywhere, then, in the presence of the flux tube, the solution of the Schrödinger equation can be written as

$$\psi(\vec{r}) = e^{-i\chi_{\vec{R}_0}(\vec{r})} \xi_0(\vec{r} - \vec{R}_0), \quad (13.35)$$

with

$$\chi_{\vec{R}_0}(\vec{r}) \equiv \frac{2\pi}{\Phi_0} \int_{\vec{R}_0}^{\vec{r}} d\vec{r}' \cdot \vec{A}(\vec{r}'), \quad (13.36)$$

where the line integral is evaluated along some path lying inside the box. Because  $\vec{A}$  has no curl anywhere inside the box, the integral is independent of the path over which it is evaluated and hence  $\chi(\vec{r})$  is well-defined everywhere inside the box. (We do not require  $\chi$  globally, only inside the box.)

As we have emphasized several times, we can multiply  $\psi(\vec{r})$  by an arbitrary (global) phase factor  $e^{i\theta}$  and still have a valid solution. Furthermore, the phase  $\theta(\vec{R}_0)$  could be a function of the location in parameter space (but *not* a function of the real-space coordinate  $\vec{r}$ ). Different choices of  $\theta(\vec{R}_0)$  correspond to different gauge choices for the Berry connection in parameter space. Different choices of  $\chi(\vec{r})$  correspond to different electromagnetic gauge choices. Let us choose the gauge for the Berry connection by requiring that  $\psi(\vec{r}) = e^{i\theta(\vec{R}_0)} e^{-i\chi_{\vec{R}_0}(\vec{r})} \xi_0(\vec{r} - \vec{R}_0)$  be real and positive at some point  $(\vec{R}_0 + \vec{\Delta})$  in the interior of the box (where  $\xi_0(\vec{\Delta}) \neq 0$ ). Without loss of generality, we assume that  $\xi_0(\vec{r} - \vec{R}_0)$  is everywhere real and that  $\xi_0(\vec{\Delta})$  is positive. Then we have very simply  $\theta(\vec{R}_0) = +\chi_{\vec{R}_0}(\vec{R}_0 + \vec{\Delta})$ .

Using

$$\nabla_{\vec{R}_0} \theta(\vec{R}_0) = \nabla_{\vec{R}_0} \chi_{\vec{R}_0}(\vec{R}_0 + \vec{\Delta}) = \frac{2\pi}{\Phi_0} \left\{ \vec{A}(\vec{R}_0 + \vec{\Delta}) - \vec{A}(\vec{R}_0) \right\}, \quad (13.37)$$

we find that the Berry connection is then

$$\vec{A}(\vec{R}_0) = i \langle \psi | \nabla_{\vec{R}_0} | \psi \rangle \quad (13.38)$$

$$= -\frac{2\pi}{\Phi_0} \vec{A}(\vec{R}_0 + \vec{\Delta}) + i \int d^3\vec{r} \xi_0(\vec{r} - \vec{R}_0) \nabla_{\vec{R}_0} \xi_0(\vec{r} - \vec{R}_0) \quad (13.39)$$

<sup>11</sup> It should be clear by now that, in this example, the parameters  $\vec{\mathcal{R}}$  describing the Hamiltonian,  $H$ , are simply  $\vec{R}_0$ , the three real-space coordinates of (the corner of) the box.

<sup>12</sup> Note that, while  $\vec{A}$  has no curl outside the excluded region, its line integral around a closed loop with winding number  $n \neq 0$  is finite. There is no continuous function  $\chi(\vec{r})$  which is globally well-defined (single-valued), but we will not require this.

$$= -\frac{2\pi}{\Phi_0} \vec{A}(\vec{R}_0 + \vec{\Delta}) + \frac{i}{2} \int d^3\vec{r} \nabla_{\vec{r}} \xi_0^2 (\vec{r} - \vec{R}_0) \quad (13.40)$$

$$= -\frac{2\pi}{\Phi_0} \vec{A}(\vec{R}_0 + \vec{\Delta}), \quad (13.41)$$

where we have used the fact that the integrand in the second term in Eq. (13.40) is a total derivative so that the integral vanishes (because  $\xi_0$  vanishes on the boundary of the box).

If we now slowly vary the parameter  $\vec{R}_0$  in a loop such that  $(\vec{R}_0 + \vec{\Delta})$  encircles the flux tube with winding number  $n = +1$ , we have

$$\gamma = \oint d\vec{R}_0 \cdot \vec{A}(\vec{R}_0) = - \oint d\vec{R}_0 \cdot \frac{2\pi}{\Phi_0} \vec{A}(\vec{R}_0 + \vec{\Delta}) = -2\pi \frac{\Phi}{\Phi_0}. \quad (13.42)$$

Thus, the Berry phase is nothing but the Aharonov–Bohm phase in this simple case, and the origin of the Berry flux is the physical magnetic flux confined in the physical flux tube.<sup>13</sup>

### 13.3 Spin-1/2 Berry Phase

As another simple example, let us consider a spin-1/2 system, whose most general Hamiltonian (up to a constant which is irrelevant here<sup>14</sup>) takes the form

$$H = \vec{h} \cdot \vec{\sigma} = \sum_{j=1}^3 h_j \sigma_j, \quad (13.43)$$

where  $\sigma_j$  is the  $j$ th Pauli matrix. Of course, the Hamiltonian for all two-level systems can be described this way. In this case the parameter space of  $\vec{h}$  is the familiar 3D Euclidean space<sup>15</sup>  $E_3$ . As will soon be clear, it is convenient to write the position vector in parameter space,  $\vec{h}$ , using polar coordinates:

$$\vec{h} = (h_1, h_2, h_3) = h(\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta). \quad (13.44)$$

The two eigenstates with energies  $\pm h$  are

$$|-\rangle = \begin{pmatrix} \sin(\theta/2)e^{-i\varphi} \\ -\cos(\theta/2) \end{pmatrix}, \quad |+\rangle = \begin{pmatrix} \cos(\theta/2)e^{-i\varphi} \\ \sin(\theta/2) \end{pmatrix}. \quad (13.45)$$

**Exercise 13.4.** Prove that the expressions in Eq. (13.45) do indeed represent eigenstates obeying  $H|\pm\rangle = \pm h|\pm\rangle$ . Further show that the spin polarization is parallel (anti-parallel) to the magnetic field  $\langle \pm | \vec{\sigma} | \pm \rangle = \pm \vec{h}/h$ .

It is interesting to note that, while the eigenvalues depend on the magnitude of  $h$ , the eigenstates depend only on the orientation of the “quantization axis,” which is determined by two of the three parameters of  $\vec{h}$ , namely  $\theta$  and  $\varphi$ , that form the space of  $S^2$  (the surface of the unit sphere). It is easy to calculate the Berry connection and curvature on the  $S^2$  space, say, for the ground state:

<sup>13</sup> The minus sign in the expression for the Berry phase simply reflects the negative charge of the electron.

<sup>14</sup> Even if the constant is one of the parameters that varies as one moves around in parameter space, it still contributes only a dynamical phase, not a geometric phase.

<sup>15</sup> More precisely, it is a punctured version of  $E_3$  in which the origin  $\vec{h} = \vec{0}$  has been removed. This is necessary because the energy-level spacing (the excitation gap)  $2|\hbar|$  vanishes at the origin.

$$\mathcal{A}_\theta^- = \langle - | i \partial_\theta | - \rangle = 0; \quad (13.46)$$

$$\mathcal{A}_\varphi^- = \langle - | i \partial_\varphi | - \rangle = \sin^2(\theta/2); \quad (13.47)$$

$$\omega_{\theta\varphi}^- = \partial_\theta \mathcal{A}_\varphi^- - \partial_\varphi \mathcal{A}_\theta^- = \frac{1}{2} \sin \theta. \quad (13.48)$$

Thus the Berry flux through a small element of  $S^2$  is

$$\omega_{\theta\varphi}^- d\theta d\varphi = \frac{1}{2} \sin \theta d\theta d\varphi = \frac{1}{2} d\Omega, \quad (13.49)$$

where  $d\Omega = \sin \theta d\theta d\varphi$  is the solid angle subtended by the element when viewed from the origin. As we will discuss further below in Eq. (13.55), this is a remarkable result which tells us that the “magnetic” field (Berry curvature) is a constant everywhere on the surface of a sphere in parameter space. The integral of the Berry curvature over the entire surface is thus

$$\int \omega_{\theta\varphi}^- d\theta d\varphi = \frac{1}{2} \int d\Omega = 2\pi, \quad (13.50)$$

and the Berry phase of a loop on  $S^2$  is

$$\gamma_- = \frac{1}{2} \int_S d\Omega = \frac{1}{2} \Omega_S, \quad (13.51)$$

or one-half of the solid angle subtended by the trajectory, as viewed from the center of the sphere (and with a sign determined by the handedness of the trajectory).

It is important to understand that, in writing Eq. (13.45), we have implicitly made a particular gauge choice for the Berry connection. More generally, we could have written

$$|- \rangle = \begin{pmatrix} \sin(\theta/2) e^{-i\varphi} \\ -\cos(\theta/2) \end{pmatrix} e^{i\Theta_-(\theta, \varphi)} \quad (13.52)$$

$$|+ \rangle = \begin{pmatrix} \cos(\theta/2) e^{-i\varphi} \\ \sin(\theta/2) \end{pmatrix} e^{i\Theta_+(\theta, \varphi)}, \quad (13.53)$$

where  $\Theta_\pm(\theta, \varphi)$  are two distinct and arbitrary (smooth) functions of the position in parameter space. It is straightforward to show, however, that the Berry curvature remains unaffected by this gauge change in the Berry connection.

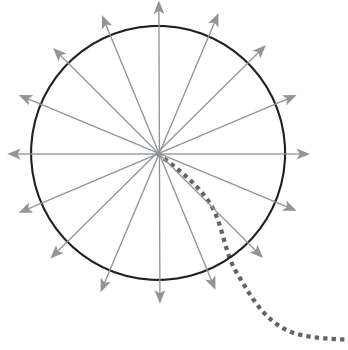
**Exercise 13.5.** Prove that the Berry curvature for adiabatic transport of the states in Eqs. (13.52) and (13.53) is independent of the gauge choice determined by the arbitrary (smooth) function  $\Theta_\pm(\theta, \varphi)$ .

It is easy to generalize our results for spin-1/2 to the general case of a spin  $s$  coupled to a Zeeman field.<sup>16</sup> One finds that the Berry phase and curvature is simply proportional to the size  $s$  of the spin:

$$\gamma_- = s \int_S d\Omega = s \Omega_S, \quad (13.54)$$

**Exercise 13.6.** Derive Eq. (13.54) for arbitrary spin angular momentum  $s$ . Hint: use the fact the angular momentum components are the generators of rotations and that they obey the commutation algebra  $[S^x, S^y] = i\hbar S^z$  and cyclic permutations thereof.

<sup>16</sup> As a side note, we point out that the Zeeman coupling  $\vec{h} \cdot \vec{S}$  is the most general possible Hamiltonian for spin-1/2. This is not true for higher spins.



**Figure 13.2** A monopole that radiates either Berry or magnetic flux (solid lines with arrows). Dirac proposed realizing a magnetic monopole in the form of a semi-infinite solenoid (dotted line) carrying a single quantum of magnetic flux  $hc/e$ . The total magnetic flux emanating from the surface of the sphere is  $hc/e$ , provided that we neglect the singular contribution from the infinitely narrow solenoid. This can be neglected because it produces an Aharonov–Bohm phase shift of  $\pm 2\pi$  and has no observable physical effects. This **Dirac monopole** thus acts like a point magnetic charge with no string attached.

Going back to the Cartesian coordinate parameter space of  $\vec{h}$ , it is clear that the Berry curvature seen by the lower (adiabatic) energy eigenstate  $|-\rangle$  is

$$\vec{b}_- = s \frac{\vec{h}}{h^3}, \quad (13.55)$$

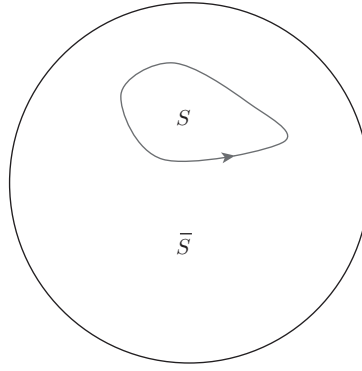
which has the form of a magnetic field generated by a **magnetic monopole** with charge  $s$  located at the origin of parameter space; see Fig. 13.2. This magnetic field’s magnitude is equal to  $s$  times the local **Gaussian curvature** of the surface of a sphere with radius  $h$ , and its direction is the local normal direction of this surface. Adiabatic evolution assumes the existence of an excitation gap, which in this model is  $\epsilon_+ - \epsilon_- = 2|\vec{h}|$ . We thus see that the gap vanishes at the origin of parameter space and we can think of the degeneracy at that point “radiating” Berry curvature  $\vec{b}_-$  just as a magnetic monopole is a point source of magnetic field.

**Exercise 13.7.** Calculate the Berry curvature of the state  $|+\rangle$ , and show that it is the opposite of that of  $|-\rangle$ ; thus the sum of the Berry curvatures of these two states is zero everywhere in parameter space.

The first observation of the spin Berry phase was made by the Pines group [92] using nuclear magnetic resonance techniques to drive the effective magnetic field through a small loop and then use a kind of interferometry to compare the phases acquired by the ground and excited states. The same experiment has also now been done using manipulation of a superconducting quantum bit acting as an artificial two-level atom [93], and the ability to implement “geometric phase gates” plays a prominent role in the nascent field of quantum information processing.

Our discussion would not be complete without pointing out that, for a closed loop on the sphere  $S^2$ , there is an ambiguity as to which part of the sphere is interior to the path and which part is exterior. As can be seen in Fig. 13.3, a small loop with right-hand<sup>17</sup> orientation enclosing region  $S$  can also be viewed as a left-hand-oriented loop enclosing a large region  $\bar{S}$  complementary to  $S$ . Thus there are

<sup>17</sup> That is, if the fingers of your right hand curl in the direction of the motion around the loop, your thumb is pointing radially outward away from the surface.



**Figure 13.3** Illustration of an oriented path on the sphere  $S^2$ . The path can be viewed as having a right-handed orientation enclosing region  $S$  or a left-handed orientation enclosing the complementary region  $\bar{S}$ .

two different possible values for the Berry phase,  $\gamma_- = s\Omega_S$  and  $\bar{\gamma}_- = -s\Omega_{\bar{S}}$  (where we are using a convention in which the solid angle is always taken to be positive). In order for the Berry phase to be mathematically well-defined, we require that these two methods of calculation give the same answer:

$$\gamma_- = \bar{\gamma}_- \mod (2\pi). \quad (13.56)$$

Using the geometric fact that the total solid angle of the sphere is  $4\pi$ , we know that

$$\Omega_S + \Omega_{\bar{S}} = 4\pi, \quad (13.57)$$

and thus

$$\gamma_- - \bar{\gamma}_- = s(4\pi) = 0 \mod (2\pi). \quad (13.58)$$

We thus see that our self-consistency condition in Eq. (13.56) is true, provided that the spin  $s$  is an integer multiple of  $1/2$ .

This remarkable topological result is a beautiful way to explain the quantization of spin in non-relativistic quantum mechanics, if we use Eq. (13.54) or Eq. (13.55) as the *definition* of the spin  $s$ . Because the Berry curvature in parameter space is like that of a magnetic monopole, the topological argument for the quantization of spin is closely related to the Dirac argument for the quantization of the magnetic charge of monopoles. One way to create an effective monopole, illustrated in Fig. 13.2, is to have a semi-infinitely long solenoid which abruptly terminates at the position where we want the monopole to be located. In order for the magnetic flux emanating from the end of the solenoid to act like the field from a point (magnetically charged) particle, the “Dirac string” (i.e. the solenoid) has to be undetectable. This will be the case provided that the flux inside the solenoid is an integer multiple of the flux quantum  $hc/e$  (or  $h/e$  in SI units) so that an electron circling the solenoid will pick up an Aharonov–Bohm phase of  $\pm 2\pi$  which is equivalent to zero phase shift as if there were no string. (The product of the charge quantum  $e$  and the (SI) flux quantum  $h/e$  being Planck’s constant guarantees that the Aharonov–Bohm phase is  $h/\hbar = 2\pi$ .) Dirac further argued, by reversing the logic, that quantization of electric charge is due to the presence of magnetic monopoles.<sup>18</sup>

<sup>18</sup> Although they are allowed in certain quantum field theories, there is no convincing evidence for the existence of magnetic monopoles at the time of writing.

### 13.3.1 Spin–Orbit Coupling and Suppression of Weak Localization

As an example of another physically observable effect of the Berry phase, let us consider how it affects electron backscattering and localization in the presence of a random disorder potential. As we discussed in Chapter 11, backscattering is enhanced by constructive quantum interference between different scattering processes that are related to each other by time-reversal transformation; see Fig. 11.1 and in particular Fig. 11.3. However, electron spin was not considered there. Neglecting spin would be appropriate if the spin state does *not* change during the scattering processes. This is true for pure potential scattering in the absence of spin–orbit coupling, but is no longer the case for pure potential scattering in the presence of spin–orbit coupling or for scattering due to spin-dependent forces. In the presence of spin–orbit coupling, the spin of the energy eigenstates depends on momentum. Hence each potential scattering that changes the momentum may be accompanied by a corresponding change in spin state. As a result, the electron spin precesses through *different* paths in spin space (or the unit sphere parameterized by  $\theta, \varphi$ ) for the different sequences of scattering processes illustrated in Figs. 11.3 and 11.1. For the case of scattering from a smoothly varying potential, the evolution of the spin can be treated as adiabatic motion on the Bloch sphere, and the phase *difference* between these two paths is precisely

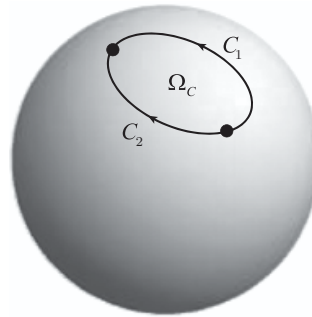
$$\gamma = \gamma_{C_1} - \gamma_{C_2} = \gamma_C = \Omega_C/2, \quad (13.59)$$

where  $C_1$  and  $C_2$  represent these two paths in spin space (see Fig. 13.4 for an illustration), and  $C = C_1 - C_2$  is a *closed loop* with  $C_1$  and  $C_2$  as its segments (with proper orientations). We thus find that spin–orbit coupling induces a non-zero phase difference among different backscattering processes, whose origin is nothing but the Berry phase associated with spin. Its effect is very similar to that of a magnetic field which induces an Aharonov–Bohm phase difference between time-reversal-related paths (as illustrated in Fig. 11.5), and thus *suppresses* weak-localization effects.

As an example of this, let us consider an extreme form of spin–orbit coupling, namely a Hamiltonian with Dirac (instead of Schrödinger) kinetic energy in 2D:

$$H = v_F \vec{\sigma} \cdot \vec{p} + V(\vec{r}), \quad (13.60)$$

where  $v_F$  is the Fermi velocity. This Hamiltonian describes electrons near the Dirac points in graphene (where, as described in Section 7.5,  $\vec{\sigma}$  is a pseudospin corresponding to the A, B sublattice degree of freedom) or the surface states of a topological insulator (to be discussed in Chapter 14). For the case of the Hamiltonian in Eq. (13.60), the direction of spin (or pseudospin) in an energy eigenstate is always parallel or anti-parallel to the momentum and thus  $\vec{k}$ . As a result, for the two scattering



**Figure 13.4** Illustration of the spin Berry phase difference between different paths ( $C_1$  and  $C_2$ ) that an electron spin traverses. This phase difference suppresses enhanced backscattering and weak localization, in the presence of spin–orbit coupling.

processes illustrated in Fig. 11.3, the loop  $C$  is precisely the equator and thus  $\gamma_C = \pi$ , resulting in a complete *cancelation* of backscattering! The resulting enhancement of diffusion is known as weak *anti-localization*. A magnetic field which introduces an additional phase difference between these two processes would suppress this anti-localization effect by ruining this perfect cancelation; this results in *positive* magnetoresistance, in contrast to what we saw in Chapter 11, where spin–orbit coupling was absent.

The Dirac equation calculation was presented here because of its simplicity – the spin direction in the plane-wave energy eigenstates is exactly parallel (or anti-parallel) to the momentum. This is a good description for the graphene (pseudospin) band structure. For more general Schrödinger bands with spin–orbit coupling, the quantitative computations are more complex, but the essential physics of the (real) spin Berry phase contribution to the backscattering amplitude is the same.

### 13.4 Berry Curvature of Bloch Bands and Anomalous Velocity

Owing to the (magnetic) translation symmetry, an eigenfunction of the Hamiltonian in Eq. (12.106) may be written as

$$\psi_{n\vec{k}}(\vec{r}) = e^{i\vec{k}\cdot\vec{r}} u_{n\vec{k}}(\vec{r}), \quad (13.61)$$

where  $u_{n\vec{k}}(\vec{r})$  satisfies (generalized) periodic boundary conditions:

$$\tilde{T}_{\vec{a}_j} u_{n\vec{k}}(\vec{r}) = e^{i\phi_{\vec{a}_j}(\vec{r})} u_{n\vec{k}}(\vec{r} + \vec{a}_j) = u_{n\vec{k}}(\vec{r}). \quad (13.62)$$

We can solve for  $u_{n\vec{k}}(\vec{r})$  and the corresponding band dispersion by combining Eq. (13.62) with the  $\vec{k}$ -dependent Schrödinger equation it satisfies:

$$h(\vec{k}) u_{n\vec{k}}(\vec{r}) = \epsilon_n(\vec{k}) u_{n\vec{k}}(\vec{r}), \quad (13.63)$$

where

$$h(\vec{k}) = \frac{1}{2m} \left[ \vec{p} + \hbar\vec{k} + \frac{e}{c} \vec{A}(\vec{r}) \right]^2 + V(\vec{r}). \quad (13.64)$$

In this formulation  $\{\vec{k}\}$  becomes a set of *parameters* of the Hamiltonian  $h(\vec{k})$ . We can thus define the corresponding Berry curvature for each band (with band index  $n$ ) in this  $\vec{k}$ -space. Because this space has the same number of components as the spatial dimension, the curvature tensor (in 3D<sup>19</sup>) can be expressed in a compact vector form (recall the discussion leading to Eq. (13.19)):

$$\vec{b}_n(\vec{k}) = i \langle \partial_{\vec{k}} u_{n\vec{k}} | \times | \partial_{\vec{k}} u_{n\vec{k}} \rangle. \quad (13.65)$$

Before getting into a description of the physical effects associated with (non-zero)  $\vec{b}_n(\vec{k})$ , we briefly comment on the geometry and topology of  $\vec{k}$ -space. We know that, as the lattice momentum/wave vector,  $\vec{k}$  is defined inside the first Brillouin zone (1BZ). Naively, one would think the 1BZ is a segment of a line in 1D, a parallelogram or hexagon in 2D, and some kind of polygon in 3D. However, due to the periodic structure in reciprocal space,  $\vec{k}$ s that differ by a reciprocal lattice vector should be identified as the *same* point. As a consequence, opposite edges or faces of 1BZ should be identified with each other, and a 1BZ in  $d$ -dimensions corresponds to a  $d$  dimensional torus,  $\mathcal{T}_d$ . In 1D this is just a ring or circle. While their geometry depends on details of the lattice structure and the magnetic field, topologically all 1BZs with the same dimensionality are equivalent.

<sup>19</sup> For the case of 2D where  $\vec{k}$  lies in the  $XY$  plane,  $\vec{b}_n$  has only a  $\hat{z}$  component and hence is effectively a scalar.

### 13.4.1 Anomalous Velocity

As we warned the reader in Section 8.2, our derivation of the semiclassical equations of motion for Bloch-band wave packets was not quite correct due to a subtle Berry phase effect in certain band structures. It is now time to address this point. We will show in this subsection that non-zero Berry curvature  $\vec{b}_n(\vec{k})$  yields a correction to the (quantum-mechanical) average velocity of Bloch states, in the presence of a uniform or slowly varying (on the scale of a wave packet) external electric field.

Recall that we showed in Chapter 7 that

$$\langle \psi_{n\vec{k}} | \vec{v} | \psi_{n\vec{k}} \rangle = \langle u_{n\vec{k}} | \frac{1}{\hbar} \frac{\partial h(\vec{k})}{\partial \vec{k}} | u_{n\vec{k}} \rangle = \frac{1}{\hbar} \frac{\partial \epsilon_n(\vec{k})}{\partial \vec{k}}. \quad (13.66)$$

An external perturbing field *modifies* the state  $|\psi_{n\vec{k}}\rangle$ , which can lead to a correction to  $\langle \vec{v} \rangle$ . In the following, we consider the effect of a weak uniform electric field, and treat it using first-order perturbation theory in a manner very similar to Exercise 12.7. The perturbation is

$$H' = e\vec{E} \cdot \vec{r}, \quad (13.67)$$

and we have<sup>20</sup> (to first order in  $\vec{E}$ )

$$|\delta \psi_{n\vec{k}}\rangle = e\vec{E} \cdot \sum_{n', \vec{k}'} \frac{|\psi_{n'\vec{k}'}\rangle \langle \psi_{n'\vec{k}'} | \vec{r} | \psi_{n\vec{k}} \rangle}{\epsilon_{n\vec{k}} - \epsilon_{n'\vec{k}'}} \quad (13.68)$$

$$= e\vec{E} \cdot \sum_{n', \vec{k}'} \frac{|\psi_{n'\vec{k}'}\rangle \langle \psi_{n'\vec{k}'} | [\vec{r}, H] | \psi_{n\vec{k}} \rangle}{(\epsilon_{n\vec{k}} - \epsilon_{n'\vec{k}'})^2} \quad (13.69)$$

$$= i\hbar e\vec{E} \cdot \sum_{n', \vec{k}'} \frac{|\psi_{n'\vec{k}'}\rangle \langle \psi_{n'\vec{k}'} | \vec{v} | \psi_{n\vec{k}} \rangle}{(\epsilon_{n\vec{k}} - \epsilon_{n'\vec{k}'})^2}. \quad (13.70)$$

In the calculation of the matrix element  $\langle \psi_{n'\vec{k}'} | \vec{v} | \psi_{n\vec{k}} \rangle$ , it is easy to see that the contributions from each unit cell (in real space) are identical except for a unit-cell-position ( $\vec{R}$ )-dependent phase factor  $e^{i(\vec{k}-\vec{k}')\cdot\vec{R}}$ . The matrix element thus vanishes unless<sup>21</sup>  $\vec{k} = \vec{k}'$ . This allows us to make the simplification

$$\langle \psi_{n'\vec{k}'} | \vec{v} | \psi_{n\vec{k}} \rangle = \langle u_{n'\vec{k}} | \frac{1}{\hbar} \frac{\partial h(\vec{k})}{\partial \vec{k}} | u_{n\vec{k}} \rangle \delta_{\vec{k}'\vec{k}} \quad (13.71)$$

and removes the sum over  $\vec{k}'$  in Eq. (13.70), so that we are left with a single summation  $n' \neq n$ . From now on we focus on  $|u_{n\vec{k}}\rangle$ , and Eq. (13.70) reduces to

$$|\delta u_{n\vec{k}}\rangle = ie\vec{E} \cdot \sum_{n' \neq n} \frac{|u_{n'\vec{k}}\rangle \langle u_{n'\vec{k}} | (\partial h(\vec{k}) / \partial \vec{k}) | u_{n\vec{k}} \rangle}{(\epsilon_{n\vec{k}} - \epsilon_{n'\vec{k}})^2}. \quad (13.72)$$

As a result, we find (to first order in  $\vec{E}$ )

$$\begin{aligned} \langle \vec{v} \rangle &= \langle u_{n\vec{k}} | \frac{1}{\hbar} \frac{\partial h(\vec{k})}{\partial \vec{k}} | u_{n\vec{k}} \rangle + \langle \delta u_{n\vec{k}} | \frac{1}{\hbar} \frac{\partial h(\vec{k})}{\partial \vec{k}} | u_{n\vec{k}} \rangle + \langle u_{n\vec{k}} | \frac{1}{\hbar} \frac{\partial h(\vec{k})}{\partial \vec{k}} | \delta u_{n\vec{k}} \rangle \\ &= \frac{1}{\hbar} \frac{\partial \epsilon_n(\vec{k})}{\partial \vec{k}} + \vec{v}_a(n, \vec{k}), \end{aligned} \quad (13.73)$$

<sup>20</sup> Experts will recognize that  $\vec{r}$  is not, strictly speaking, a well-defined operator in a space of plane-wave-like states. We escape this mathematical subtlety (without proof) by transforming the matrix element to that of the velocity operator.

<sup>21</sup> Strictly speaking, we should write  $\vec{k} = \vec{k}' + \vec{G}$  for any reciprocal lattice vector  $\vec{G}$ . However, we can safely restrict  $\vec{k}$  and  $\vec{k}'$  to the 1BZ because we are allowing for different band indices in our expressions.

where the “anomalous velocity,” which is a correction to Eq. (13.66) due to electric-field-induced band mixing (i.e. interband coherence), is

$$\vec{v}_a(n, \vec{k}) = \frac{ie}{\hbar} \sum_{n' \neq n} \frac{\langle u_{n\vec{k}} | (\partial h(\vec{k}) / \partial \vec{k}) | u_{n'\vec{k}} \rangle [\vec{E} \cdot \langle u_{n'\vec{k}} | (\partial h(\vec{k}) / \partial \vec{k}) | u_{n\vec{k}} \rangle] - \text{c.c.}}{(\epsilon_{n\vec{k}} - \epsilon_{n'\vec{k}})^2} \quad (13.74)$$

$$= \frac{ie}{\hbar} \vec{E} \times \left[ \sum_{n' \neq n} \frac{\langle u_{n\vec{k}} | (\partial h(\vec{k}) / \partial \vec{k}) | u_{n'\vec{k}} \rangle \times \langle u_{n'\vec{k}} | (\partial h(\vec{k}) / \partial \vec{k}) | u_{n\vec{k}} \rangle}{(\epsilon_{n\vec{k}} - \epsilon_{n'\vec{k}})^2} \right] \quad (13.75)$$

$$= \frac{e}{\hbar} \vec{E} \times \vec{b}_n(\vec{k}), \quad (13.76)$$

where we used Eq. (13.22) in the last step.

Recognizing that a weak electric field  $\vec{E}$  induces semiclassical time-evolution (or a “velocity” in momentum space) for the lattice momentum as in Eq. (8.14), we can write Eq. (13.76) in the following suggestive form:

$$\vec{v}_a(n, \vec{k}) = - \left( \frac{d\vec{k}}{dt} \right) \times \vec{b}_n(\vec{k}). \quad (13.77)$$

Notice that the RHS takes a form very similar to the Lorentz force, except that  $d\vec{k}/dt$  is the velocity in *momentum* (instead of real) space, and the Berry curvature  $\vec{b}_n(\vec{k})$  can be viewed as magnetic field (or flux density), also in momentum space.<sup>22</sup>

Combining the contribution from the anomalous velocity with the equations of motion derived in Chapter 8, we obtain the following corrected equations that are now highly symmetric:

$$\hbar \frac{d\vec{R}}{dt} = \nabla_{\vec{k}} \epsilon_n(\vec{k}) - \hbar \left( \frac{d\vec{k}}{dt} \right) \times \vec{b}_n(\vec{k}); \quad (13.78)$$

$$\hbar \frac{d\vec{k}}{dt} = -\nabla_{\vec{R}} V(\vec{R}) - \frac{e}{c} \left( \frac{d\vec{R}}{dt} \right) \times \vec{B}(\vec{R}). \quad (13.79)$$

As we see under the exchange of (wave-packet) position  $\vec{R}$  and lattice momentum  $\vec{k}$ , the band energy  $\epsilon_n(\vec{k})$  (which is the momentum-dependent energy) exchanges with the external potential  $V(\vec{R})$  (which is the position-dependent energy), while the Berry curvature exchanges with the magnetic field. We would not have such a beautiful duality between real and momentum spaces without the anomalous velocity term.<sup>23</sup>

While Eq. (13.76) derived above is correct, strictly speaking, in the presence of the  $H'$  term (Eq. (13.67)), the Hamiltonian no longer respects (magnetic) lattice translation symmetry, thus its eigenstates can no longer be characterized by a lattice wave vector  $\vec{k}$ . Another problem with the derivation above is that  $H'$  in Eq. (13.67) is unbounded, and thus, strictly speaking, cannot be treated as a perturbation. To avoid these issues, in the following we use an additional time-dependent vector potential to model the electric field, as we did in Section 8.2.2:

$$\vec{E} = -\frac{1}{c} \frac{\partial \vec{A}'}{\partial t}; \quad \vec{A}' = -c \vec{E} t. \quad (13.80)$$

Since  $\vec{A}'$  is independent of  $\vec{r}$ , lattice symmetry is respected. The price to pay is that we now need to deal with a time-dependent version of the Hamiltonian (13.64):

<sup>22</sup> Recall that a real magnetic field can be viewed as Berry curvature in real space, as we learned in Section 13.2.

<sup>23</sup> This is similar to the lack of electromagnetic duality without the displacement current term in Maxwell’s equations.

$$h(\vec{k}, t) = h[\vec{k}(t)] = \frac{1}{2m} \left[ \vec{p} + \hbar \vec{k}(t) + \frac{e}{c} \vec{A}(\vec{r}) \right]^2 + V(\vec{r}), \quad (13.81)$$

where

$$\vec{k}(t) = \vec{k}_0 + \frac{e}{\hbar c} \vec{A}' = \vec{k}_0 - \frac{e \vec{E} t}{\hbar}, \quad (13.82)$$

from which we immediately obtain

$$\hbar \dot{\vec{k}} = -e \vec{E}, \quad (13.83)$$

which is a special case of Eq. (8.14). The weak-electric-field limit is thus the adiabatic limit, where  $\vec{k}$  varies slowly with time. However, even in this limit there is a small admixture of states in different bands (see Eq. (8.37)) that is proportional to  $\vec{E}$ ; this coherent admixture is responsible for the anomalous velocity, to which we now turn.

We now formulate the calculation of the anomalous velocity more generally in the context of adiabatic time-evolution and corresponding transport. As in Section 13.1, we consider a Hamiltonian  $H(\vec{\mathcal{R}}) = H(\mathcal{R}_1, \mathcal{R}_2, \dots)$  that depends on a set of parameters  $(\mathcal{R}_1, \mathcal{R}_2, \dots)$ . We further introduce generalized velocity operators

$$V_\mu = \frac{\partial H(\vec{\mathcal{R}})}{\partial \mathcal{R}_\mu} = \partial_\mu H. \quad (13.84)$$

Now let  $\vec{\mathcal{R}} = \vec{\mathcal{R}}(t)$  vary slowly in time, and consider the adiabatic time-evolution of a state  $|\psi_n(t)\rangle$  that will remain very close, but *not* identical, to the  $n$ th (non-degenerate) instantaneous eigenstate of  $H(\vec{\mathcal{R}})$ ,  $|n[\vec{\mathcal{R}}(t)]\rangle$ , whose eigenvalue is  $\epsilon_n[\vec{\mathcal{R}}(t)]$ . To single out the anomalous contribution to  $V_\mu$  from the adiabatic evolution of  $|\psi_n(t)\rangle$ , we introduce

$$\tilde{H}(\vec{\mathcal{R}}) = H(\vec{\mathcal{R}}) - \epsilon_n(\vec{\mathcal{R}}), \quad (13.85)$$

thus

$$V_\mu = \frac{\partial \epsilon_n(\vec{\mathcal{R}})}{\partial \mathcal{R}_\mu} + \frac{\partial \tilde{H}(\vec{\mathcal{R}})}{\partial \mathcal{R}_\mu}, \quad (13.86)$$

and the last term above gives rise to the anomalous velocity:

$$V_\mu^a = \langle \psi_n(t) | \frac{\partial \tilde{H}(\vec{\mathcal{R}})}{\partial \mathcal{R}_\mu} | \psi_n(t) \rangle. \quad (13.87)$$

We now consider the time-evolution of  $|\psi_n(t)\rangle$ . After factoring out the dynamical phase factor,

$$|\psi_n(t)\rangle = |\tilde{\psi}_n[\vec{\mathcal{R}}(t)]\rangle \exp \left[ -\frac{i}{\hbar} \int_0^t dt' \epsilon_n(t') \right], \quad (13.88)$$

the ket  $|\tilde{\psi}_n[\vec{\mathcal{R}}(t)]\rangle$  satisfies a (reduced) Schrödinger equation

$$i \hbar \frac{\partial}{\partial t} |\tilde{\psi}_n[\vec{\mathcal{R}}(t)]\rangle = \tilde{H}[\vec{\mathcal{R}}(t)] |\tilde{\psi}_n[\vec{\mathcal{R}}(t)]\rangle. \quad (13.89)$$

Of course, using  $|\tilde{\psi}_n[\vec{\mathcal{R}}(t)]\rangle$  (instead of  $|\psi_n(t)\rangle$ ) in Eq. (13.87) yields the same result. Now, taking  $\partial_\mu = \partial/\partial \mathcal{R}_\mu$  on both sides of Eq. (13.89), and leaving the  $t$ -dependence of  $\vec{\mathcal{R}}$  implicit, we have

$$i \hbar \partial_\mu [|\partial_t \tilde{\psi}_n(\vec{\mathcal{R}})\rangle] = [\partial_\mu \tilde{H}(\vec{\mathcal{R}})] |\tilde{\psi}_n(\vec{\mathcal{R}})\rangle + \tilde{H}(\vec{\mathcal{R}}) [|\partial_\mu \tilde{\psi}_n(\vec{\mathcal{R}})\rangle], \quad (13.90)$$

from which we obtain, using Eq. (13.87),

$$V_\mu^a = -\langle \tilde{\psi}_n(\vec{\mathcal{R}}) | \tilde{H} | \partial_\mu \tilde{\psi}_n(\vec{\mathcal{R}}) \rangle + i \hbar \langle \tilde{\psi}_n(\vec{\mathcal{R}}) | \partial_\mu | \partial_t \tilde{\psi}_n(\vec{\mathcal{R}}) \rangle \quad (13.91)$$

$$= i \hbar [\langle \partial_t \tilde{\psi}_n(\vec{\mathcal{R}}) | \partial_\mu \tilde{\psi}_n(\vec{\mathcal{R}}) \rangle - \langle \partial_\mu \tilde{\psi}_n(\vec{\mathcal{R}}) | \partial_t \tilde{\psi}_n(\vec{\mathcal{R}}) \rangle]. \quad (13.92)$$

In the last step we used the fact that

$$i\hbar\langle\tilde{\psi}_n|\partial_t\tilde{\psi}_n\rangle=\langle\tilde{\psi}_n|\tilde{H}|\tilde{\psi}_n\rangle=O[(\partial_t\mathcal{R})^2] \quad (13.93)$$

(see Eq. (8.37)) and is thus negligible in the adiabatic limit. Also, using the fact that in the adiabatic limit we have  $|\tilde{\psi}_n\rangle\rightarrow|n(\vec{\mathcal{R}})\rangle$ , and

$$\frac{\partial}{\partial t}=\sum_v\left(\frac{\partial\mathcal{R}_v}{\partial t}\right)\partial_v, \quad (13.94)$$

we obtain

$$V_\mu^a=i\hbar\sum_v[\langle\partial_vn(\vec{\mathcal{R}})|\partial_\mu n(\vec{\mathcal{R}})\rangle-\langle\partial_\mu n(\vec{\mathcal{R}})|\partial_vn(\vec{\mathcal{R}})\rangle](\partial_t\mathcal{R}_v) \quad (13.95)$$

$$=-\hbar\sum_v\left(\frac{\partial\mathcal{R}_v}{\partial t}\right)\omega_{\mu v}, \quad (13.96)$$

from which Eq. (13.76) follows as a special case where  $\vec{\mathcal{R}}=\vec{k}$ . We note that the derivation above was formulated entirely within a single band, and we did *not* need to invoke the existence of other bands explicitly.

Equation (13.76) helps establish some important symmetry properties of the Bloch-band Berry curvature  $\vec{b}_n(\vec{k})$ . Under time-reversal transformation, we have  $\vec{v}\rightarrow-\vec{v}$ ,  $\vec{k}\rightarrow-\vec{k}$ , and  $\vec{E}\rightarrow\vec{E}$ . Hence, in the presence of **time-reversal symmetry**, we have

$$\vec{b}_n(\vec{k})=-\vec{b}_n(-\vec{k}). \quad (13.97)$$

Under space inversion or parity transformation, we have  $\vec{v}\rightarrow-\vec{v}$ ,  $\vec{k}\rightarrow-\vec{k}$ , and  $\vec{E}\rightarrow-\vec{E}$ . Thus, in the presence of inversion symmetry, we have

$$\vec{b}_n(\vec{k})=\vec{b}_n(-\vec{k}). \quad (13.98)$$

Notice that, as we discussed earlier, the Berry curvature vector  $\vec{b}$  has the same properties as an ordinary magnetic field  $\vec{B}$ . It is odd under time reversal and even under parity (i.e. it is a pseudovector). As a result,  $\vec{b}_n(\vec{k})=\vec{0}$  everywhere when both symmetries are present. On the other hand, in the absence of either symmetry we generically have non-zero  $\vec{b}_n(\vec{k})$  and  $\vec{v}_a(n,\vec{k})$ , and the latter contributes to transport. Since the anomalous velocity is proportional to the vector (cross) product of the electric field and the Berry curvature, it is therefore *perpendicular* to both. As a result, it plays a particularly important role in the Hall effect and other related transport phenomena (e.g. the anomalous Hall effect in ferromagnets [94]).

## 13.5 Topological Quantization of Hall Conductance of Magnetic Bloch Bands

In the presence of a magnetic field, time-reversal symmetry is broken, thus  $\vec{b}_n(\vec{k})$  and  $\vec{v}_a(n,\vec{k})$  are (generically) non-zero; this leads to the Hall effect. For the case of a uniform magnetic field (and no periodic potential) discussed earlier in Chapter 12,  $\vec{v}_a(n,\vec{k})$  is nothing but the  $\vec{E}\times\vec{B}$  drift velocity we encountered back in Chapter 8. Equation (13.76) allows us to discuss the Hall effect and in particular, its quantization under more general settings. For a filled magnetic band in 2D, we have the Hall current density from electrons in that band:

$$\vec{J}_n=-\frac{e}{A}\sum_{\vec{k}\in\text{1BZ}}\vec{v}_a(n,\vec{k})=\sigma_{xy}^n\hat{z}\times\vec{E}, \quad (13.99)$$

where  $A$  is the area of the system. Thus

$$\sigma_{xy}^n = \frac{e^2}{\hbar A} \sum_{\vec{k} \in \text{1BZ}} b_n(\vec{k}) = \frac{e^2}{h} \frac{1}{2\pi} \int_{\vec{k} \in \text{1BZ}} b_n(\vec{k}) d\vec{k}. \quad (13.100)$$

We note that in 2D  $\vec{b}_n(\vec{k})$  always points in the  $\hat{z}$  direction, and can thus be viewed as a scalar (with both signs). Equation (13.100) suggests that the Hall conductance of a filled band is proportional to the *total* Berry curvature of the band. We are going to show below that

$$\frac{1}{2\pi} \int_{\vec{k} \in \text{1BZ}} b_n(\vec{k}) d\vec{k} = C_n \quad (13.101)$$

is an integer known as the (first) **Chern number**, as long as the band is not degenerate with other bands for any  $\vec{k}$ . This leads to the quantization of the Hall conductance in units of  $e^2/h$ .

Before proving Eq. (13.101), let us discuss a related result in geometry, which is more familiar and in a simpler setting. This is the famous **Gauss–Bonnet theorem**, which states that the total curvature of a closed 2D surface  $M$  is quantized in units of  $2\pi$ :

$$\frac{1}{2\pi} \int_M K dA = 2 - 2g_M, \quad (13.102)$$

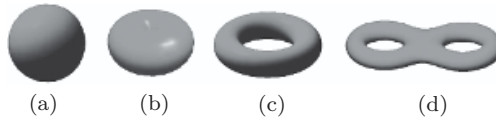
where  $K$  is the local Gaussian curvature and  $g_M$  is the **genus** or “number of handles” of the surface  $M$ .<sup>24</sup> A few examples are illustrated in Fig. 13.5. A physicist’s “proof” of Eq. (13.102) can be found in Appendix D.

Equation (13.102) is significant as it tells us the total curvature of a 2D surface is a quantized *topological invariant* of the surface. That is, smooth changes of its geometry, which change the curvature  $K$  *locally*, do *not* change  $g_M$  which is a *global* or *topological* property of the surface. Similarly, a smooth change of the band structure changes the local Berry curvature, but cannot change the (quantized) Chern number and the Hall conductance. Thus  $\sigma_{xy}$  is a global property of the magnetic band, and its quantization has a topological origin similar to the Gauss–Bonnet theorem.

As another warm-up, let us inspect the total Berry curvature of a closed (2D) surface  $M$  in the (3D) parameter space of the ground state of the spin-1/2 (or two-level) problem discussed in Section 13.3:

$$C_- = \frac{1}{2\pi} \int_M \vec{b} \cdot d\vec{S}. \quad (13.103)$$

From Gauss’ theorem we find that  $C_-$  depends only on the Berry monopole charge enclosed in  $M$ :  $C_- = 1$  if  $M$  encloses the origin (which is where the monopole is located, see Fig. 13.6(a)), and  $C_- = 0$  otherwise (Fig. 13.6(b)). We thus find that  $C_-$ , which is analogous to the Chern number, takes two possible quantized values. Since  $C_-$  is *independent* of the shape or geometry of  $M$ , it is a topological invariant of the surface  $M$ . Note that the surfaces with  $C_- = 0$  and  $C_- = 1$  are topologically distinct in this case because, in order to connect the two smoothly, one needs to have



**Figure 13.5** A few examples of closed 2D surfaces. (a) A sphere. (b) A genus-zero closed surface, which is topologically equivalent to a sphere. (c) A torus, with genus one. (d) A genus-two torus.

<sup>24</sup> More precisely, the genus of a connected, orientable surface is the maximum number of cuts along non-intersecting closed simple curves that can be made without breaking the object into disconnected pieces.

surfaces that contain the origin (Fig. 13.6(c)). As discussed in Section 13.3, the origin  $\vec{h} = 0$  is a singular point where the Hamiltonian and the gap between the two states vanish, which should be removed from the parameter space. As a result, surfaces of the type illustrated in Fig. 13.6(c) are not allowed, leading to two distinct topological classes of closed surfaces.<sup>25</sup>

Now let us prove Eq. (13.101), following Thouless *et al.* [88]. We write the left-hand side as

$$\frac{1}{2\pi} \int_{\vec{k} \in \text{1BZ}} \omega_n(\vec{k}) dk_x dk_y = \frac{1}{2\pi} \oint \vec{A}_n(\vec{k}) \cdot d\vec{k} \quad (13.104)$$

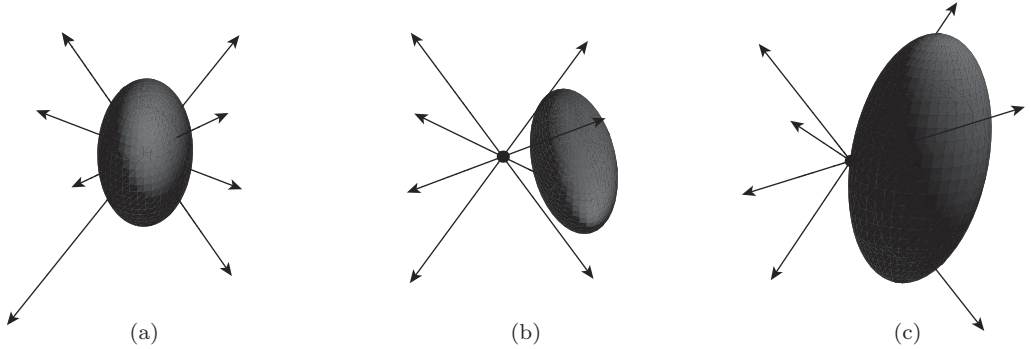
$$= \frac{i}{2\pi} \oint d\vec{k} \cdot \left[ \int d^2\vec{r} u_{n\vec{k}}^*(\vec{r}) \nabla_{\vec{k}} u_{n\vec{k}}(\vec{r}) \right], \quad (13.105)$$

where the loop integral in  $\vec{k}$  space is around the boundary of the first Brillouin zone (1BZ); see Fig. 13.7. Note that the opposite edges of the 1BZ differ from each other by a reciprocal lattice vector  $\vec{G}$ , and are thus *equivalent*. Naively one would think this equivalence means

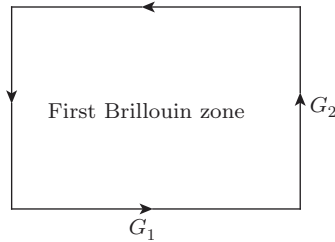
$$\psi_{n\vec{k}}(\vec{r}) = \psi_{n, \vec{k} + \vec{G}}(\vec{r}) \quad (13.106)$$

and thus

$$u_{n\vec{k}}(\vec{r}) = e^{-i\vec{G} \cdot \vec{r}} u_{n, \vec{k} + \vec{G}}(\vec{r}). \quad (13.107)$$



**Figure 13.6** A closed surface in the Hamiltonian parameter space of a two-level system that: (a) encloses the origin; (b) does not enclose the origin; and (c) intersects the origin. The origin is a singular point where the level spacing vanishes. It is the source of the Berry flux (represented by arrows), very much like a magnetic monopole, which is the source of magnetic flux.



**Figure 13.7** Illustration of integration of Berry connection around the boundary of the first Brillouin zone (1BZ). Notice that opposite edges of the 1BZ differ from each other by a reciprocal lattice vector and are thus equivalent.

<sup>25</sup> The reader is encouraged to ponder the similarity and difference between the situation here and that in the Aharonov–Bohm effect. Does the point singularity here lead to distinct topological classes of closed loops?

If that were the case we would find that the integral of Eq. (13.105) vanishes due to the cancelation of contributions from opposite edges of the 1BZ. However, in general, as with all quantum symmetries, we should only expect  $\psi_{\vec{k}}(\vec{r})$  to return to itself under  $\vec{k} \rightarrow \vec{k} + \vec{G}$  up to a phase factor; this implies

$$\psi_{n\vec{k}}(\vec{r}) = e^{-i\theta_n(\vec{k})} \tilde{\psi}_{n\vec{k}}(\vec{r}), \quad (13.108)$$

with  $\tilde{\psi}_{n\vec{k}}(\vec{r}) = \tilde{\psi}_{n,\vec{k}+\vec{G}}(\vec{r})$  being a periodic function in reciprocal space. Obviously, if  $\theta_n(\vec{k})$  is a constant, Eq. (13.105) yields zero. For non-trivial  $\theta_n(\vec{k})$  we have an additional contribution due to the  $\vec{k}$ -dependence of  $\theta_n$ :

$$\frac{i}{2\pi} \oint d\vec{k} \cdot \left[ \int d^2\vec{r} u_{n\vec{k}}^*(\vec{r}) \nabla_{\vec{k}} u_{n\vec{k}}(\vec{r}) \right] = \frac{1}{2\pi} \oint d\vec{k} \cdot \nabla_{\vec{k}} \theta_n(\vec{k}) = C_n, \quad (13.109)$$

thus proving Eq. (13.101). In the above we used the fact that since  $e^{-i\theta_n(\vec{k})}$  is single-valued, the change of  $\theta_n(\vec{k})$  around a loop must be an integer multiple of  $2\pi$ .<sup>26</sup> Bands having non-zero Chern numbers are topologically non-trivial, and are often referred to as **Chern bands** in the literature.

In the presence of time-reversal symmetry, Eq. (13.97) guarantees that  $C_n = 0$  for all bands. However, as we will see in Chapter 14, this does *not* mean that all these bands are topologically trivial. Indeed, in the presence of spin-orbit coupling we can have topologically non-trivial bands because of the coupling of the spin degree of freedom to the momentum. As we will discuss in Chapter 14, these bands are characterized by a set of topological quantum numbers that can take only two possible values, 0 or 1. Such bands can exist both in 2D and in 3D crystals, and, as for the Chern-band case studied here, the gaps between bands with different topological quantum numbers generically close at the boundary of the sample; more on this point later.

### 13.5.1 Wannier Functions of Topologically Non-trivial Bands

In Section 7.4.2 we demonstrated that it is possible, using Bloch states of a Bloch band, to construct complete and orthonormal localized orbitals known as Wannier functions. This procedure can also be carried out for magnetic Bloch bands, including those carrying non-zero Chern numbers. Here we show that, while this is true, it is *not* possible to have exponentially localized Wannier functions for such topologically non-trivial bands. In the following we present a very simple argument due to Thouless [95]. Let us assume that such exponentially localized Wannier functions do exist. Then one can use them to reconstruct the Bloch states:

$$|\Psi_{n\vec{q}}\rangle = \frac{1}{\sqrt{N}} \sum_j e^{i\vec{q} \cdot \vec{R}_j} |\chi_{nj}\rangle, \quad (13.110)$$

where  $|\chi_{nj}\rangle$  is the Wannier function at site  $j$ . Exponential localization guarantees the absolute convergence of the summation over  $j$ . This immediately implies  $|\Psi_{n\vec{q}}\rangle = |\Psi_{n\vec{q}+\vec{G}}\rangle$ , thus the Chern number must be zero, as discussed earlier. Thus, to have a non-zero Chern number, the summation over  $j$  above *cannot* be absolutely convergent, which implies that the Wannier function has a decay that is slower than  $1/r^2$  at large distance  $r$  from its center.<sup>27</sup>

<sup>26</sup> The mathematical origin of this quantization is very similar to the quantization of vorticity of a vortex in a superfluid or superconductor (to be discussed in Chapter 19), except here the vorticity is in *momentum* space instead of in real space.

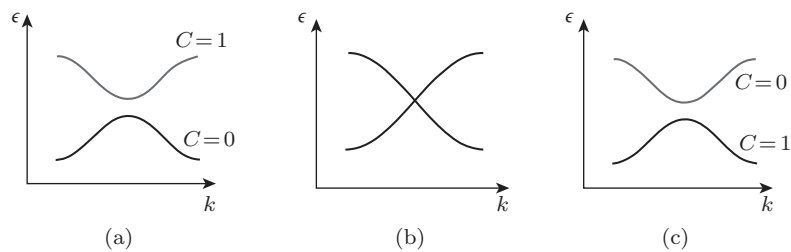
<sup>27</sup> Note that this discussion applies to 2D systems, which are the only ones that have band structures that can be labeled by the (first) Chern number.

Physically, this reflects the fact that such topologically non-trivial Bloch states are *intrinsically non-local*. We have already seen one aspect of such non-locality in *real space*, namely that the band spectra are very sensitive to the presence of edges or boundaries; the presence (or absence) of edges is a topological property of the system in which the electrons live. We also saw that these edge states are robust, *not* associated with any symmetries (or their breaking), and cannot be destroyed by disorder (unless it is so strong as to destroy the state in the bulk, for example, the quantum Hall effect itself). On the other hand, ultra-local Wannier functions cannot sense and thus be sensitive to the (change of) topology of the system. As we will see through more examples in later chapters, such sensitivity to system topology (e.g. dependence of the ground state degeneracy on the genus of the manifold in which the system lives) is a hallmark of topological phases.

### 13.5.2 Band Crossing and Change of Band Topology

Being an integer, the Chern number and thus the Hall conductance *cannot* change continuously with changes of band structure; the quantization is topologically protected. A change in Chern number can occur only when the band gap vanishes and two bands become degenerate at one point in the first Brillouin zone; see Fig. 13.8. This is also referred to as a band crossing. At such a degeneracy point there is ambiguity as to which state belongs to which band, and thus the local Berry curvature is undefined there.<sup>28</sup> This ambiguity or singularity is also reflected in Eq. (13.22), where the energy denominator vanishes. Typically such a degeneracy/crossing occurs on tuning one external parameter, and the Chern numbers of the two bands each change by  $\pm 1$  before and after the crossing, with their *sum* remaining unchanged. In the following we discuss such band crossings in some detail.

When two bands come close together in some region of the 1BZ we can ignore the presence of other bands that are far away in energy (their effects can be treated perturbatively if necessary), and reduce the band structure calculation to a two-level problem similar to that discussed in Section 13.3. We immediately see that, in order for the gap in a two-level system to vanish, all three components of the (effective) magnetic field  $\vec{h}$  that couple to the pseudospin  $\vec{\sigma}$  acting on the two-level Hilbert space (see Eq. (13.43)) must vanish. However, in 2D we have only two independent components of the lattice momentum  $\vec{k}$ , and thus generically we will *not* encounter a band crossing, unless we *fine-tune* one more parameter in the Hamiltonian, which we call  $m$ .<sup>29</sup> Thus we have  $\vec{h} = \vec{h}(k_x, k_y, m)$ . Without loss



**Figure 13.8** Change of Chern numbers through band degeneracy/crossing. (a) Before crossing. (b) Degeneracy or crossing between two bands occurs at one point in the first Brillouin zone. (c) After crossing, the Chern numbers of the two bands involved change (typically by  $\pm 1$ ), with their sum unchanged.

<sup>28</sup> The reader should note that the change of genus in 2D surfaces through a singularity as illustrated in Fig. D.1(b), where the local Gauss curvature is not well-defined at the point at which the two arms join, is similar to the band-crossing point in Fig. 13.8(b).

<sup>29</sup> As we will learn in Section 14.4, the situation is very different in 3D, where we can have such crossing *without* fine-tuning, leading to fascinating topological physics.

of generality, we may choose the crossing point to be at  $k_x = k_y = m = 0$  by proper parameterization, with the minimal (non-zero) band gap at  $k_x = k_y = 0$  for  $m \neq 0$ . By defining the direction of  $\partial \tilde{h} / \partial m$  to be the  $\hat{z}$  direction in the pseudospin space and choosing its magnitude to be 1, and using a proper linear transformation in the  $(k_x, k_y)$  space (which alters its geometry but does not affect its topology), we can bring the two-level Hamiltonian (13.43) to the form

$$h(\vec{k}) = v(\pm k_x \sigma_x + k_y \sigma_y) + m \sigma_z, \quad (13.111)$$

or

$$h = v(\pm p_x \sigma_x + p_y \sigma_y) + m \sigma_z, \quad (13.112)$$

where  $v$  is a velocity scale.<sup>30</sup> We thus encounter the 2D Dirac Hamiltonian again, with a band crossing that occurs when the mass  $m$  is tuned to zero. As we learned in Section 13.3, the crossing point is a “magnetic” monopole carrying  $2\pi$  Berry flux. Thus the Chern number of each band changes by  $\pm 1$  when the crossing occurs. Also, because the Berry flux is opposite for the two levels, the sum of their changes is zero. The change of band topology is accompanied by (or caused by!) a change of the sign of the Dirac mass, which is similar to what we learned in Section 7.6 for a 1D band.

The reader should note the similarity between the change of Chern number here and the change of total Berry flux through a 2D closed surface in the 3D parameter space of the spin-1/2 Hamiltonian illustrated in Fig. 13.6. Here the 1BZ for any fixed  $m$  (Fig. 13.8(b)) is a closed surface (a torus) in the 3D parameter space  $(k_x, k_y, m)$ , and  $m = 0$  corresponds to the special case that this surface contains the singular point  $\vec{h} = 0$  of Fig. 13.6(c), where the total Berry flux through the surface (the Chern number in the present case) changes discontinuously.

### 13.5.3 Relation Between the Chern Number and Chiral Edge States: Bulk–Edge Correspondence

The facts discussed above about topological protection can be used to make very powerful and general arguments about the guaranteed existence of gapless modes at the edges of (free-electron) materials whose band structure contains gapped bands with non-trivial topological quantum numbers. Suppose that the Hamiltonian contains a parameter  $\lambda$ . Further suppose that when  $\lambda < 0$  the band structure has the form shown in Fig. 13.8(a), and that, when  $\lambda > 0$ , the band structure has the form shown in Fig. 13.8(c). Now imagine an interface between two materials, one with  $\lambda < 0$  and one with  $\lambda > 0$ . As a first approximation, imagine that the interface is not sharp but gradual and can be modeled by having  $\lambda$  vary smoothly in space from negative to positive. If the spatial variation is sufficiently slow, we can define a local band structure which varies in space. In this “adiabatic” interface limit it is clear that the gap has to close at the interface. This result remains generically robust even in the limit of a sharp interface. Furthermore, if, instead of an interface between two materials, we simply have a material with an edge (e.g. the second material is the vacuum), the same argument applies. The bulk bands are topologically non-trivial but the vacuum “bands” are topologically trivial. Hence the gap generically has to close at the edge.

<sup>30</sup> Without the proper linear transformation in the  $(k_x, k_y)$  space one would generically encounter a rank-two velocity tensor  $v^{\mu\nu}$  describing how  $(k_x, k_y)$  are coupled to  $(\sigma_x, \sigma_y)$ :  $h = \sum_{\mu, \nu=x,y} v^{\mu\nu} k_\mu \sigma_\nu + m \sigma_z$ , whose role is very similar to the rank-two effective-mass tensor in the effective-mass approximation. The transformation thus consists of first diagonalizing this velocity tensor (exactly the same as is often done to the effective mass tensor), and then performing a scale transformation to make the two velocity eigenvalues identical in magnitude, thus reducing the velocity tensor to a single number  $v$ . The two cases with the  $\pm$  sign in front of  $\sigma_x$  correspond to the cases of the two velocity eigenvalues having the same or opposite signs, giving rise to two different chiralities of the Dirac fermions. The meaning of chirality will be discussed in connection with Eq. (13.120).

To prove the robustness of the gapless edge/interface modes, and understand their dispersion relations, we may model the electronic states near the edge/interface using the Dirac Hamiltonian (13.112), with a *spatially dependent* mass that changes sign at the edge/interface. As we learned in Section 7.6 in 1D, this leads to a zero-energy state bound to the 0D edge/interface. Here the edge/interface is 1D, and the bound state develops a chiral dispersion along the edge/interface; see Exercise 13.8.

After these quite general arguments, let us return to the specific case of Landau levels as examples of topologically non-trivial bands with  $C \neq 0$ . When the potential  $V(\vec{r}) = 0$ , but in the presence of a uniform external magnetic field, we have seen that each Landau level is a (special type of) magnetic Bloch band, and each has Chern number  $C = 1$ . Correspondingly, they each carry one branch of gapless edge modes at the edge. When  $V(\vec{r}) \neq 0$  we can (depending on the potential) have magnetic Bloch bands with non-zero Chern number  $C \neq 1$ . We will show below that, for each band that carries non-zero Chern number  $C_n \neq 0$ , there must be  $|C_n|$  branches of chiral edge modes. It will turn out that the chirality (or direction of propagation) is determined by the sign of  $C_n$ .

To proceed, we consider a rectangular system with size  $L_x \times L_y$ , impose (magnetic) periodic boundary conditions along the  $\hat{y}$  direction, and leave the boundary conditions along the  $\hat{x}$  direction open. Then the geometry of the system is that of a cylinder (wrapped around the  $\hat{y}$  direction), and contains two edges. See Fig. 13.9 for an illustration. To induce a Hall current along the  $\hat{x}$  direction, we insert a *time-dependent* magnetic flux through the hole of the cylinder, which results in an electric field along the  $\hat{y}$  direction through Faraday's law:

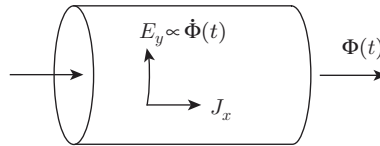
$$E_y = \frac{1}{cL_y} \frac{d\Phi(t)}{dt}. \quad (13.113)$$

The resultant *bulk* Hall current moves charge from one edge to the other. We consider a specific case in which the flux  $\Phi$  increases slowly (and thus adiabatically) from 0 to one flux quantum  $\Phi_0$  over a long period  $T$ , and calculate the amount of charge transferred from one edge to another:

$$Q = \int_0^T dt \int_0^{L_y} dy J_x = \sigma_{xy}^n \int_0^T dt \int_0^{L_y} dy E_y = \sigma_{xy}^n \Phi_0 / c = C_n e. \quad (13.114)$$

In the above we consider only the contribution from band  $n$  that is fully occupied in the bulk, and find that precisely  $C_n$  electrons have been transferred. We note that, after  $\Phi$  has changed from 0 to one flux quantum  $\Phi_0$ , the Hamiltonian of the 2DEG has returned to the original one (up to a gauge transformation<sup>31</sup>):

$$H(\Phi = 0) \cong H(\Phi = \Phi_0), \quad (13.115)$$



**Figure 13.9** Illustration of the Laughlin and Halperin construction for flux insertion through a cylinder. Not shown is the large magnetic field pointing radially outward normal to the surface of the sample that induces the quantum Hall effect. See Refs. [96, 97].

<sup>31</sup> Again, the physics is that the flux tube produces a phase shift of  $2\pi$  for a particle encircling it, and this is physically unobservable.

where  $\cong$  stands for equivalence. The spectra of  $H(\Phi = 0)$  and  $H(\Phi = \Phi_0)$  are identical (cf. Exercise 10.1), and so are the corresponding eigenstates up to a gauge transformation. Thus the charge transfer we found in Eq. (13.114) must be the result of changes of occupation numbers of states localized at the edge. Furthermore, since the whole process is adiabatic, the states at  $t = 0$  and  $t = T$  must have an infinitesimal energy difference in the thermodynamic limit. This means that the states whose occupation numbers have changed must be at the Fermi energy, even though the Fermi energy lies in a band gap in the bulk. This immediately implies the existence of gapless edge states, whose spectra must look like that in Fig. 12.4 for  $C_n = 1$ . For general  $C_n \neq 0$  there are  $|C_n|$  branches of chiral edge modes, and for  $C_n < 0$  the velocities of these edge modes are directed oppositely to the velocities of those with  $C_n > 0$ . For the whole system we need to add up the Chern numbers of all *occupied* bands (those below the Fermi energy in the bulk) to obtain the total Chern number:

$$C = \sum_{n \in \text{occupied}} C_n, \quad (13.116)$$

and precisely  $C$  electrons are transferred from one edge to the other during this adiabatic process. We thus have  $|C|$  gapless chiral modes (that cross the Fermi energy) at the edge.<sup>32</sup>

The above argument shows that adiabatic addition of a flux quantum through the cylinder illustrated in Fig. 13.9 results in transfer of an integer number of electrons from one edge to the other. This implies that the Hall conductivity is quantized in integer multiples of  $e^2/h$  as discussed in Chapter 12 and confirms that this precise quantization is topologically protected. One obvious remark is that the reader should not forget that zero is an integer! The system might simply be insulating and have zero Hall conductivity. To distinguish this from the integer quantum Hall states with  $C \neq 0$  but nevertheless having Fermi energy in a bulk band gap, the latter are also referred to as **Chern insulators** as they would be classified as band insulators in the transitional classification scheme of Chapter 7 (that does not take into account the topology of bands).

The other less obvious remark is that our entire discussion has been based on the idea that the occupation numbers of individual single-particle states are well-defined (at least in some Fermi-liquid sense; see Section 15.11). When we study the fractional quantum Hall effect in Chapter 16 we will see that strong electron–electron interactions can produce a strikingly different result, namely that the topological quantum numbers can be rational fractions! In fact, if we simply repeat the Laughlin–Halperin construction of threading a flux quantum through the cylinder, we find that it is not an electron which is transferred from one edge to the other, but rather a fractionally charged quasiparticle. We will further see that this is deeply connected to the fact that the ground state degeneracy of the interacting many-body system with (magnetic) periodic boundary conditions depends on the genus of the manifold on which the electrons move – a result which is clearly of topological origin.

**Exercise 13.8.** Consider a 2D, two-component Dirac Hamiltonian with an  $x$ -dependent mass:

$$H = v_F(\sigma_x p_x + \sigma_y p_y) + m(x)\sigma_z, \quad (13.117)$$

with  $m(x \rightarrow \pm\infty) = \pm m_0$ , with  $m_0 > 0$ . Using the translation-invariance along the  $y$  direction, energy eigenstates can be chosen to be plane waves along the  $y$  direction with wave vector  $k_y$ . (i) Show that for  $k_y = 0$  there is a single zero-energy bound state localized near  $x = 0$ . Hint: the eigenvalue equation for this zero mode is closely related to those of the SSH model

<sup>32</sup> In principle we may also have  $C_+$  forward-propagating modes and  $C_-$  backward-propagating modes, with  $C_+ - C_- = C$ . However, backscattering due to disorder would normally localize pairs of counter-propagating modes, leaving us with only the  $|C|$  remaining (unpaired) chiral modes that are topologically protected.

discussed in Section 7.6. In particular, see Exercise 7.23. To reveal the similarity, it is useful to choose a representation of the Pauli matrices in which  $\sigma_y$ , instead of  $\sigma_z$ , is diagonal. This representation will be particularly useful for the next question. (ii) Now consider generic values of  $k_y$ , and show that this zero-energy bound state becomes a dispersive branch of states, which are precisely the edge/interface states between two integer quantum Hall states (or Chern insulators) whose Chern numbers differ by one. Find the edge-state dispersion relation. (iii) How does the dispersion/chirality change for  $m_0 < 0$ ?

## 13.6 An Example of Bands Carrying Non-zero Chern Numbers: Haldane Model

As an example of bands carrying non-zero Chern numbers, we discuss a model introduced by Haldane in 1988 [98]. This is basically a tight-binding model on the honeycomb lattice. As we learned in Section 7.5, when only nearest-neighbor hopping  $t$  is included, the two bands are degenerate at two Dirac points  $K$  and  $K'$ , and there is no band gap. In this case the system has both time-reversal  $\Theta$  and inversion (or parity,  $\mathcal{P}$ ) symmetries, so the Berry curvature is zero everywhere (except at  $K$  and  $K'$  where it is not defined). Time-reversal  $\Theta$  symmetry also guarantees the Hall conductance is zero. To have well-defined Chern numbers we need to open up a band gap. To have non-zero Chern numbers and Hall conductance we need to break  $\Theta$  symmetry.

The easiest way to open a band gap is to introduce an energy difference  $2m$  between the A and B sublattices, which breaks  $\mathcal{P}$  but not  $\Theta$  (see Exercise 7.19). In this case the Hamiltonian in momentum space becomes

$$h(\vec{k}) = \begin{pmatrix} m & -tf(\vec{k}) \\ -tf^*(\vec{k}) & -m \end{pmatrix}, \quad (13.118)$$

with band dispersions

$$\epsilon_{\pm}(\vec{k}) = \pm \sqrt{m^2 + t^2 |f(\vec{k})|^2}, \quad (13.119)$$

and a band gap of  $2|m|$  is opened up at  $K$  and  $K'$ . In this case the Chern numbers are well-defined but zero for both bands, as guaranteed by  $\Theta$  symmetry. It is instructive to understand how this comes about in detail, as the breaking of  $\mathcal{P}$  allows non-zero local Berry curvature in the band, even though the overall Chern number vanishes.

For simplicity let us assume  $|m| \ll |t|$  (the quantization of the Chern numbers is of course independent of the magnitude of  $|m|$ ). In this case, the effect of  $m$  is important only near the Dirac points, where  $|f(\vec{k})|$  is small. There  $h(\vec{k})$  reduces to *massive* Dirac Hamiltonians (note the similarity to Eq. (13.112), except that here the chirality is opposite for  $K$  and  $K'$ ):

$$H_{K,K'} = v_F(\pm\sigma_x p_x + \sigma_y p_y) + m\sigma_z, \quad (13.120)$$

where the plus sign applies to the  $K$  Dirac point and the minus sign applies to the  $K'$  Dirac point. Solving these Hamiltonians gives rise to the pseudospin configurations of the conduction band near  $K$  and  $K'$  as illustrated in Fig. 13.10 (assuming  $m > 0$ ). We see that near  $K$  and  $K'$  the pseudospin wraps around the upper hemisphere, and each configuration gives rise to integrated Berry curvature of  $\pi$  in magnitude (recall the two-level system discussed in Section 13.3), but in opposite directions; in other words the Berry curvatures are of opposite sign near  $K$  and  $K'$ . Thus, as guaranteed by  $\Theta$  symmetry, the total Chern number vanishes:

$$C_+ = \frac{1}{2} - \frac{1}{2} = 0. \quad (13.121)$$

Similarly  $C_- = 0$ .

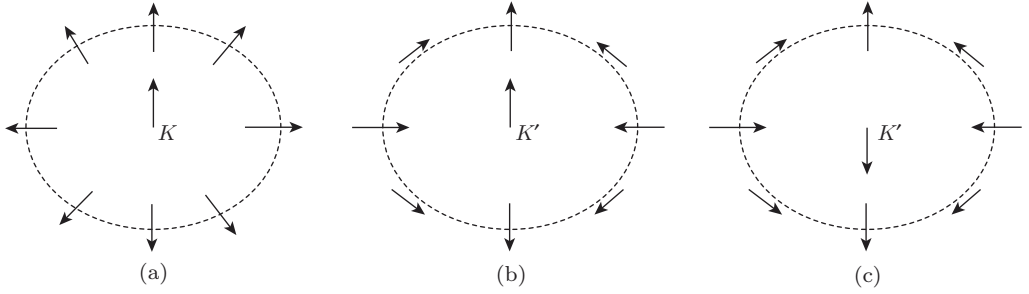
In order to have non-zero Chern numbers, we need to have a situation where the Berry curvatures are of the same sign near  $K$  and  $K'$ . This would be accomplished if the mass were of opposite signs at  $K$  and  $K'$ :

$$H'_{K,K'} = v_F(\pm\sigma_x p_x + \sigma_y p_y) \pm m'\sigma_z, \quad (13.122)$$

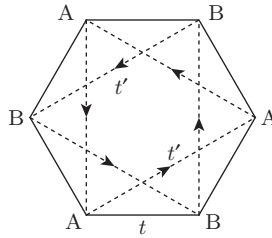
or equivalently, in a notation that unifies  $K$  and  $K'$ ,

$$H' = v_F(\tau_z\sigma_x p_x + \sigma_y p_y) + m'\tau_z\sigma_z, \quad (13.123)$$

where we have introduced another set of Pauli matrices  $\vec{\tau}$  that acts on the two-level valley or Dirac point space, with  $\tau_z = \pm 1$  for  $K$  and  $K'$ , respectively.<sup>33</sup> Now let us discuss how to get such a mass term from the microscopic lattice Hamiltonian. First of all, notice that it is diagonal in the sublattice index. The simplest term of this kind would be an on-site energy, but that just gives the regular mass term in Eq. (13.120). The other possibility is a hopping term between the sites in the *same* sublattice, the simplest of which is next-nearest-neighbor hopping; see Fig. 13.11. Furthermore, to have non-zero Chern number and Hall conductance, such a term must break  $\Theta$  symmetry; this is achieved by having



**Figure 13.10** Pseudospin configurations of the conduction band near the Dirac points. (a) At the  $K$  point, in the presence of either regular mass or Haldane mass. The pseudospin is up at the  $K$  point and winds in the  $xy$  plane further away. (b) At the  $K'$  point, in the presence of regular mass. The pseudospin is up at the  $K'$  point but winds in the opposite direction around the equator further away. (c) At the  $K'$  point, in the presence of the Haldane mass, which causes the pseudospin to point down at the  $K'$  point.



**Figure 13.11** Next-nearest-neighbor hopping on the honeycomb lattice. The arrows indicate the direction of hopping, with hopping-matrix element  $t'$ , which is *different* from the matrix element  $t'^*$  for hopping in the opposite direction.

<sup>33</sup> The  $\vec{\tau}$  matrices are independent of  $\vec{\sigma}$ , which acts on the pseudospin or sublattice space.

an imaginary next-nearest-neighbor hopping  $t' = i|t'|$ . With complex hopping-matrix elements, the corresponding direction of hopping must be specified, as hopping over the same bond in the opposite direction has a matrix element which is the complex conjugate of the original one.

In the presence of  $t'$ , we have

$$h(\vec{k}) = \begin{pmatrix} |t'|f'(\vec{k}) & -tf(\vec{k}) \\ -tf^*(\vec{k}) & -|t'|f'(\vec{k}) \end{pmatrix}, \quad (13.124)$$

with

$$f'(\vec{k}) = \sin(\vec{k} \cdot \vec{a}_1) - \sin(\vec{k} \cdot \vec{a}_2) - \sin[\vec{k} \cdot (\vec{a}_1 - \vec{a}_2)]. \quad (13.125)$$

For  $|t'| \ll |t|$ , the Hamiltonian reduces to Eq. (13.123), with

$$m' = -\frac{3\sqrt{3}}{2}|t'|. \quad (13.126)$$

The two bands are topologically non-trivial, with

$$C_{\pm} = \pm 1. \quad (13.127)$$

**Exercise 13.9.** Use Eqs. (13.120) and (13.123) to calculate the local Berry curvature near  $K$  and  $K'$ , in the presence of regular and Haldane masses, respectively, and show that the Chern numbers of the two bands are zero for the former, and  $\pm 1$  for the latter.

## 13.7 Thouless Charge Pump and Electric Polarization

We started this chapter by considering adiabatic time evolution, and found the presence of Berry phases in such processes. Thus far our discussion of the importance of the Berry phase and Berry curvature has been limited to charge transport by static Hamiltonians, for example in the (quantization of the) Hall conductivity. We also found that, due to the presence of the anomalous velocity, which is closely related to the Berry curvature of a Bloch band, a completely filled band *can* transport Hall current. The Hall effect exists in 2D and in 3D. A natural question is, can a completely filled 1D band give rise to charge transport? In this section we show that the answer is yes, via what is known as the **Thouless charge pump** [99, 100], which is a *time-dependent* but adiabatic process.

Consider a Hamiltonian that depends on a single parameter  $\lambda$  in a periodic manner:

$$H(\lambda) = H(\lambda + 1). \quad (13.128)$$

Let  $\lambda$  be time-dependent:

$$\lambda(t) = t/\tau, \quad (13.129)$$

as a result of which  $H$  is time-dependent and periodic with period  $\tau$ . Since the band of interest is completely filled, and we assume that it has a finite gap to other bands, we expect the time-evolution of the system to be adiabatic in the limit  $\tau \rightarrow \infty$ .

Now consider  $H$  to be a 1D periodic lattice Hamiltonian (generalization to higher dimensions is straightforward) with the Bloch wave vector  $k$  being a good quantum number. We then have a 2D parameter space for the  $k$ -dependent Hamiltonian  $h(k, \lambda)$ . Owing to the periodicity (13.128), the topology of this parameter space is a torus and identical to that of a 2D Brillouin zone. Owing to the time-dependence (13.129), a Bloch state picks up an anomalous velocity (see Eq. (13.96))

$$v_a^n(k) = -\omega^n(k, \lambda)/\tau, \quad (13.130)$$

where  $n$  is the band index, and  $\omega$  is the (only component of) Berry curvature in the  $(k, \lambda)$  space. Now consider the amount of charge that moves through the system in a period  $\tau$ , due to electrons completely filling band  $n$ :

$$Q = \int_0^\tau dt \langle J(t) \rangle = \frac{-e}{2\pi} \int_0^\tau dt \int_{\text{1BZ}} v_a^n(k) dk = e \int_0^1 d\lambda \int_{\text{1BZ}} \omega^n(k, \lambda) dk = eC_n, \quad (13.131)$$

where  $C_n$  is an integer-valued Chern number defined by the equation above. It should be clear that the origin of the quantization is identical to that of the Chern number of 2D bands, since here  $\lambda$  plays a role identical to that of the second momentum component. We thus find that a completely filled 1D band can also transport charge, provided that we have a suitable *time-dependent* Hamiltonian (or potential). For periodic time-dependence, such transport is quantized in an integer number of electrons per cycle. Such a system is known as a Thouless charge pump.

It is worth noting not only that the quantization of a Thouless charge pump has the same mathematical/topological origin as the quantization of the Hall conductivity, but also that they are physically equivalent to certain degree. This is because the Hall current along the  $x$  direction can be viewed as the result of pumping by a (linearly) time-dependent  $A_y$ , as seen in Eq. (13.80). Even though  $A_y$  is not periodic in time, the resultant Hamiltonian is periodic, modulo a so-called large gauge transformation; see Exercise 13.10.

Actual experiments realizing charge pumps can be quite different than the simple pedagogical example given above, and indeed do not necessarily follow the Thouless model of adiabatic pumping. **Electron “turnstile”** consist of a set of voltage-biased metallic islands connected by tunnel junctions [101]. If the voltage biases are cyclically modulated, charge can be driven through the turnstile in a controlled manner using an external bias. These devices rely on strong Coulomb interactions to fix the number of electrons allowed through the system at one per cycle. However, there is irreversible relaxation of the electrons as they move downhill in energy through the device (generating heat along the way).

Reversible **electron pumps** were subsequently developed [102]. These pump charge through the device in a direction that depends only on the direction of motion around the closed loop in parameter space and not on the externally applied bias (i.e. they can pump electrons “uphill” as well as “downhill”), in contrast to the turnstile. Electron turnstiles and pumps are interesting for metrological purposes because they can convert the cycle frequency  $f$  to current  $I = ef$  with accuracies of  $\sim 10^{-8}$  [103]. However, the internal states of the mesoscopic normal-metal islands are not gapped, and so the pumping is only quasi-adiabatic – becoming closer and closer to adiabatic only in the limit of zero pumping frequency. As we will learn in Chapter 20, superconductivity can gap out all the low-energy excited states of a metallic island, and a superconducting charge pump should therefore easily be able to enter the adiabatic pumping regime. Unfortunately, such devices are plagued by “quasiparticle poisoning” caused by broken Cooper pairs.<sup>34</sup>

The quantum dots [104] discussed in Section 9.5.3 do not involve a 1D spatially periodic system. They do, however, involve more than one control parameter (gate voltage), a necessary (but not sufficient) condition if the system is to be driven through a closed loop in parameter space that contains (or may contain) a net Berry flux which will lead to charge pumping. Optical lattices can be controlled by varying the relative phase and amplitude of different laser beams. Because there is only a single state per lattice site (in a given band), these systems can come much closer to the truly adiabatic case envisioned by Thouless for ground-state pumping [105, 106].

<sup>34</sup> Cooper pairs and quasiparticles will be introduced in Chapter 20.

### 13.7.1 Modern Theory of Electric Polarization

Electric polarization plays an important role in the electrodynamics of solids and the electrostatics of ferroelectric materials. The modern theory of polarization resolves various computational difficulties and ambiguities by extending the picture of Thouless pumping to consider adiabatic time evolution over a finite time interval, resulting in a final Hamiltonian and ground state at  $t$  with  $H(\lambda) = H(t/\tau) \neq H(0)$ . As we demonstrate below, we can use the Berry curvature to calculate the *change of electric polarization* of the system. We again illustrate this in 1D, where things are particularly simple because polarization can be represented as a scalar (instead of a vector).

We know that the derivative (or gradient) of the local polarization  $P(x)$  is related to the local charge density [18, 19]:

$$\partial_x P(x, t) = -\rho(x, t). \quad (13.132)$$

From the continuity equation

$$\partial_t \rho(x, t) = -\partial_x J(x, t) \quad (13.133)$$

we obtain

$$\partial_x [\partial_t P(x, t) - J(x, t)] = 0. \quad (13.134)$$

In a spatially periodic system we have<sup>35</sup>

$$\partial_t P(x, t) = J(x, t), \quad (13.135)$$

and integrating over  $t$  leads to

$$\Delta \langle P \rangle = \int_0^t dt' \langle J(t') \rangle = \sum_n \frac{-e}{2\pi} \int_0^t dt' \int_{\text{1BZ}} v_a^n(k) dk \quad (13.136)$$

$$= e \sum_n \int_0^\lambda d\lambda' \int_{\text{1BZ}} \omega^n(k, \lambda') dk. \quad (13.137)$$

In the above, the average is over  $x$ , or a unit cell if the system is strictly periodic, and the summation is over fully occupied bands. We thus find that the change of polarization  $P$  is determined by the Berry curvature  $\omega^n(k, \lambda)$ . It turns out that the absolute value of  $P$  is not well-defined due to (among other reasons) boundary charge contributions; only its difference between two states that are adiabatically connected to each other is well-defined and measurable. For further discussions of this subtle point see Ref. [91]. This modern formulation of the electric polarization is important in the study of the electrodynamics of solids and in such physical phenomena as ferroelectricity. It is also very helpful in removing ambiguities in the numerical computation of polarizations within band-structure calculations that study small system sizes using periodic boundary conditions.

**Exercise 13.10.** Consider a 2D system with finite size  $L_y$  along the  $y$  direction, and impose periodic boundary conditions on the wave function such that

$$\psi(x, y + L_y) = \psi(x, y). \quad (13.138)$$

This describes a physical situation illustrated in Fig. 13.9. Naively, one could “gauge away” a constant vector potential along the  $y$  direction via  $A'_y = A_y + d\phi(y)/dy$ . (i) Find a  $\phi(y)$

<sup>35</sup> This neglects possible effects associated with charge build-up at the boundaries. In 3D the corresponding equation is  $\nabla \cdot (\partial_t \vec{P} - \vec{J}) = 0$ , which allows for an additional divergence-free contribution in the equation below. We do not discuss these subtleties here.

that makes  $A'_y = 0$ . (ii) Show that for generic values of  $A_y$ , after this gauge transformation the transformed wave function no longer satisfies the boundary condition (13.138). (iii) Find the special values of  $A_y$  such that, after the gauge transformation, the transformed wave function still satisfies the boundary condition (13.138), and find the corresponding values of the flux through the hole of the cylinder in Fig. 13.9. Such gauge transformations are often referred to as large gauge transformations. For these values of  $A_y$  the Hamiltonian is gauge-equivalent to that with  $A_y = 0$ . (iv) Discuss the relation between these results and those of Exercise 10.1.